

Master Thesis

Evaluating Multimodal LLMs for Word-Level Intelligibility Classification and Phonetic Explanation

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Abstract

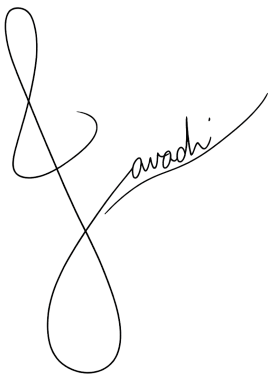
Reliable word-level assessment of second-language pronunciation calls not only for a judgment of whether a spoken word is intelligible, but also for an explanation of that judgment that a learner can act on. This thesis asks whether speech-enabled multimodal large language models can meet both demands. Three such models are evaluated on a word-level corpus of L2 English speech across six prompt conditions and a transcription-augmented variant; a single configuration is selected on a development set and evaluated once on a test set, its labels are compared with an earlier ASR-based system on the same corpus, and a sample of its explanations is annotated for feedback quality, acoustic grounding, and pedagogical safety. The results give a mixed picture. Prompted directly and without any task-specific fine-tuning, the best model reaches roughly the same precision as the ASR pipeline on the not-intelligible class - the class that matters most for feedback - while accepting far more unintelligible words, so that the pipeline remains marginally stronger overall. No single prompt is best across the three models. The explanations, however, seem to track the label the model has chosen rather than the recording: most are not grounded in audible evidence, many are unsafe to show a learner, and the specific phonetic contrasts at issue are rarely named. A correct label is therefore not, on its own, evidence of a grounded decision. The thesis contributes both an empirical account of these models on the task and an evaluation approach that treats acoustic grounding as a separate axis rather than assuming it follows from label accuracy.

Declaration of Authorship

I, Masoumeh Javadi Zarnaghi, declare that this thesis, titled *Evaluating Multimodal LLMs for Word-Level Intelligibility Classification and Phonetic Explanation* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a Master degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Date: 31 July 2026



Signed:

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Chapter 1

Introduction

1.1 What Counts as Correct Pronunciation?

This thesis was developed as part of an internship at the Centre for Global English at Vrije Universiteit Amsterdam, in the context of the MOOC *English Pronunciation in a Global World*, a course that has attracted more than 139,000 learners worldwide FutureLearn (2026). The course is built on the idea of English as a lingua franca. Rather than measuring a learner against a native speaker, the aim is to set the speaker's linguistic background aside and ask only whether their English works as a shared, global means of communication, that is, whether it can be understood. What this project ultimately seeks is a way of assessing pronunciation that follows this same principle, at a scale where it can actually support learners. Before such a system can be built, however, one prior question has to be settled: on what basis should a learner's speech be judged?

The assessment of second-language pronunciation rests on a choice of criterion: should a learner's speech be judged by how close it sounds to a native speaker, or by whether a listener understands it? Research on L2 speech has largely settled this question in favor of the second criterion. Munro and Derwing (1995) showed that accentedness and intelligibility are independent dimensions: speech can be heavily accented and still be fully intelligible. Intelligibility, in their sense, is the extent to which a speaker's intended message is actually recovered by a listener, and it is this, rather than closeness to a native model, that has become the dominant goal in pronunciation teaching (Levis, 2022). In this view, an accent is not an error to be removed, but a feature of linguistic diversity; what matters is communicative success, not native-likeness.

The same conclusion is reached from a different direction within the framework of English as a lingua franca. Jenkins (2000) says that in international interaction, where speakers of different first languages are now more likely to use English with one another than with a native speaker, imitating native norms is unnecessary and can even mislead communication, since some native features become obstacles to mutual understanding. For this reason, the lingua-franca approach does not treat the native speaker as the reference point against which everyone else falls short; it treats linguistic diversity as an asset rather than a deficiency. The distinction between accentedness, comprehensibility, and intelligibility, and how this distinction behaves once assessment moves down to the level of a single word, is further developed in Chapter 2.

This choice of criterion has consequences, many automatic systems built to assess pronunciation still rely on Automatic Speech Recognition, which rewards proximity to

the speech patterns it was trained on, typically native and standard varieties. A system that penalizes intelligible non-native speech is therefore not just inaccurate; it tells learners that their way of speaking is less valid, even when they have been perfectly understood, which can erode their confidence. In a pedagogical setting, rejecting an intelligible word just because it differs phonetically from a native pattern works directly against the goal an intelligibility-based tool should serve.

This is also the orientation of the data used in this study. In the MOOC *English Pronunciation in a Global World* at Vrije Universiteit Amsterdam, the criterion was set as communicative success rather than native-likeness, that is, a background-agnostic approach to pronunciation whose goal is effective communication among people.

1.2 Transcription Is Not Explanation

Many automatic pronunciation assessment systems rely on Automatic Speech Recognition (ASR) models. The two previous theses developed in the context of the *English Pronunciation in a Global World* MOOC, Zou (2024) and Kuan (2025), both built systems on top of Whisper to classify word-level intelligibility. In these systems, speech is first transcribed, and the transcription is then compared with the target word.

The logic of this approach is that if the machine recognizes a word, the word must have been pronounced correctly. Cámara-Arenas et al. (2023) take this assumption as the starting point for their critique, and rightfully argue that a transcription error should not be equated with a communication failure. As Munro and Derwing argue, accentedness is not the same as unintelligibility. A system trained mostly on native-speaker data may transcribe an accented word incorrectly, while the same word is perfectly comprehensible to a human listener.

McGuire (2025), reviewing the history of Automatic Speech Recognition (ASR) development, observes that transcriptions of accented speech consistently receive higher error rates than those of unaccented speech. It would be too quick, however, to read this gap as evidence of unfair penalization on its own; non-native speakers may basically produce more deviations from the expected forms, so some increase in error rate is to be expected. The question that matters is whether the system makes more errors than the actual difficulty of the speech would justify, that is, whether it penalizes words that a human listener would have understood without trouble. Kuan (2025), in his error analysis, shows that the Whisper-based system mislabels speech produced by learners that was labeled as intelligible, particularly for words involving aspiration or vowel substitution.

An ASR system can produce a transcription, and that transcription can be used to build a system capable of returning a binary intelligible/not-intelligible judgment. However, such a system still does not tell the learner why a word failed. The learner is told a word was unintelligible with no indication of whether the problem lay in vowel quality, a missing aspiration on an initial stop, or a deleted final consonant. It is worth being precise about the nature of this gap. It is not that these systems are incapable in principle of saying more; the phonetic representations they already computed could, in principle, have been used to point toward a likely cause. Rather, they were not designed to answer that question. As deep learning advanced, ASR systems became better at detecting subtle patterns in speech, yet their output remains a transcription and a score (Ma et al., 2025). This can be used to declare a learner's pronunciation "correct" or "incorrect," but offers no account of why. The result is a real pedagogical

limitation in which neither learner nor instructor can discover what exactly caused the communication breakdown.

1.3 The Proposed Approach

Recent speech-enabled multimodal Large Language Models (LLMs), including the Qwen models and MiniCPM-o, differ from traditional ASR systems in that they can process speech and, due to abilities developed through instruction training, can in principle produce a linguistic explanation alongside a classification. This means they can do more than label a word as intelligible or not intelligible. They can also offer a possible reason for that judgment, for instance by naming a pronunciation feature that may have affected intelligibility, such as a mid vowel that was not produced clearly or a final consonant that was dropped. This capacity for phonetic reasoning is what earlier systems lacked, and it is what makes these models attractive for educational feedback.

This apparent closing of the gap brings a new question with it. Producing a fluent, phonetically phrased explanation is not the same as producing one that is actually based on the acoustic signal. The literature on these models warns that they can generate explanations that sound convincing but are not grounded in the audio (Chen et al., 2026; Madsen et al., 2024). The concern is no longer the absence of an explanation but the validity of the explanation; whether what the model says is faithful to what it actually based its decision on. Checking whether these explanations are really grounded in the audio is therefore essential before such feedback could be put in front of a learner, and it is one of the central questions this study sets out to examine.

The present study evaluates this potential on second-language learner data from the Vrije Universiteit Amsterdam MOOC *English Pronunciation in a Global World*. The evaluation is carried out at the word level, and the models are guided through prompt engineering, including structured output, Duty-Distinct Chain-of-Thought reasoning, and few-shot examples.

1.4 Why Word-Level Assessment?

This study assesses intelligibility at the level of the single word. The work is part of an internship tied to the MOOC *English Pronunciation in a Global World*. While the MOOC collects speech at higher levels such as paragraphs, the only annotated dataset available for this thesis was labeled at the word level.

This constraint turns out to suit the aims of the study well. The most important reason is that a single word isolates the acoustic signal. A word on its own carries no surrounding context, syntax, or expectation that a model could lean on, so a phonetic judgment has to come from the sound itself. The same absence of context also sharpens what is being measured because once there is no surrounding material to ease understanding, the listener effort that defines comprehensibility has little to draw on, and the judgment comes to rest on the acoustic signal much as intelligibility does. The single word is in this sense a more controlled setting for assessing intelligibility than connected speech, though the fuller relation between these notions is taken up in Chapter 2.

A further advantage is that the word level keeps phonetic analysis tractable. The prior work on this corpus moved away from matching written words and toward matching their sounds, mapping each word to a sequence of phonemes using a pronunciation dictionary (Kuan, 2025; Zou, 2024). Aligning a single word with its expected phoneme

sequence is more tractable than aligning a longer utterance, although it is not straightforward because acceptable pronunciation variation must also be represented. There are only a limited number of sounds to align, and no word boundaries or pauses need to be resolved first. This means that when something goes wrong, the breakdown can be traced to a specific sound in a specific word, for instance a dropped final consonant or a shortened vowel, rather than being embedded within longer utterances. At the sentence or utterance level, the signal has to be segmented into words before anything can be localized, and a single mispronunciation can be masked by surrounding words that the listener, or the model, still understands. Sentence-level assessment is also complicated by suprasegmental features, such as stress, pitch, rhythm, and duration, which become more complex to model and interpret in longer utterances.

1.5 Research Gap and Contributions

Prior work, especially in the form of ASR-based approaches, has shown that automatic assessment of word-level intelligibility is feasible in principle, though feasible is not the same as solved. The systems built so far reached modest performance on this task, while remaining unable to explain why communication succeeded or failed. This second limitation is what most constrains their use in teaching, since a label does not tell a learner what to correct or how.

The emergence of speech-enabled multimodal LLMs has made it possible to go beyond the label and produce linguistic explanations and phonetic feedback. This opens a new possibility for educational use. At the same time, research on the faithfulness of model explanations gives reason for caution; because an explanation that reads convincingly does not necessarily reflect the basis on which the model actually decided, and the literature warns that such models can produce fluent but ungrounded justifications. Whether this happens in the present setting is something to be examined.

What has therefore remained relatively unexplored is whether speech-enabled multimodal LLMs can, at the word level, classify intelligibility and at the same time provide phonetic explanations that are grounded in the acoustic evidence, trustworthy, and suitable for an educational setting. On this basis, the present study adopts a behavior-centered approach that probes how these models reach their decisions.

The contributions of this thesis are as follows. First, it provides a word-level evaluation of three contemporary speech-enabled multimodal LLMs on authentic L2 learner data, comparing their classification performance against prior ASR-based work. Second, it studies the effect of a systematic set of prompt-engineering conditions, including structured output, Duty-Distinct Chain-of-Thought (DDCoT) reasoning, and few-shot examples, on both classification and explanation behavior. Third, it asks whether the phonetic explanations these models produce are grounded in the acoustic signal and safe to present to learners, treating acoustic grounding, faithfulness, and calibrated uncertainty as evaluation targets. These prompting strategies, and the evaluation metrics used to compare them, are described in detail in the methodology (Chapter 3), and their theoretical background is reviewed in Chapter 2.

1.6 Research Questions and Hypotheses

Given the recent progress of speech-enabled multimodal LLMs, this thesis hypothesizes that, when guided by prompting strategies, these models can achieve a reasonable level

of performance in word-level intelligibility classification while generating phonetic explanations that are sufficiently informative for pronunciation feedback. The prompting strategies investigated include structured JSON outputs, phonetic checklists, DDCoT-inspired prompts that separate describing the perceived acoustic evidence from making the final intelligibility judgment, and few-shot examples. Because the literature reports that such models can generate fluent explanations that are not always grounded in the acoustic signal, a second hypothesis is that their explanations will not be reliably grounded by default; whether this is so, and whether prompting can improve it, is treated here as an open question to be tested.

The research question. To what extent can speech-enabled multimodal LLMs assess word-level intelligibility of L2 English speech in a reliable way, both classifying intelligibility and explaining it?

- **RQ1.** How does the classification performance of speech-enabled multimodal LLMs on word-level intelligibility compare with that of prior ASR-based systems on this task?
- **RQ2.** How do different prompt-engineering strategies compare in performance?
- **RQ3.** To what extent are the phonetic explanations these models produce grounded in the acoustic signal and safe to present to learners?

The corresponding hypotheses, derived from the literature reviewed in Chapter 2, are presented at the end of the Literature Review.

1.7 Thesis Outline

The remainder of this thesis is organized as follows. Chapter 2 reviews the relevant literature, covering the distinction between intelligibility and native-likeness, bias in speech technology, prior ASR-based work, and the explainability and grounding challenges of multimodal models. Chapter 3 describes the methodology, including the dataset, the models, the prompt conditions, the evaluation metrics, and the sanity checks. Chapter 4 reports the results, both the prompt-engineering experiments on the development set and the final evaluation of the selected configuration on the test set. Chapter 5 discusses these findings, drawing on the manual error analysis to interpret how the models behave. Chapter 6 concludes and outlines directions for future work.

Chapter 2

Literature

This chapter opens with the three concepts at the heart of pronunciation assessment: intelligibility, comprehensibility, and accentedness. Then it clarifies how they differ, which one this study takes as its criterion, and how the distinction behaves once assessment moves down to the single word. It next discusses two forms of bias that affect automatic intelligibility assessment: bias in the systems themselves and instability in the human labels against which they are evaluated. With this in place, the chapter traces how automatic systems have approached pronunciation assessment over time, from early ASR-based pipelines to the prior work on this corpus by Zou (2024) and Kuan (2025), on which this thesis builds. It then moves to recent speech-enabled multimodal LLMs, explaining how they are built and trained, how they differ from ASR systems and from text-only language models, and the source of their particular weakness, namely, a tendency to produce explanations without relying sufficiently on the audio. Before drawing these strands together, it reviews prompting as a method for shaping and probing model behavior.

2.1 Intelligibility, Comprehensibility, and Native-Likeness

A central decision in pronunciation assessment is which criterion defines a "good" pronunciation. The most intuitive answer, how closely a learner approximates a native speaker, is the wrong one, and seeing why means separating three notions that are easily conflated. In much of the earlier work on pronunciation assessment, the guiding concern was less whether the listener had actually understood the speaker's message than whether the speaker came close to a native model. Drawing on Di (2024) citing Levis (2022), this orientation reflects what Levis terms the *Nativeness Principle*: an emphasis on attaining a native-like accent and narrowing the distance between the learner's pronunciation and that idealized target. Where this assumption was operative, any deviation from the native pattern tended to be treated as an error, so that the greater the distance from the native model, the lower the score, and the smaller the distance, the higher it was.

The problem with this view begins precisely where speech meets the real world, because in practice a great many speakers are accented yet entirely understood. The foundational work of Munro and Derwing (1995) established this by separating three dimensions that everyday usage runs together. *Accentedness* is the degree to which a speaker's pronunciation departs from a native pattern; *comprehensibility* is the degree of effort a listener must invest to understand it; and *intelligibility* is the extent to which

the speaker's intended message is actually recovered by the listener. The decisive result, on which this whole line of research rests, is that these dimensions are independent rather than points on a single scale: speech can be accented and at the same time fully intelligible. Accentedness, on this account, is not in itself a measure of communicative success or failure. The same separation is taken up by later work in the same tradition; Gottardi and Silveira (2025), drawing on Derwing and Munro's *Pronunciation Fundamentals*, restate the three notions in just these terms and say that the proper goal of pronunciation teaching is intelligibility, not native-likeness.

This independence reframes what should count as an error. Against the Nativeness Principle stands what Levis calls the *Intelligibility Principle*, which discards the distance from native scale altogether and substitutes a different question: not how native the speech sounds, but whether the message gets through. As Di (2024) observes in tracing this shift, the criterion moves away from native target and toward successful communication. In the framework of English as a lingua franca, Jenkins (2000) argues that in international settings, where speakers of different first languages use English with one another rather than with native speakers, imitating native norms can even obstruct understanding, since some native features, such as elision and weak forms themselves become obstacles in these interactions. Tracing the development of Jenkins's Lingua Franca, Di (2024) describes this as a reorientation away from native models, and Walker (2010) declares that native accents such as British or American are not inherently intelligible. These lines of work converge on intelligibility, and it is this, rather than proximity to a native model, that the present study adopts.

These distinctions were developed for connected speech and require qualification when assessment is limited to isolated words. Comprehensibility is defined in terms of the listener's effort to understand speech, but in connected speech that effort is supported by contextual information. At the single-word level, this contextual support is absent, leaving the acoustic signal as the primary basis for judgment. As a result, comprehensibility and intelligibility become much more closely aligned than they are in longer utterances. Accentedness is similarly more difficult to evaluate from an isolated word, because many of the cues that contribute to accent perception, such as rhythm, and intonation are unavailable. At the word level, then, the reliability of this distinction decreases, and the three dimensions converge on whether a listener can recover the intended word from the sound that was produced.

This criterion also matches the orientation of the data used in this study, drawn from the MOOC *English Pronunciation in a Global World* at Vrije Universiteit Amsterdam, which takes communicative success rather than native-likeness as its standard.

2.2 Bias and Annotation

If intelligibility is to be judged automatically, two distinct sources of bias have to be kept in view: bias in the systems that do the judging, and instability in the human labels those systems are measured against. Both bear directly on how the evaluation in this thesis has to be read.

Automatic speech systems do not perform equally across speakers. Koenecke et al. (2020) found substantially lower transcription accuracy for Black American speakers than for white American speakers across major commercial systems from Apple, Amazon, IBM, Microsoft, and Google, and Martin and Wright (2023) report the same underperformance on African-American English. Extending the pattern to accented speech,

Prasad and Jyothi (2020) show that end-to-end models align strongly with word boundaries for North American accents but that this alignment weakens for Indian-accented English. The same pattern occurs for non-native speech; McGuire (2025), reviewing ASR development, reports a wide gap across first-language groups, with American-English speakers near the lowest error rates and some non-native groups several times higher, and Wills et al. (2023) find that Whisper's word error rate rose from around 8 percent for native children's read speech to roughly 25 percent for non-native speech. Taken together, this evidence undermines the assumption that the tool is neutral. Choices about which data a system is trained on and how it is trained can increase or reduce these disparities rather than leave them unchanged.

As already noted in the Introduction, a higher error rate on non-native speech does not on its own prove that a system is unfair. Some increase in error may reflect genuine differences in the difficulty of the speech rather than unfair penalization. For example, Issa et al. (2026), testing Whisper large-v3 on non-native Arabic speech, found that its word error rate was close to that of human listeners, and that the system's errors tracked the strength of the accent, with heavier accents drawing more errors. This can be read in two ways. If both humans and the system struggle on the same words, those words probably carried some real communicative difficulty; but it is also possible that the system penalizes accented pronunciations beyond the difficulty humans faced. The evaluation conducted in this thesis cannot cleanly separate these two possibilities. Either way, what bias on the system side establishes is that an automatic judgment is not a neutral instrument, and its errors are unlikely to be evenly distributed across the speakers it serves.

The second source of bias sits on the human side, and it is easy to overlook because automatic systems are usually evaluated against human labels that are treated as ground truth. They are not. Plank (2022) says that human label variation is a pervasive reality because people bring different perspectives to the same data, and disagreement among annotators is a normal feature of labeled datasets. This variation has been traced to several distinct sources, among them task ambiguity (Tamkin et al., 2023) and the beliefs and identities that annotators themselves bring to a labeling task (Sap et al., 2022). For the present study the consequence is specific and important. Experts can agree that a given word was unintelligible while disagreeing about *why*; one may attribute the breakdown to vowel quality, another to a deleted final consonant. The phonetic cause of a breakdown is therefore partly interpretive, and a model asked to produce such explanations is being held against a target on which human experts do not fully converge. This has a direct methodological consequence, taken up in Chapter 3: a disagreement between the model and a human label cannot automatically be read as a model error, and the evaluation has to be designed with the reliability of the annotations in mind.

2.3 The Evolution of Pronunciation Assessment and Prior Work on This Corpus

Early Automatic speech recognition systems, from the 1990s onward, relied on statistical models that turned acoustic patterns into text or estimated pronunciation quality from a handful of predefined indicators (Ma et al., 2025). At that stage a system was considered successful if it could detect that the learner had produced the target word. The task has never been an easy one, and it remains difficult today. With the maturing of

ASR, assessment came to lean on transcription; if a learner's word was transcribed incorrectly, this was read as a sign of mispronunciation or reduced intelligibility. ASR-based assessment is still a common and for many purposes, preferred approach. Gottardi and Silveira (2025) describe ASR being used for classroom pronunciation activities, and the two prior theses on this corpus, Zou (2024) and Kuan (2025), both build intelligibility-classification systems on top of Whisper models.

The reason transcription is such an attractive proxy is that it does not require direct analysis of the acoustic signal. Inspecting the signal directly demands expertise in acoustic phonetics; converting speech to text and comparing it against the target word is far faster and more scalable. But the convenience rests on an assumption worth making explicit that if the machine recognizes a word, the word must have been pronounced well enough. Cámara-Arenas et al. (2023), following Thomson (2012), use transcription as a proxy for pronunciation quality; a well-pronounced word is recognized, whereas a poorly pronounced one is not. However, ASR systems are designed to produce the most likely transcription, not to assess pronunciation. Part of what a transcription discards is the paralinguistic layer of speech. Fucci et al. (2024) note that the signal also carries information such as intonation, loudness, and rhythm, and when assessment is reduced to text, these layers, which can matter for human understanding, are thrown away. As a result, a transcription-based score indicates whether the system recognized the word, rather than whether a human listener understood it. A system may therefore fail to recognize a word that a human listener would have understood.

It is against this background that the two prior studies on the *English Pronunciation in a Global World* corpus should be read. Zou (2024) built a word-level intelligibility-classification system on top of Whisper. Rather than relying only on exact text matching, Zou converted both the target word and Whisper's transcription into phoneme sequences using CMUdict[1], which allowed homophones and phonetic similarity to be handled more appropriately. She then combined this phonetic comparison with Whisper's confidence scores and decoding temperature, prioritizing precision and $F_{0.5}$ for the *not_intelligible* class. The results showed that Whisper outputs could be repurposed for binary word-level intelligibility classification without changing the model architecture or performing additional training. Building on Zou's system, Kuan (2025) focused on errors in which isolated target words were transcribed as short phrases. Kuan continually pre-trained Whisper on single-word recordings to reduce these errors and then applied the same general classification design introduced by Zou, including phonetic conversion, confidence thresholds, and temperature settings selected to prioritize precision and $F_{0.5}$. Continual pre-training lowered the word error rate, with the largest reduction, approximately 6.5 percent, observed for the medium model, and also improved intelligibility classification, particularly for the *not_intelligible* class. Kuan's adapted system scored a precision of roughly 0.58 and a recall of roughly 0.73 on the *not_intelligible* class; competent enough to show the task is workable, but well short of the reliability a pedagogical tool would need. A direct comparison with the present system is given in the results chapter; the comparison is indicative only, since the two systems were evaluated on different test splits. An important innovation of this prior work was the move away from matching written words and toward matching their sounds. Text-string matching

[1]The Carnegie Mellon University Pronouncing Dictionary (CMUdict) is a publicly available English pronunciation lexicon that maps words to their phonemic transcriptions using the ARPAbet phoneme set, a standard symbol set for the sounds of English. It is used to convert written words into standardized phoneme sequences.

has difficulty with homophones. If a model transcribes *meat* as *meet*, letter matching would mark it unintelligible even though the two are pronounced identically. To avoid this, Kuan (2025), following Zou (2024), mapped both the target and the recognized word onto phoneme sequences using CMUdict. For example, the word *cat* maps to the three-sound sequence `K AE T`; by converting both the target word and the recognized word into sequences of this kind, the two can be compared sound by sound rather than letter by letter, which is what lets the comparison see that *meat* and *meet*, spelled differently, are in fact the same string of sounds. This shifted the comparison from the letter level to the phoneme level. They did not simply trust the raw transcription; both also drew on the model's prediction confidence and on these phonetic representations to refine the judgment, a detail that matters for the critique that follows.

Even with these refinements, the underlying limitation persisted, and Cámara-Arenas et al. (2023) locate its source. The property that makes an ASR system good at recognition, a language model that weights candidates by their probability and frequency, is also what leads it to register an expected word even when the pronunciation was off. Their illustration is the word *din*; because a single-word utterance of *din* is improbable in ordinary language, the system pushes it down its candidate list and returns the more frequent *then*, a recognition driven by linguistic probability rather than by the acoustic input. They conclude that improving speech recognition alone is not enough to achieve reliable pronunciation assessment. The issue is not that ASR uses probabilities, since every statistical model does, but that an objective optimized for transcription rewards the lexically expected answer, which is not the same thing as faithfully reflecting what was pronounced. This is exactly why Zou and Kuan turned to prediction confidence and phonetic representations.

This same mechanism shows up concretely in Kuan's error analysis. Words such as *pat* and *pit* drew many false rejections (False Positives), clearly pronounced tokens marked unintelligible, and the source was variation in the aspiration of voiceless stop consonants such as *p*, *t*, and *k*, which differs from speaker to speaker among L2 learners. Aspiration was always treated as an important feature in the annotation guidelines, but it was evidently difficult for annotators to judge reliably, leading to a high number of inconsistent labels for this feature. The system could therefore mark a word wrong without being able to say that aspiration was the reason (Kuan, 2025, sec. 5.2.1, p. 30).

This points to the limitation that matters most for the direction of this thesis. These systems could assign a label, but they did not produce an explanation. They could decide whether a word was intelligible, but could not tell a learner why it had failed, whether the problem lay in vowel quality, a deleted final consonant, or a mismatch in stress. This was not strictly a matter of capability because the phonetic differences they already computed could, in principle, have been turned toward an explanation. But in general, they were not designed to answer that question. In a classification-based framing the omission might seem enough, but in a pedagogical setting, knowing only that an error occurred is of limited use, since a learner needs to know which feature to work on. It is this gap, between producing a label and explaining it, that motivates the turn to models that can in principle give reasons.

2.4 Multimodal Speech Language Models

As mentioned earlier, the systems discussed so far provide no explanation for their predictions. Ma et al. (2025) note that speech foundation models such as wav2vec 2.0

capture rich acoustic features but do not represent content or vocabulary in a way that would support analytic feedback, so they are not, in themselves, able to explain a judgment. Gatto (2024) shows the same limitation through interpretability; when a system only maps an acoustic signal to text, the path from sound to decision is not clear enough, which limits both trust and pedagogical use.

Speech-enabled multimodal LLMs differ from ASR systems in architecture, and the difference matters for what they can produce. An ASR model is trained for a single objective, mapping audio to text, and is optimized to return a best-guess transcription even from noisy input. A speech-enabled multimodal LLM instead connects an audio encoder to a pre-trained language model, so that the system takes in the acoustic signal together with a text prompt and generates text in response. Chu et al. (2023) describe Qwen-Audio, which combines a pre-trained language model with an attached audio encoder and is trained across many audio tasks at once, from speech recognition and translation to sound captioning and emotion recognition. To keep these many tasks from interfering with one another, Chu et al. (2023) condition the decoder on a hierarchy of labels, and they show that adding a word-level timestamp recognition task, which aligns the acoustic signal more finely with temporal labels, strengthens the model's ability to ground its output in the signal itself. Because the language component is large and pre-trained on text, such a model can in principle do more than classify; it can name a phonetic feature, such as a shortened vowel or a deleted final consonant, and present it as a justification for a judgment.

This is also where these models differ from text-only LLMs. A text-only model cannot receive audio input, whereas a speech-enabled model can. The capacity to produce an explanation is therefore new, but it comes with a risk that ASR systems never had the chance to pose; that the explanation may be fluent and phonetically phrased while not actually derived from the audio input received by the model. It is worth being precise here, since it is easy to overstate what these models offer. Producing an explanation does not make a model transparent. Like ASR systems and text LLMs, these models remain black boxes whose internal decision process is not directly observable. What changes is not that the model becomes interpretable, but that the absence of any explanation is replaced by an explanation whose validity then has to be checked. If a model reports that a word was unintelligible "because a long vowel was shortened," the statement may be accurate, or it may be a guess; Gatto (2024) marks this as the distinction between *plausibility*, whether an explanation seems reasonable to a human reader, and *faithfulness*, whether it reflects the model's actual basis for the decision.

There is also an architectural reason to expect this risk. It is tempting to assume that in a multimodal model text and audio carry equal weight, as though the system had learned from the outset to treat both as equally important. In practice, many of today's multimodal models are at heart text language models to which an audio capability was added afterwards: Chu et al. (2023) explain that the linguistic backbone was pre-trained before audio processing was introduced. The order in which a model is trained has a consequence, since a system trained primarily on text can be expected, when deciding, to lean more readily on lexical cues than on acoustic evidence. Chen et al. (2026) make this hypothesis explicit, arguing that because most contemporary audio LLMs are adapted from text models and then fine-tuned on paired audio-text data, they inherit a structural bias toward lexical information, with the acoustic signal relegated to second place. The empirical evidence for this lexical bias is examined in Section 2.4.2.

2.4.1 The Models Evaluated in This Study

This study evaluates three open-source speech-enabled multimodal LLMs: Qwen2-Audio-7B-Instruct, Qwen2.5-Omni-7B, and MiniCPM-o-2.6. All three follow the general pattern described above, an audio encoder connected to a pre-trained language model, but they differ in scale, in the range of inputs they accept, and in how audio was integrated, and these differences are worth setting out because they bear on the behavior observed later.

Three considerations guided this selection. The first is openness and cost. All three models are open-source and can be downloaded and run locally without a commercial API, which is important because routing learner data through a third-party service raises privacy concerns. The second is scale. At roughly seven to eight billion parameters, these models are large enough to bring real linguistic ability to bear, yet small enough to run on the GPUs available for this project, so that all three could be evaluated under the same practical constraints. The third, and most important is that each model places a large text-pre-trained language model at its center with audio perception attached around it. Beyond these shared traits, the three were chosen to span a small but informative range. Qwen2-Audio is audio-only, whereas Qwen2.5-Omni and MiniCPM-o were additionally trained on vision, which allows a first look at whether broader multimodal training changes how a model attends to speech; and MiniCPM-o differs from the Qwen models in that it combines separately trained components for different modalities, whereas the Qwen models adopt a more unified multimodal architecture. Comparing these models makes it possible to examine how differences in modality support and model design influence their behavior, while keeping the overall task the same.

Qwen2-Audio-7B-Instruct. Developed by Alibaba, Qwen2-Audio (Chu et al., 2024) connects an audio encoder to the Qwen-7B language model and is, of the three, the most narrowly focused because it accepts audio and text as input and produces text only. Its design illustrates the architectural point made above, a language backbone with audio capability attached, particularly clearly. Notably, its developers changed how the model is told which task to perform. The earlier Qwen-Audio architecture conditioned training on a hierarchy of special task tags: fixed label tokens marking, for instance, the task type and the language of each training example, so that the model learned to associate each combination of tags with a specific behavior. In Qwen2-Audio these tags were replaced by natural-language prompts, in which the task is described in plain text, much as a user would phrase an instruction. This simplifies the training setup and brings pre-training closer to how the model is actually queried at inference time. In addition, they applied Direct Preference Optimization (DPO) to improve the factuality of the model's responses and its adherence to intended behavior. The instruct variant used here is the instruction-tuned model, oriented toward following textual instructions about an audio input, which is exactly the mode this study relies on.

Qwen2.5-Omni-7B. Also from Alibaba, Qwen2.5-Omni (Xu et al., 2025) belongs to the "Omni" line, and its defining difference from Qwen2-Audio is breadth of modality; it perceives not only audio and text but also images and video, and it can generate both text and natural speech. To manage the two output streams without interference, it uses what its authors call a *Thinker-Talker* architecture, in which a language-model

Feature	Qwen2-Audio	Qwen2.5-Omni	MiniCPM-o
Developer	Alibaba	Alibaba	OpenBMB
Parameters	7B	7B	∼8B
Input modalities	audio, text	audio, image, video, text	audio, image, video, text
Language backbone	Qwen-7B	Qwen2.5 (Thinker)	Qwen2.5-7B
Audio encoder	Whisper-large-v3	Whisper-large-v3	Whisper-medium
Distinctive feature	audio-specialized	unified omni architecture	modular architecture

Table 2.1: The three speech-enabled multimodal LLMs evaluated in this study. All three audio encoders are derived from OpenAI's Whisper, differing only in size: the two Qwen models build on Whisper-large-v3, while MiniCPM-o uses the smaller Whisper-medium. Only the audio-to-text capability of each is used here.

component (the "Thinker") handles text while a separate component (the "Talker") produces speech. For the present study only its audio-to-text capability is used, but its broader multimodal training distinguishes it from the audio-only Qwen2-Audio.

MiniCPM-o-2.6. Developed by OpenBMB, MiniCPM-o-2.6 (Yao et al., 2024) is the largest of the three at roughly 8B parameters and is built by assembling several specialized components in an end-to-end fashion: a SigLip vision encoder, a Whisper-medium audio encoder, a ChatTTS-based speech decoder, and, as its language backbone, Qwen2.5-7B. Its make-up is a concrete illustration of the structural point developed earlier in this chapter, that these models are, at their core, text language models to which perception in other modalities was later attached. Like Qwen2.5-Omni it handles vision, audio, and text; for this study, again, only the audio-to-text path is used.

Two observations from Table 2.1 matter for what follows. The first is modality breadth: Qwen2-Audio is audio-only, whereas both Qwen2.5-Omni and MiniCPM-o were trained on vision as well, which may affect how each attends to the acoustic signal. The second is a shared architectural feature: all three place a large, text-pretrained language model at their center, with audio perception added around it. This is precisely the configuration that, as discussed in Section 2.4.2, gives reason to check whether a model's phonetic explanations are grounded in the audio input rather than driven primarily by expectations from the language component.

2.4.2 Acoustic Grounding and the Faithfulness of Explanations

The capacity to produce an explanation raises a question that did not arise for systems that only output a label. When a model gives a justification for its judgment, on what basis can we be sure that the reason is drawn from the acoustic signal? This is the problem of acoustic grounding, and it is the point at which the plausibility-faithfulness distinction introduced above becomes concrete.

Chen et al. (2026) address this directly with LISTEN (Lexical vs. Acoustic Speech Test for Emotion in Narratives), a benchmark designed to test whether an audio language model genuinely relies on the acoustic signal when making its predictions. The test is built around a separation of the two kinds of information. In some items the words spoken are emotionally neutral while the tone of voice clearly carries an emotion, and in others there are no informative words at all, so that any correct judgment can only come from the acoustic signal, the prosody like the pitch, rhythm, and tone of

the voice. If a model relies on the words, it will fail exactly where the words are uninformative and only the sound carries the answer. Evaluating several state-of-the-art models this way, Chen et al. (2026) report a consistent *lexical dominance*, a systematic tendency to give textual content priority over acoustic substance. When the words are neutral or absent, the models retreat to a safe default answer, and in the purely acoustic conditions their performance falls close to chance, that is, barely better than random guessing. Receiving the audio, in other words, is not the same as relying on it.

The same pattern has been documented more broadly. In a survey of large audio language models, Luo et al. (2026) group these failures under what they call the *acoustic-semantic gap* and name two that are directly relevant here. In *modality neglect*, a model leans on lexical cues so heavily that replacing the audio with silence or noise changes its output only negligibly; in *textual dominance*, the text acts as an interpretive anchor that overrides the acoustic evidence. They report that on a benchmark built around text-audio conflict, accuracy drops sharply while the model's confidence stays high, a combination that is especially troubling for educational use, where a confident but ungrounded judgment is precisely the kind of feedback that can mislead a learner.

If a model can reach a judgment without truly using the audio, then the explanation it offers for that judgment is open to the same doubt. Madsen et al. (2024) describe this as a gap between plausibility and faithfulness. A language model can produce a self-explanation that reads as convincing to a human yet does not correspond to the process that actually produced the decision, amounting to post-hoc rationalization rather than an account of the evidence. An explanation that a word was unintelligible "because a final consonant was dropped" may therefore be correct, or it may be a guess assembled from what the model expected the word to be, and, from the text alone, the two cannot be told apart.

This is why acoustic grounding, rather than transcription accuracy, is the more revealing object of study here, and why the evaluation in this thesis cannot stop at whether a classification is correct. It has to ask a second question: whether the model's judgment, and the explanation it attaches to that judgment, are actually anchored in the sound the learner produced. How this is operationalized through prompts that probe the model's reliance on the audio and through checks on the explanations it generates, is described in Chapter 3, and the evidence these probes generate is examined in the discussion.

2.4.3 Confidence, Uncertainty, and Calibration

A model's expressed confidence is useful only insofar as it is calibrated: the confidence a model reports for a given answer should track the probability that the answer is correct. Geng et al. (2024) organize existing calibration approaches into four families: logit-based methods that derive confidence from token-level generation probabilities, methods that prompt the model to verbalize its confidence in words, methods that measure semantic consistency across multiple sampled responses, and self-verification methods in which the model judges whether its own generated answer is correct. P(True)-style approaches belong to this last family: the model is prompted to assess the probability that a candidate answer is true, producing a self-verification signal that is distinct from the confidence attached to the original answer.

Calibration changes over the course of training. Luo et al. (2025) show that pre-trained language models are reasonably well calibrated, and that post-training, particularly Reinforcement Learning from Human Feedback (RLHF), degrades this calibration:

post-trained models assign high confidence to correct and incorrect outputs alike. This follows from what post-training optimizes for. RLHF trains a model to produce fluent, preferred-sounding responses, and this objective is separate from tracking the model's own uncertainty; a model's verbalized confidence therefore reflects its post-training history as much as its underlying knowledge.

Kuan et al. (2026) test several ways of measuring a model's uncertainty, including P(True), across multiple audio-aware language models. One family, which they call semantic-level methods, queries the model repeatedly and then checks whether the resulting answers agree in meaning: a model whose answers are semantically stable is treated as confident, while one whose answers shift in meaning is treated as uncertain. This is distinct from token-level probability, which reflects how strongly the model committed to a particular wording rather than to the underlying answer.

On tasks that specifically test whether a model can be trusted, such as hallucination detection, which of these methods works best depends heavily on the model and the benchmark: a method that performs well on general reasoning tasks can behave unpredictably on a trust-focused one.

Word-level intelligibility classification is exactly this kind of trust-focused task: the present study relies on a model's self-reported confidence and its uncertainty flag as signals of whether its judgment can be trusted. Kuan et al. (2026)'s findings suggest these signals should not be assumed to behave the same way here as they do on general audio-reasoning benchmarks.

A separate question from how well such signals are calibrated is why a model should be offered an explicit way to express uncertainty. In a setup that forces the model to commit to one of the two labels, without an option to abstain, the model must produce a label even when the acoustic evidence is weak. As a result, uncertainty cannot be expressed directly and may instead appear as a confident prediction. The literature on uncertainty quantification treats this as a safety problem: in high-stakes applications, an estimate of how much an answer can be trusted is what allows risk to be managed (Liu et al., 2025; Kapoor et al., 2024). Allowing a model to qualify or withhold its answer is therefore a mechanism for selective prediction, in which downstream use depends on the model's confidence in its own prediction (Tao et al., 2024). Recent work goes further and treats the expression of doubt as a property to be trained for, rewarding models for admitting uncertainty where their accuracy does not warrant confidence (Bani-Harouni et al., 2026). An uncertainty option also benefits the people who rely on the model's predictions. Steyvers et al. (2024) showed that users tend to overestimate a model's confidence when its outputs do not communicate uncertainty. They further found that calibrated verbal expressions of uncertainty reduce this mismatch between perceived and actual confidence. In educational settings, where the model's feedback is intended for learners, feedback that cannot express uncertainty invites exactly this kind of overtrust.

For the present task there is an additional reason. As Section 2.2 established, word-level intelligibility judgments carry inherent ambiguity, and human annotators themselves disagree on a nontrivial share of cases. A model denied any way of marking such cases would be forced to render every ambiguous item as a confident binary decision, making ambiguity in the signal indistinguishable from model failure. Giving the model an explicit uncertainty flag, separate from its graded confidence, creates space for the model to behave conservatively where the evidence is weak. Whether the models actually use this channel informatively is an empirical question taken up in the error

analysis.

This literature treats a model's expressed confidence and its explicit uncertainty signal as two separate measurements. RLHF degrades calibration regardless of how hard the task is, and P(True)-style methods perform differently from one benchmark to the next. Section 3.4.4 describes how the present study's output schema separates these two signals.

2.5 Prompting as a Method for Probing Model Behavior

The previous section established that speech-enabled multimodal LLMs may produce fluent explanations. The question that remains is how this behavior can be studied, and possibly improved, without retraining the models. The answer this thesis takes up is prompting. In recent work the prompt is treated as an instrument for shaping a model's response process and exposing what it relies on when it decides. Three strands of these works are relevant.

The first is chain-of-thought prompting. Wei et al. (2022) showed that asking a model to produce intermediate reasoning steps, rather than jumping straight to an answer, reduces its reliance on surface cues and improves performance on tasks that require reasoning. A weakness of language models is precisely this tendency to settle on a conclusion prematurely, on the basis of prior expectation. Chain-of-Thought is one way of counteracting it. In multimodal settings the risk is sharper, because the model can lean on the text and never really engage the other modality. Zheng et al. (2023, pp. 1-2) address this with Duty-Distinct Chain-of-Thought (DDCoT), a prompting strategy designed to separate different stages of the response-generation process. It is based on two principles that the authors describe as "letting everyone do their jobs" and "keeping critical thinking". The first separates the recognition stage, which identifies what is in the input, from the interpretation stage, which interprets its meaning. The second encourages the model to explicitly express uncertainty when the available evidence is insufficient to support a confident decision. By decomposing the task this way, DDCoT is designed to stop the model from inferring the answer before it has actually attended to the signal. For the present study, this separation is valuable not only because it may improve accuracy but because it makes the model's reliance on the audio easier to inspect.

The second strand is structured output. Requiring a model to answer in a fixed format places the prediction, the claimed evidence, the stated confidence, and any markers of uncertainty in separate fields. These structured answers allow each component to be examined independently, separating the prediction from its justification. This is especially useful given the plausibility-faithfulness gap discussed above. Madsen et al. (2024) caution that self-explanations often act as post-hoc rationalizations. While they may appear plausible and convince humans of a prediction, they do not necessarily reflect the model's actual reasoning process. A structured output is therefore treated here as a tool for observing and dissecting model behavior, not as a guarantee that the behavior is correct.

The third and final strand is the use of few-shot examples, providing a small number of worked cases in the prompt to demonstrate the intended format and response structure, which has been shown to shape model behavior without any change to the model's weights (Brown et al., 2020). Taken together, these strategies treat the prompt as a controlled intervention whose effect on the model's behavior, and on the ground-

edness of its explanations, can be measured. The specific prompt conditions built from these ideas, and how they are implemented and compared in this study, are described in Chapter 3.

2.6 From the Literature to the Present Study

The literature reviewed in this chapter motivates a small set of expectations that the rest of this thesis sets out to test. These expectations are based on three main findings from the literature. First, word-level intelligibility is a feasible but challenging task on which ASR-based systems have achieved only modest performance. Second, unlike these systems, speech-enabled multimodal LLMs can provide phonetic explanations alongside their judgments. Third, both empirical evidence and architectural considerations raise questions about whether these explanations are really grounded in the audio. Based on these findings, three broad hypotheses are considered in relation to the research questions introduced in Chapter 1.

H1: Classification performance. Because these models process the audio directly and bring a large pre-trained language component to bear, they are expected to handle word-level intelligibility classification and to be at least competitive with prior ASR-based work on this corpus. Since they can draw on the acoustic signal and on broad linguistic knowledge at once, rather than reducing the judgment to a transcription match, there is reason to expect them to match or improve on that earlier ceiling.

H2: Effect of prompting. Since these models are guided entirely through their prompt rather than retrained, the way the task is framed is expected to matter. Prompting strategies that impose structure and separate perception from decision, such as structured output, a phonetic checklist, a DDCoT-style split, and few-shot examples, are expected to shape and, in the better cases, improve classification behavior relative to a plain baseline prompt.

H3: Grounding of explanations. The most uncertain most important expectation concerns whether the models' explanations are grounded in the audio. The hope is that careful prompting, by directing the model's attention to the acoustic signal, can encourage explanations that rest on what was heard. At the same time, the lexical dominance that Chen et al. (2026) report and the gap between plausibility and faithfulness that Madsen et al. (2024) describe give real reason for concern that phonetically phrased justifications may be produced even where the audio does not warrant them. Because these two possibilities cannot be told apart from the fluency of the output, grounding is not something this study is willing to assume in either direction; it is probed directly, and this is treated as the central open question the evaluation is designed to answer.

Whether each of these expectations holds is examined in the results and taken up in the discussion.

Chapter 3

Methodology

3.1 Dataset

3.1.1 Corpus Origin

The data for this study comes from the MOOC *English Pronunciation in a Global World*, offered by Vrije Universiteit Amsterdam's Centre for Global English. Recordings were produced by L2 English learners under normal, unsupervised home conditions; no researcher was present during recording, and learners used their own devices to record and submit their audio. As a result, the audio is not acoustically uniform, and some recordings contain background noise that was not filtered out during collection. This preserves the recordings original acoustic conditions and allows the dataset to reflect naturalistic recording conditions.

Reliable metadata on speaker demographics is not available. English proficiency level is similarly not reliably recorded; a self-report survey was introduced in later course iterations, but its response rate was too low to treat as representative. For these reasons, this study does not make any claim that depends on speaker nationality, first-language background, or proficiency level.

The number of recording attempts per learner is also unknown. Learners could re-record and choose which take to submit, so the dataset may include a degree of self-selection. Some learners likely submitted a practiced or preferred take, while others may have submitted a first or incomplete attempt. This was not tracked during data collection.

It is also worth noting that this dataset was not originally collected for machine learning purposes. The recordings were first and foremost pedagogical materials, produced as part of learners pronunciation practice within the MOOC; their use for model training and evaluation in this thesis, as in the prior work of Zou (2024) and Kuan (2025) on the same corpus, is a secondary use of data collected for a different original purpose. All data were anonymized prior to use, and learners consented to data use for research purposes at the point of course enrollment.

3.1.2 Word List

The word list used across the corpus consists of 53 target words, predominantly following a consonant-vowel-consonant cluster pattern (e.g. *pat*, *board*, *seedy*).[1] The list was

[1]Kuan (2025) works from a 52-word list; the exact subsets are not identical

adapted by Dr. Laura Rupp from an existing word list. Table 3.1 lists the full set.

Table 3.1: Distribution of gold intelligibility labels across the 53 target words in the test set.

Word	*N*	Intelligible	Not intelligible	% Intelligible
bard	158	46	112	29.1
bay	162	127	35	78.4
bear	159	75	84	47.2
bee	160	144	16	90.0
beer	160	88	72	55.0
bird	157	54	103	34.4
board	152	44	108	28.9
boat	161	101	60	62.7
boot	161	103	58	64.0
bout	159	70	89	44.0
boy	161	135	26	83.9
buy	160	126	34	78.8
caught	159	36	123	22.6
city	156	112	44	71.8
cot	157	102	55	65.0
daft	161	111	50	68.9
dance	158	124	34	78.5
doll	158	94	64	59.5
fair	156	80	76	51.3
farther	160	67	93	41.9
father	161	93	68	57.8
fern	156	92	64	59.0
fir	157	61	96	38.9
fur	158	66	92	41.8
half	158	109	49	69.0
hat	158	133	25	84.2
knows	155	90	65	58.1
mate	152	124	28	81.6
meat	153	93	60	60.8
meet	154	118	36	76.6
nose	157	125	32	79.6
pat	160	108	52	67.5
paul	157	57	100	36.3
pause	154	47	107	30.5
paw	157	37	120	23.6
paws	152	32	120	21.1
pet	161	131	30	81.4
pit	161	113	48	70.2
plate	75	53	22	70.7
pole	157	88	69	56.1
pool	159	75	84	47.2
poor	155	50	105	32.3

Word	N	Intelligible	Not intelligible	% Intelligible
pore	156	84	72	53.8
pot	160	112	48	70.0
pour	155	72	83	46.5
pull	154	75	79	48.7
put	160	112	48	70.0
putt	160	40	120	25.0
seedy	158	61	97	38.6
tide	154	62	92	40.3
tied	155	58	97	37.4
wait	87	69	18	79.3
weight	155	125	30	80.6

3.1.3 Recovering Labels and Building the Splits

There was no separate metadata file distinguishing the development and test splits. The gold label and the target word were reconstructed directly from the filenames. A filename such as `series01-s00000-board-0.mp3` was parsed so that the final segment (`0` or `1`) gives the gold label and the segment before it gives the target word.

Annotation of the gold labels was carried out separately, under the supervision of Dr. Laura Rupp, using a phonetic feature-based annotation scheme developed specifically for this word list. The scheme covers six feature types: *rhoticity* (whether /r/ is pronounced when written), *vowel length* (whether a vowel is shortened or lengthened relative to the target), *vowel quality* (whether the wrong vowel sound is produced), *consonant substitution and aspiration* (whether a consonant is replaced by a different one, or an initial stop lacks the expected aspiration), and *extra or missing sounds* (sounds inserted or dropped at the start, middle, or end of the word). Inter-annotator agreement on this scheme was 78.9% (pre-adjudication), calculated as full agreement across *all* features for a given word; the lower figure of 66.3% (also pre-adjudication) reflects agreement restricted to files where at least one annotator flagged a mistake, i.e. cases where a phonetic issue was actually detected by someone. Here we note only that the binary intelligible / not_intelligible label used in this thesis is derived from that underlying feature annotation. This same feature vocabulary later reappears in the prompt design (Section 3.4): the fixed set of phonetic issue labels the models are constrained to choose from is drawn directly from this annotation scheme.

The development set contains 2,001 samples (1,128 intelligible, 873 not_intelligible). The test set contains 8,196 samples (4,604 intelligible, 3,592 not_intelligible). The development set was used for all prompt-engineering experiments, the diagnostic probe (Section 3.5), and few-shot example selection. The test set was used only once, at the very end, for the final evaluation of the chosen system.

3.1.4 Per-Word Distribution

Table 3.1 shows this distribution in full. Although the overall test-set base rate is 56.2% intelligible, this balance is not uniform across individual words; some words are almost always pronounced intelligibly by the learners in this corpus (e.g. *bee*, at 90%), while others are almost always flagged as not_intelligible in the gold labels (e.g. *paws* or *caught*, both around 77–79%). This distribution is relevant when interpreting per-word

model behavior. For some words, one gold label occurs far more often than the other; a model that predicts the majority label regardless of the acoustic signal will therefore appear accurate on that word, without necessarily using the audio at all. This should not be confused with a separate phenomenon, *prediction collapse*, in which the model predicts the same label for every occurrence of a word irrespective of the underlying gold-label distribution. For *pull*, for instance, the gold labels are close to balanced, yet the model predicted not_intelligible for every sample. This behavior is examined further in Chapter 5.

3.2 Task Definition

The task is a binary, word-level intelligibility classification. For every audio sample, the model must decide whether an average listener could identify the intended target word from the recording; the goal is intelligibility, not native-like pronunciation scoring, and the model is explicitly instructed to rely only on the acoustic evidence present in the recording. For every sample, the model must choose between two labels, `intelligible` and `not_intelligible`, must report its confidence and flag cases of genuine uncertainty, and must also produce a short explanation for its decision (which, as prompt complexity increases, is explicitly constrained to specific phonetic features - see Section 3.4, where the full output schema is introduced). The unit of analysis is the single spoken word; this reflects the structure of the annotated dataset made available for this internship-based project, in which only word-level labels were provided. This constraint's advantages for isolating the acoustic signal from surrounding context are discussed in the Introduction.

3.3 Models

Three open-source multimodal speech models were evaluated: Qwen2-Audio-7B-Instruct,[2] Qwen2.5-Omni-7B,[3] and MiniCPM-o-2.6.[4] All three range from roughly 7 to 8 billion parameters; each combines an audio encoder with a large language model backbone, though the specifics of how audio is integrated differ across the three. A detailed technical comparison of their architectures, training, and modality support is given in the Literature Review; here only the methodological consequences of these differences are described.

Two considerations guided this selection. First, all three models are open-source and can be run locally without routing learner audio through a third-party commercial API, which matters given the privacy sensitivity of L2 learner speech data. Second, at 7–8B parameters, the models were large enough to bring meaningful linguistic and acoustic capability to bear while remaining small enough to run on the GPU resources available for this project.

Each model was loaded using its own architecture-specific implementation rather than a generic text-only class, so that its audio encoder and multimodal processing pipeline were correctly initialized. Because each model family required a different version of the underlying `transformers` library, the three models were never run in the same session; each model's dependencies were installed, the runtime was restarted, and only then was that model evaluated, before the process was repeated for the next

[2] `https://huggingface.co/Qwen/Qwen2-Audio-7B-Instruct`
[3] `https://huggingface.co/Qwen/Qwen2.5-Omni-7B`
[4] `https://huggingface.co/openbmb/MiniCPM-o-2_6`

model. Inference settings were kept identical and deterministic across models wherever the architecture allowed. Generation used greedy decoding (`do_sample=False`), so that outputs are reproducible. The token budget allowed for generation was kept small (`max_new_tokens=220`), sized to accommodate the full six-field output described in Section 3.4.4.

An earlier fourth model, Qwen3-Omni, was evaluated during initial grounding checks (Appendix B) but was not included in the main prompt-condition matrix: its substantially higher computational cost was not justified by the limited benefit observed in initial prompt testing, and running the six-condition matrix with this model was beyond the available computational resources for this project. The inference code for all three models, together with the prompt-construction and parsing pipeline, is available in a repository[5] for reproducibility and future access.

3.4 Prompt Conditions

Six prompt conditions (C1-C6) were designed to test whether specific, controlled changes to the prompt affect model behavior on the task defined in Section 3.2. There is a single guiding principle for designing prompts; in every pairwise comparison between conditions, exactly one variable changes. This makes it possible to attribute any observed difference in model behavior to that specific intervention. All conditions were run identically across all three models on the same shared development-set subset.

Every condition shares the same base instruction, described in Section 3.2. The six conditions differ only in whether a phonetic checklist is added, whether a perception-before-decision structure (DDCoT) is added, and whether few-shot examples are added. Table 3.2 summarizes the resulting matrix, and Table 3.3 lists the specific research question each pairwise comparison is designed to answer.

Table 3.2: The six prompt conditions used in this study.

Condition	Shot type	Strategy
C1	Zero-shot	Structured baseline
C2	Zero-shot	Phonetic checklist prompt
C3	Zero-shot	Duty-Distinct Chain-of-Thought (perception before decision)
C4	Few-shot	C1 + examples
C5	Few-shot	C2 + examples
C6	Few-shot	C3 + examples

C1 serves as the reference condition against which every other condition is compared. It contains the base instruction and the required JSON output format (Section 3.4.4), but no phonetic checklist and no perception-before-decision structure. Any feature added to C1 in later conditions is added exactly once, and its effect is read off the corresponding row of that table.

3.4.1 Phonetic Checklist (C2, C5)

C2 and C5 extend the base instruction with a phonetic checklist generated dynamically for each target word, drawing on the same feature vocabulary introduced in the

[5] `https://github.com/massoumehjavadi-cloud/speech-llm-for-intelligibility-classification`

Table 3.3: Pairwise comparisons

Comparison	Research question
C1 vs. C2	Does adding explicit phonetic guidance improve intelligibility assessment?
C1 vs. C3	Does separating acoustic perception from the decision strengthen acoustic grounding?
C1 vs. C4	Does few-shot learning improve the baseline?
C2 vs. C5	Does few-shot learning strengthen checklist-based prompting?
C3 vs. C6	Does few-shot learning strengthen DDCoT?

annotation scheme described in Section 3.1.3 - for example `rhoticity_missing`, `vowel_length_shortening`, `vowel_quality_substitution`, `initial_consonant_missing_aspiration`, `final_consonant_substitution`, and the family of `extra_sound_*` and `missing_sound_*` labels -. Rather than reusing a single fixed template across all fifty-three words, the checklist names the phonetic dimensions most likely to matter for the specific word in question, together with a small set of illustrative minimal-pair contrasts. For the target word *pat*, for instance, the checklist points to aspiration, vowel quality, the final consonant, and lists *bat*, *pet*, and *pit* as contrasts a mispronunciation might be confused with; for *board*, the relevant dimensions and contrasts are different, since the two words carry different phonetic risks. The checklist functions as attention guidance rather than a scoring rubric since it tells the model where to listen closely, and the model is asked to describe only what is actually audible in the recording.

When the model predicts `not_intelligible` under C2 or C5, its explanation is required to begin with one of a small, fixed set of phonetic issue labels. The intent behind this design choice is to keep the model's explanatory vocabulary anchored to the same categories a human phonetic annotator would use. Restricting the model to a fixed label set narrows the space of labels available, but does not guarantee that a chosen label is phonetically accurate, or that the model actually detected the feature it names rather than reading it off the checklist. Whether the labels produced under this constraint are trustworthy is therefore an empirical question, addressed through the manual annotation described in Section 3.6 and examined further in the error analysis.

3.4.2 Duty-Distinct Chain-of-Thought (C3, C6)

C3 and C6 add a lightweight version of Duty-Distinct Chain-of-Thought, a prompting strategy that separates a model's perception of its input from the decision it draws from that perception; the theoretical motivation for this separation, and its origin in Zheng et al.'s work on multimodal reasoning, is discussed in the Literature Review. For the present method, the practical consequence is that the model is required to work through a fixed three-step procedure before it is permitted to commit to a label.

In the first step, the model is instructed to attend only to the acoustic signal and to briefly state what is clearly audible in the recording, describing only evidence that is directly supported by the audio itself. In the second step, it is explicitly told not to infer pronunciation details that are not clearly audible, and not to rely on the written target word when constructing this acoustic description; the written target word is available to the model elsewhere in the prompt, but the description produced at this step is meant to reflect only what was heard, independent of what the model has been told to expect. Only in the third step, once this acoustic description has been produced, is the model

permitted to decide whether the spoken word can reasonably be understood as the target word. Two additional constraints tie the decision back to the perception step: the final decision must be consistent with the evidence already recorded in the description, and the model is instructed not to introduce any new evidence at the decision stage that was not already stated during acoustic description.

This staged procedure targets a specific failure mode observed under unstructured prompting. When a model receives the audio and the written target word simultaneously and is asked directly for a verdict, it can reach that verdict by reading the target word and assuming correct pronunciation, then generating a justification after the fact. Requiring the model to produce a target-word-independent description of what it hears, before it is permitted to decide, forces a checkpoint between perception and judgment that this shortcut cannot bypass.

This ordering does not prove that the model's decision is grounded in the audio. A model can produce an acoustic description that sounds plausible and internally consistent without being supported by the signal. What the procedure provides is a separation between the model's stated evidence and its final label, which can then be checked against each other. This separation is exploited in the error analysis (Chapter 5), where the consistency between a model's stated `audio_evidence` and its `predicted_label` is examined as one indicator of whether its explanations are faithful to its own stated perception, in the sense discussed in the Literature Review.

3.4.3 Few-Shot Examples (C4, C5, C6)

C4, C5, and C6 add a fixed block of eight examples to the base instruction used in C1, C2, and C3 respectively, without changing anything else about the prompt. Each example gives a target word together with the complete JSON object the model is expected to produce for that word, illustrating both the intended decision style and the exact output format.

To select these eight examples, the researcher started from a larger pool of twenty development-set recordings, balanced across the two gold labels. All recordings were reviewed manually. For the intelligible examples, recordings with relatively clear audio quality and minimal background noise were preferred, so that the pronunciation could be evaluated from the speaker's production. For the not intelligible examples, recordings were selected in which the pronunciation error was clearly audible while the recording itself remained sufficiently clear. The aim was not to choose the easiest or the most extreme cases, but to minimize ambiguity introduced by recording quality and ensure that the examples provided clear acoustic evidence for their assigned labels.

The eight selected examples cover the target words *mate*, *boy*, *pool*, *caught*, *nose*, *paws*, *pull*, and *seedy*, split evenly between the two gold labels. The JSON object shown for each example follows the same schema described in Section 3.4.4, with placeholder values for `audio_evidence` (e.g. "Manual listening required before final few-shot use") rather than a full phonetic description, since the purpose of the block is to demonstrate format and decision style, not to supply the model with realistic acoustic-evidence text it might otherwise imitate verbatim.

The few-shot block is processed before the actual target recording in the sense that it appears earlier in the prompt, but it is not a separate inference step; the model receives the instruction, the few-shot block, and the new audio together in a single forward pass, and produces one JSON object in response. The instruction explicitly tells the model to evaluate the new audio independently and not to copy an example label unless the

new audio actually supports it, since without this instruction a model could in principle default toward reproducing whichever label appears most often in the examples rather than judging the new recording.

3.4.4 Output Schema

Every condition requires the model to return its response as a single JSON object instead of unstructured free text. This allows responses to be parsed and scored automatically without manual coding. Section 3.4.5 describes how this JSON is recovered from the raw model output, including cases where the model does not comply exactly with the requested format.

The schema requires six keys. `predicted_label` holds the binary decision itself, restricted to `intelligible` or `not_intelligible`. `confidence` takes one of three values, `low`, `medium`, or `high`. `short_reason` and `audio_evidence` together provide a justification for the decision. `uncertainty_flag` is a binary flag, `yes` or `no`.

The `confidence` field is a graded self-assessment of the label the model has already committed to: given that the model has settled on `intelligible` or `not_intelligible`, how strongly does it stand behind that specific answer. The `uncertainty_flag` field, in contrast, is not about the strength of a commitment already made, but about whether the underlying acoustic evidence is sufficient to support committing to an answer at all; it gives the model a way to signal that the evidence is weak, ambiguous, or otherwise insufficient, rather than forcing it into a confident-sounding guess. Because these two fields measure different things, a model can report high confidence in a label while simultaneously flagging `uncertainty_flag` as `yes` - for instance, if it judges its best guess more likely correct than not, while still recognizing that the evidence for that guess is thin.

Whether the `confidence` field actually reflects prediction correctness is a separate question from requiring the model to produce it. The literature on LLM calibration shows that confidence scores cannot automatically be interpreted as reliable indicators of correctness (Section 2.4.3). RLHF, the post-training procedure used to align model outputs with human preferences, is known to reduce calibration because it encourages models to produce responses that humans prefer, rather than responses whose stated confidence accurately reflects the model's uncertainty. As a result, post-trained models may assign similarly high confidence to both correct and incorrect predictions.

Several verification-based methods, such as P(True), attempt to estimate whether a model's own answer is likely to be correct by asking the model to evaluate it. However, previous work shows that their effectiveness varies across models and benchmarks, especially on trustworthiness-oriented tasks. Word-level intelligibility classification belongs to this category, since this thesis evaluates not only whether the predicted label is correct, but also whether the model's reported confidence and uncertainty flag provide reliable signals about when its judgment should be trusted.

Taken together, these findings mean that a mismatch between `confidence` and `uncertainty_flag` - a model reporting high confidence while also flagging uncertainty - is informative, and is examined as a potential sign of miscalibration in Chapter 5, alongside the more basic question of whether `confidence` values track the model's actual accuracy at all.

The prompt does not give the model explicit criteria for choosing among the three confidence levels or between the two uncertainty values. The only guidance offered in the prompt is a single general instruction, given before the target word and schema are

introduced: if the acoustic evidence is weak, ambiguous, incomplete, noisy, or difficult to interpret, the model should adopt a conservative decision style and note the uncertainty in `audio_evidence`. This instruction is not tied to either field specifically, and it offers no operational threshold for what counts as weak or ambiguous evidence.

The two fields are also not generated at the same point in the response. Because the schema fixes the order of keys as `predicted_label`, `confidence`, `short_reason`, `audio_evidence`, `uncertainty_flag`, and `possible_irrelevant_answer_risk`, and because generation is autoregressive, `confidence` is produced immediately after the label itself, before any justification has been written. `uncertainty_flag`, by contrast, is produced only after `short_reason` and `audio_evidence` have already been generated. This means `confidence` is committed to with comparatively little of the response already in place, while `uncertainty_flag` is generated with the model's own stated reasoning and evidence already available to condition on. Consequently, the model assigns its confidence before articulating its reasoning, whereas the uncertainty flag is generated after that reasoning is already available in the context. Together with the absence of explicit selection criteria, this difference in generation order may contribute to the mismatches between the two fields examined in Chapter 5.

Separating `short_reason` from `audio_evidence` was motivated by the requirements of the DDCoT conditions described above; it makes it possible to check, after the fact, whether the acoustic evidence the model claims to have observed actually matches the label and justification it goes on to give. `possible_irrelevant_answer_risk`[6] is a binary flag the model sets when it judges that its own answer may not directly address the task as intended - for example, if the recording appears to contain a different word entirely, contains no discernible speech, or if the model suspects its response may not be usable for the intended evaluation.

3.4.5 Output Parsing

Because the raw text returned by the models was not always well-formed JSON, parsing was kept separate from inference, and treated as a distinct stage of the pipeline. The raw output of every model call was saved exactly as produced, before any interpretation was applied to it. If a bug or limitation is later discovered in the parser or in a downstream metric, the models do not need to be rerun, since the original responses are preserved unmodified.

The parser itself operates in three stages, attempted in a fixed order. It first attempts a JSON parse using `json.loads`. If this fails, it attempts to recover the response as a Python-literal dictionary using `ast.literal_eval`, which handles a common failure mode in which a model returns dictionary-style text using single quotation marks instead of the double quotation marks required by the JSON specification. If both of these fail, a final regular-expression fallback attempts to extract at minimum the `predicted_label` and `confidence` fields directly from the raw text, on the reasoning that even a response that fails to conform to any structured format may still contain these two fields in a recoverable form. Every row of output was tagged with which of these three methods succeeded - `strict_json`, `python_literal_fallback`, or `regex_field_recovery` - so that the proportion of each outcome could later be reported and compared across models and conditions; these proportions are reported in Chapter 4.

[6]The field is spelled `possible_irrelevent_answer_risk` in the underlying code and raw model outputs; the misspelling is preserved there for consistency with the logged data, though the correctly spelled form is used in prose throughout this thesis.

To make the format concrete, a well-formed response to condition C6 for the target word *put* takes the following form:

```
{
  "predicted_label": "intelligible",
  "confidence": "medium",
  "short_reason": "The pronunciation is understandable.",
  "audio_evidence": "The initial consonant is clearly audible.",
  "uncertainty_flag": "no",
  "possible_irrelevant_answer_risk": "no"
}
```

This example illustrates the full six-key schema described in Section 3.4.4: a binary label, a graded confidence value, a separated justification and evidence field, and the two flags for genuine uncertainty and possible task irrelevance.

3.5 C6+ (Transcription-Augmented)

C6+ is a diagnostic extension of C6, used to probe more directly whether a model's decision is actually grounded in the acoustic signal. It shares everything with C6 - the same base instruction, the same DDCoT structure, the same eight-example few-shot block, and the same parser - with one addition: before committing to a label, the model must first report the single word it hears in the recording, based only on the acoustic signal, in a `heard_word` field. This field is placed first in the JSON object, ahead of `predicted_label` and every other key.

The position of this field matters because generation is autoregressive; whatever the model writes first is generated with the least prior context, and everything it writes afterward can be conditioned on it. Placing `heard_word` first means the transcription is produced before the model has committed to an intelligibility judgment, which gives it a degree of independence from that judgment that a transcription placed later in the response would not have. But this independence is partial; the target word is still shown in the prompt, so the transcription is not blind to what the model has been told to expect, and it is not possible to fully separate what the model heard from what it was primed to hear. What generation order provides is a soft asymmetry in which direction of influence is more plausible, not a guarantee that the transcription is acoustically pure.

C6+ was run on the full development set, across all three models, under identical settings; it was not run on the test set and does not appear in the main results tables reported in Chapter 4. Because C6+ isolates a transcription that is generated before the label, it is used specifically to compare what a model reports having heard against what it goes on to decide, and any systematic divergence between the two is treated as evidence bearing on acoustic grounding.

3.6 Manual Annotation Protocol

Classification accuracy alone does not indicate whether the explanations a model produces are trustworthy. To address the second part of the research question, which concerns the quality of the model's explanations rather than the correctness of its labels, 300 samples were drawn from the frozen test-set predictions of the chosen system, MiniCPM-o-2.6 under C6, and manually annotated by the researcher.

The sample is balanced across four groups of 75 examples each, defined by the model's test-set predictions relative to the gold labels: false rejections (FP under the positive class `not_intelligible`), missed problems (FN), true rejections (TP), and true acceptances (TN). Within each group, the examples were selected to cover a range of target words, confidence levels, uncertainty flags, and word-difficulty levels (easy, medium, and hard), based on the per-word accuracy reported in Section 3.1.4. In addition, no more than six examples of the same target word were included in any group. These constraints were intended to prevent the sample from being dominated by a small number of words or by only easy or difficult cases.

The two correct-prediction groups, TN and TP, were included, because a correct label does not by itself mean that the model arrived at the right answer for the right reason. A model may classify a word correctly while offering an explanation that is not supported by the acoustic evidence, a possibility discussed in the literature review. Examining only the FP and FN groups would miss this entirely, since those groups contain only incorrect predictions; including TN and TP makes it possible to ask whether correct labels are also accompanied by acoustically grounded explanations, rather than treating a correct label as sufficient on its own.

Annotation was carried out through a custom interface, one sample at a time. For each item the interface displayed the associated metadata and the model's output, played back the corresponding audio recording, and presented the three judgment axes together, so that every judgment was made while listening to the recording. The three axes, and the categories available on each, are set out in Table 3.4. Every axis also offered a "cannot determine" option so that no judgment was forced in genuinely unclear cases.

This annotation is reported as qualitative, diagnostic evidence for the second part of the research question, alongside the quantitative classification metrics in Chapter 4.

All 300 samples were annotated by a single researcher, without a second annotator or an inter-annotator agreement check. This differs from the two-annotator agreement protocol used for the original phonetic feature labels described in Section 3.1.3, and is acknowledged as a limitation of this qualitative analysis.

3.7 Evaluation Metrics

Three kinds of measure are reported: accuracy, Macro-$F_{0.5}$, and per-class precision, recall, and $F_{0.5}$ for both the `intelligible` and `not_intelligible` classes. Accuracy measures the proportion of correctly classified samples but does not distinguish how errors are distributed across the two classes. Macro-averaging gives equal weight to each class, preventing performance on the larger class from dominating the score. Throughout this thesis, Macro-$F_{0.5}$ is treated as the primary evaluation metric.

In any precision/recall analysis, one class must be designated the positive class, since precision and recall are defined with respect to it. Here, the positive class is not_intelligible; it is the error class, and the one the system is specifically built to catch. Under this choice, a False Positive is a word whose gold label is intelligible but which the model labeled not_intelligible - in a teaching context, this means telling a learner that a perfectly understandable word was wrong, for no real reason. A False Negative is a word whose gold label is not_intelligible but which the model labeled intelligible - a real pronunciation problem that the model let slip through unflagged.

F0.5 weights precision more heavily than recall. Since precision is defined as TP /

Table 3.4: The manual annotation scheme: the three judgment axes and the categories available for each.

Axis	Categories
Feedback quality	*Appropriate and sensible* - the explanation is correct and useful for the word. *Appropriate but incomplete* - correct as far as it goes, but missing part of the picture. *Incorrect but relevant* - the stated reason is wrong, but still plausible for that word. *Irrelevant* or *Unsupported* - the reason does not relate to the audio, or is not supported by the evidence.
Grounding quality	*Clearly based on audible evidence* - the explanation reflects what is actually in the recording. *Partly based on audible evidence* - some acoustic basis, mixed with unsupported claims. *Mainly text-based* - the explanation is driven by the identity or spelling of the target word rather than the recording; it describes what the word should sound like rather than what is actually in the audio. *Not supported by the audio* - the explanation makes an acoustic claim that the recording does not bear out; it asserts audible evidence that is not actually there.
Pedagogical safety	*Safe* — the feedback could be shown to a learner as is. *Partly safe* — usable only with reservations. *Unsafe* — the feedback would mislead or confuse a learner.

Each axis additionally offered a *cannot determine* option for cases the annotator could not resolve.

(TP + FP), the quantity that most directly hurts precision is the FP count. Weighting precision more heavily than recall therefore means that F0.5 penalizes FPs more than it penalizes FNs. This weighting reflects a pedagogical judgment: in a learning setting, telling a learner that a clearly understandable word is wrong is judged more harmful than quietly missing a pronunciation error, since a false rejection risks damaging a learner's confidence and pushing them toward chasing native-like norms they do not need, whereas a missed problem simply leaves an existing issue unflagged for the moment. A metric that rewards models for keeping FPs low is therefore rewarding exactly the conservative, cautious labeling behavior that a pedagogically-oriented system should have.

Macro-averaging both classes prevents a model from improving its apparent performance simply by collapsing onto the majority class. A model that always predicts intelligible would be penalized in the not_intelligible class's F0.5 score and therefore in the macro average, even though such a model could otherwise look strong on accuracy alone. Macro-F0.5 is therefore used as the main measure of system health across both classes; where the discussion turns specifically to the pedagogical stakes of the task, the analysis narrows to the F0.5 and precision of the not_intelligible class specifically, since those numbers most directly reflect how well a system avoids false rejections in particular.

Alongside these aggregate scores, per-class precision, recall, and F0.5 are reported separately for the `intelligible` and `not_intelligible` labels. The per-class metrics are used throughout Chapter 4, since it is the per-class numbers that reveal asymmetries

such as a model achieving strong precision on one class while collapsing on the other.

Chapter 4

Results

This chapter reports the results of the experiments run under the conditions set out in the methodology. Section 4.1 reports an initial prompt evaluation used to narrow down the models and prompt conditions carried into the main experiments. Section 4.2 reports the six prompt conditions on the development set and identifies the configuration that was carried forward, which is where the second research question is addressed. Section 4.3 reports the evaluation of the best system on the test set, and Section 4.4 compares the selected system against the earlier ASR-based system on the same corpus, which is where the first research question is addressed. The small acoustic-grounding sanity checks conducted before the main evaluation are reported separately in Appendix B, because they are diagnostic controls rather than part of the primary results. The analysis of performance by individual words, of the explanations the models produced, and of how the three models relate to one another is reported in the error analysis in Chapter 5. Throughout this chapter the positive class is `not_intelligible`. Everything here is reported as a raw outcome; what these outcomes mean, and why the models might behave this way, is taken up in the discussion.

4.1 Initial Prompt Evaluation

Before settling on the setup used for the rest of this chapter, a short run was carried out first, mostly to see which models would run reliably and how they behaved when simply asked for a label with no further guidance. This was a quick first pass; the models were given a prompt, with no required output format and no reasoning steps, and their answers were looked at only to get a rough sense of whether the pipeline worked end-to-end and whether each model was usable at all. This initial run also showed which models were reliable enough to carry forward into the main experiments. Qwen3-Omni, the largest of the models at 30 billion parameters, ran but was far too slow to keep in the main comparison. It took on average around 62.6 seconds to process a single sample, two to three orders of magnitude more than the smaller models, which made it prohibitively expensive to run across the full prompt-condition matrix. Its behavior on this first pass was in any case weak, since it leaned heavily toward the `intelligible` label, assigning it to roughly 93% of the 1,000 samples, and as a result it recovered only a small fraction of the real problems, with a recall on unintelligible speech of about 0.12. Between the cost of running it and how little it caught, there was little to justify carrying it further. The models that kept running into errors were not simply dropped. They were looked at closely and run repeatedly, adjusting how each one was called

and how its output was handled, until a clean and reproducible pipeline could be built around them. Beyond confirming that these remaining models worked end-to-end, and that they already differed noticeably in how readily they assigned each label, nothing in this initial experiment is treated as a result in its own right; the evaluation proper begins with the structured prompt conditions in Section 4.2.

4.2 General Performance

The main experiments of this thesis focused on the three models that were carried forward from the initial experimentation step (see Section 4.1): MiniCPM-o-2.6, Qwen2.5-Omni-7B, and Qwen2-Audio-7B-Instruct, each run across all six prompt conditions on the development set of 2,001 samples. This is the part of the study that speaks to the second research question, since it is where the effect of the prompt design is actually measured, and it is also the stage at which the configuration used for the final test-set evaluation was chosen.

Table 4.1 reports accuracy, macro-averaged $F_{0.5}$, and then the precision, recall, and $F_{0.5}$ for each of the two classes separately, so that the two sides of the model's behavior can be read off directly. The macro-$F_{0.5}$ is used as the headline figure because, as explained in the methodology, this task treats a false rejection (intelligible words flagged as not intelligible) as more costly than a missed problem, and $F_{0.5}$ weights precision accordingly; the per-class columns are given alongside it because the averaged score, on its own, can hide the fact that a model is doing well on one class only by abandoning the other.

Table 4.1: Development-set performance for each model and prompt condition, with per-class precision, recall, and $F_{0.5}$. The highest macro-$F_{0.5}$ is shown in bold.

				intelligible			*not_intelligible*		
Model	Cond	Acc	Macro-$F_{0.5}$	P	R	$F_{0.5}$	P	R	$F_{0.5}$
MiniCPM-o-2.6	C1	0.597	0.538	0.594	0.901	0.637	0.614	0.204	0.438
MiniCPM-o-2.6	C2	0.568	0.410	0.569	0.963	0.620	0.543	0.057	0.201
MiniCPM-o-2.6	C3	0.550	0.552	0.623	0.511	0.597	0.488	0.601	0.507
MiniCPM-o-2.6	C4	0.590	0.494	0.584	0.950	0.633	0.661	0.125	0.356
MiniCPM-o-2.6	C5	0.600	0.516	0.590	0.954	0.638	0.705	0.142	0.393
MiniCPM-o-2.6	C6	0.609	**0.580**	0.614	0.824	0.647	0.593	0.332	0.513
Qwen2.5-Omni-7B	C1	0.560	0.436	0.568	0.924	0.615	0.479	0.090	0.256
Qwen2.5-Omni-7B	C2	0.562	0.437	0.569	0.926	0.617	0.485	0.090	0.258
Qwen2.5-Omni-7B	C3	0.566	0.463	0.572	0.913	0.618	0.512	0.118	0.307
Qwen2.5-Omni-7B	C4	0.579	0.515	0.584	0.877	0.626	0.550	0.195	0.403
Qwen2.5-Omni-7B	C5	0.586	0.497	0.583	0.932	0.630	0.611	0.139	0.364
Qwen2.5-Omni-7B	C6	0.598	0.550	0.599	0.870	0.638	0.593	0.246	0.462
Qwen2-Audio-7B-Instruct	C1	0.558	0.446	0.566	0.912	0.612	0.481	0.104	0.279
Qwen2-Audio-7B-Instruct	C2	0.436	0.340	0.500	0.058	0.197	0.432	0.926	0.483
Qwen2-Audio-7B-Instruct	C3	0.522	0.486	0.560	0.710	0.585	0.427	0.279	0.386
Qwen2-Audio-7B-Instruct	C4	0.531	0.529	0.593	0.534	0.580	0.466	0.526	0.477
Qwen2-Audio-7B-Instruct	C5	0.505	0.510	0.587	0.412	0.541	0.451	0.625	0.478
Qwen2-Audio-7B-Instruct	C6	0.535	0.523	0.584	0.614	0.589	0.464	0.432	0.457

The configuration that reaches the highest macro-$F_{0.5}$ is MiniCPM-o-2.6 under con-

dition C6, and it is this pairing that was carried into the test-set evaluation in Section 4.3. Beyond that best pair, the improvement from one prompt to the next is not steady or uniform across the models. As expected, there is a natural trade-off between precision and recall here: for MiniCPM, the conditions that push recall on the `not_intelligible` class upward are exactly the ones that pull its accuracy down, so that the prompt which looks strongest on the headline figure is not the one that flags the most problems. Across the three models, C6 was the strongest condition for two of them, MiniCPM-o-2.6 (macro-$F_{0.5}$ of 0.580) and Qwen2.5-Omni-7B (0.550), matching the expectation that the fullest prompt would help most. Qwen2-Audio-7B-Instruct behaves differently: its highest macro-$F_{0.5}$ comes from C4 (0.529), with C6 close behind (0.523), while C1 already scores higher than C2 even though C2 adds the phonetic checklist, so that more instruction does not translate cleanly into better behavior. The models also do not respond to the same prompt in the same way, which is visible in how differently the per-class recall shifts down each block of Table 4.1. The question of why the prompts act so unevenly is returned to in the discussion; here it is enough to note that no single condition is uniformly best, and that the configuration chosen for the test set was selected on the macro-$F_{0.5}$ criterion set out in advance.

4.2.1 Class Prediction Patterns

One recurring pattern in Table 4.1 is particularly important for interpreting the per-class results. Under several conditions, the model predicts almost every sample as belonging to the same class. As a result, recall for that class becomes very high, while recall for the other class falls close to zero. MiniCPM under C2 is the clearest case, assigning `intelligible` to almost every sample and so recovering only a handful of the real problems, but the same tendency appears, in the opposite direction, for Qwen2-Audio under C2, where the model swings the other way and labels most samples `not_intelligible`. The effect is not confined to one model or one label, then, but is a recurring failure mode across the matrix. Because this class prediction pattern also plays out at the level of individual words, where whole words are pushed entirely into one class, it is examined in full, with the per-word counts, in the error analysis (Chapter 5); for now, some of the per-class figures in the table above reflect these prediction patterns.

If the conditions are compared directly rather than considered model by model, the pattern becomes clearer. One way to see it is to look at the distance between the two per-class recall figures in each row: when a model is discriminating between the classes, the recall values for the two labels tend to be relatively close, whereas a large gap between them indicates a strong preference for one label. In the second case, the model has leaned onto one label. Read this way, condition C2 is the condition under which all three models lean most strongly toward one label. For each model, it produces the widest recall gap of any condition. It is therefore the prompt to which all three models respond most similarly, and also the one that most reduces the balance between the two classes. However, the direction of that lean is not shared; MiniCPM and Qwen2.5-Omni both fall onto `intelligible` under C2, whereas Qwen2-Audio falls the opposite way, onto `not_intelligible`, so the same prompt drives different models toward opposite ends.

The three models also differ in how far they move away from this behavior. For Qwen2.5-Omni across all six of its conditions the recall on `intelligible` stays high and the recall on `not_intelligible` stays low. Its results therefore differ only in the degree of the lean and never in its direction. Even its strongest condition remains visibly

tilted toward one class. MiniCPM predicts `intelligible` more often in five of its six conditions and the single exception is C3, where the two recalls come close together and the lean disappears. Qwen2-Audio is the most mobile of the three, ranging from a near total lean onto `not_intelligible` under C2 to the most balanced result in the table under C4, where its two per-class recalls sit almost on top of one another. Only a small number of the eighteen model-condition combinations show an acceptable level of balance. They do not appear under one particular prompt, but are spread across different models and conditions. The results show how the predictions are distributed; the per-word patterns behind them, and the possible reasons for those patterns, are discussed later in the error analysis.

4.2.2 Output Parsing Success Rates

A separate practical question is how closely each model followed the requested output format. As described in the methodology, parsing was kept as a stage separate from inference: the raw output of each model call was preserved and then passed through the parser's fixed sequence of recovery steps. Table 4.2 reports, for every model and condition, the share of the 2,001 development samples whose response was returned in a strictly schema-conforming form, together with the parsing route that dominated for that model.

Table 4.2: Rate of strictly schema-conforming output on the development set for each model and prompt condition, with the dominant parsing route. Schema conformance means the response directly satisfied the requested JSON format and value constraints; responses failing this test could still contribute to the evaluation if a usable label was recovered at a later parsing stage. The final column reports the parsing stage through which most *non-conforming* responses were recovered.

Model	C1	C2	C3	C4	C5	C6	Dominant recovery route
MiniCPM-o-2.6	1.000	1.000	0.999	1.000	1.000	1.000	Strict JSON
Qwen2.5-Omni-7B	0.927	0.974	0.998	1.000	0.998	0.951	Strict JSON
Qwen2-Audio-7B-Instruct	0.888	0.252	0.875	0.999	0.604	0.993	Python-literal fallback

Besides the conformance rates, the models also differed clearly in how their outputs were recovered, which reflects how consistently each one followed the requested format. MiniCPM is highly consistent, returning well-formed JSON directly in almost every sample of every condition. Qwen2.5-Omni shows a similar pattern, with consistently high conformance and only occasional need for field-level recovery (extracting the individual required fields from a response that is not valid JSON as a whole). Qwen2-Audio behaves much less consistently than the other two models. It often fails to produce strict JSON and instead tends to return Python-style dictionaries with single quotation marks, so its output is frequently recovered through the Python-literal stage of the parser, and in some cases through a final regular-expression fallback.

Two levels of output validity need to be kept apart. Table 4.2 reports strict schema conformance: whether the response was returned as a valid JSON object with the requested fields and values. Failing that test did not necessarily make a prediction unusable, because the Python-literal and regular-expression stages could still recover

a label. Qwen2-Audio under condition C2 is the clearest case: only 504 of its 2,001 responses were strictly schema-conforming, giving the rate of 0.252 in the table, yet a usable binary label was obtained from all 2,001, so the metrics for that cell use the full development-set support. Recovering a label says nothing about whether that label is correct; these rates describe format compliance only.

A prediction was excluded from the evaluation only when no usable `intelligible` or `not_intelligible` label could be obtained at all. This was rare: 78 of the 36,018 development-set predictions (0.22%) were excluded, 20 of them from Qwen2.5-Omni (at most nine in any single condition) and 58 from Qwen2-Audio (50 of which fall under C1). Support therefore ranges from 1,951 to 2,001 across the eighteen model-condition combinations. MiniCPM-o-2.6 retained all 2,001 predictions in every condition, including the selected C6 configuration.

For Qwen2-Audio, strict schema conformance also varies sharply across conditions: it is lowest under C2 (25%) and C5 (60%), the two phonetic-checklist conditions, and highest under C4 (100%) and C6 (99%). This is a descriptive association; the design does not isolate which prompt component produced it.

4.3 Test-Set Performance

On the basis of the macro-$F_{0.5}$ scores in Table 4.1, where MiniCPM-o-2.6 under condition C6 reached the highest performance among all model-condition pairings, that configuration was then evaluated once on the test set of 8,196 samples. Since the test set played no part in selecting the model, the prompt, or any threshold, the results reported here provide an independent estimate of how well the selected system generalizes to unseen recordings. It is important to note, however, that because the dataset split was performed at the recording level rather than the speaker level (Kuan, 2025), recordings from the same learner may appear in both splits. The evaluation therefore measures generalization to new recording instances rather than necessarily to entirely unseen speakers.

Table 4.3 reports the outcome, with precision, recall, and $F_{0.5}$ given for each class separately, and the macro-average and overall accuracy beneath them.

Table 4.3: Test-set performance of MiniCPM-o-2.6 under condition C6.

Class	Precision	Recall	$F_{0.5}$	Accuracy
intelligible	0.613	0.792	0.642	–
not_intelligible	0.574	0.359	0.512	–
Macro average	0.593	0.575	0.577	–
Overall	–	–	–	0.602

The system does noticeably better on the `intelligible` class than on `not_intelligible`, and the gap between the two shows up most clearly in recall: it recovers a large majority of the intelligible words but only about a third of the genuine problems. In other words, many of the words originally labeled as `not_intelligible` are passed through as acceptable. Precision is closer between the two classes than recall is, which is the more relevant of the two here, since the cost structure set out in the methodology treats a false rejection (where intelligible words were predicted as not intelligible) as the more damaging error and $F_{0.5}$ weights precision accordingly.

Table 4.4: Confusion matrix for MiniCPM-o-2.6 under condition C6 on the test set.

Gold label \ Predicted label	not_intelligible	intelligible
not_intelligible	1,289	2,303
intelligible	958	3,646

Table 4.4 shows that the system identified fewer than half of the gold `not_intelligible` cases, with false negatives substantially outnumbering True Positives. By contrast, most gold `intelligible` cases were classified correctly. This pattern corresponds to the higher recall for `intelligible` and the lower recall for `not_intelligible` reported in Table 4.3.

4.3.1 Base Rate and Prediction Rate

Before these results are read any further, two numbers that sound almost the same but describe very different things need to be kept apart. The first is a property of the test set. Of the 8,196 gold labels, 4,604 are `intelligible`, which means that the majority class alone accounts for a little over half of the data. A trivial classifier that never listens to a single recording and simply predicts the majority label every time, Zero Rule, would therefore land at an accuracy of about 0.562, purely as a consequence of the class balance and without making any real decision at all. This figure is fixed by the data; and it is the bar that any system has to clear before its accuracy can be said to mean anything. The system evaluated here reaches an accuracy of 0.602, which sits above that bar but only by about four points. It does carry some information beyond the majority guess, though it would be misleading to present that margin as large.

The second number is a property of the model, not of the data, and this is where the two are most easily confused. Across the test set, the system assigned the `intelligible` label to roughly 72.6% of the samples, that is, to appreciably more of them than the gold distribution actually contains. What it records is a tendency in the model's own behavior, that it reached for `intelligible` more readily than the data warranted, so over-produced that label relative to the truth. The base rate tells us how the test set is composed; this prediction rate tells us how the model leaned when it labeled the test set.

4.4 Comparison with Prior ASR-Based Work

The first research question asks how a speech-enabled multimodal model compares, on this task, with the earlier ASR-based approach to word-level intelligibility on the same corpus, reported by Kuan (Kuan, 2025). As presented before (see Section 2.3), Kuan's best system is built on top of an adapted Whisper model, reaching the same binary labels through a transcription-plus-judge pipeline rather than a single end-to-end judgment. The configuration reported here is an adapted Large Whisper model combined with phonetic judging at a confidence threshold of 0.3, selected as the best configuration on the development set and then evaluated on the test set.

A comparison has to be read with one important qualification: the two systems were not evaluated on the same split. Both draw on the same MOOC corpus and report word-level intelligibility over broadly the same vocabulary; the two evaluations differ mainly in the dev/test split, which can lead to slightly different figures. The numbers below can therefore be set side by side to see the broad shape of the difference, but are best read as indicative rather than an exact contrast. Table 4.5 reports both systems with per-class precision, recall, and $F_{0.5}$.

Table 4.5: Comparison between the ASR-based baseline and the proposed system.

System	Class	Precision	Recall	$F_{0.5}$	Accuracy	Macro-$F_{0.5}$
Kuan (2025), Whisper + phonetic judge	*intelligible*	0.741	0.562	0.697	–	–
Kuan (2025), Whisper + phonetic judge	*not_intelligible*	0.581	0.726	0.605	∼0.665*	0.651
This study (MiniCPM-o-2.6, C6)	*intelligible*	0.613	0.792	0.642	–	–
This study (MiniCPM-o-2.6, C6)	*not_intelligible*	0.574	0.359	0.512	0.602	0.577

*Accuracy for Kuan (2025) is calculated from the reported confusion matrix.

The two systems show opposite class-wise patterns. Kuan's system recovers a clear majority of the real problems (recall 0.726 for `not_intelligible`) at the cost of a lower recall on `intelligible` (0.562); the system evaluated here does the reverse, accepting most intelligible words (recall 0.792) but recovering only 0.359 of the not-intelligible ones.

On the metric most relevant to the intended application, precision on the `not_intelligible` class, the two systems are essentially level: 0.581 for the earlier ASR-based system and 0.574 for the model evaluated here. The difference between them lies in recall, and because $F_{0.5}$ weights precision more heavily than recall, the resulting gap in $F_{0.5}$ stays moderate: 0.605 against 0.512 on `not_intelligible`, and 0.697 against 0.642 on `intelligible`. Across both classes the ASR-based system is therefore slightly stronger overall, while the two remain effectively level on the precision of the class that matters most here. Kuan's per-class $F_{0.5}$ values are recalculated here from his reported precision and recall.

4.5 Additional Analyses

Two further sets of results were produced during the evaluation, and although their detailed examination belongs to the error analysis, their outcomes are reported here. Both were described in the methodology: the transcription-augmented prompt variant C6+ (Section 3.5) and the manual annotation of model explanations (Section 3.6).

4.5.1 C6+ Prompt Extension

C6+ takes the best-performing condition, C6, and adds a single requirement to its output schema: a `heard_word` field, placed ahead of all other keys, in which the model gives its transcription of the spoken word. The prompt asks the model to base this transcription on the audio alone. The purpose of this variant was diagnostic rather than competitive; it was not run to find a better prompt, but to see whether requiring a transcription to appear before the label in the output, rather than the label on its own, changes the judgment the model produces, which relates to the question of acoustic grounding raised in the previous section. C6+ was run on the development set only,

across all three models. Table 4.6 reports the outcome for each of them next to its own C6 result, with per-class figures.

Table 4.6: Development-set performance under C6 and C6+ - all three models

				intelligible			*not_intelligible*		
Model	Cond	Acc	Macro-$F_{0.5}$	P	R	$F_{0.5}$	P	R	$F_{0.5}$
MiniCPM-o-2.6	C6	0.609	0.580	0.614	0.824	0.647	0.593	0.332	0.513
MiniCPM-o-2.6	C6+	0.583	0.452	0.577	0.971	0.628	0.683	0.081	0.275
Qwen2.5-Omni-7B	C6	0.598	0.550	0.599	0.870	0.638	0.593	0.246	0.462
Qwen2.5-Omni-7B	C6+	0.576	0.534	0.588	0.819	0.623	0.535	0.266	0.445
Qwen2-Audio-7B-Instruct	C6	0.535	0.523	0.584	0.614	0.589	0.464	0.432	0.457
Qwen2-Audio-7B-Instruct	C6+	0.592	0.566	0.608	0.776	0.636	0.550	0.354	0.495

For MiniCPM, requiring the model to write down what it heard before judging caused its recall on the `not_intelligible` class to fall sharply, from 0.332 to 0.081, with the number of problems it flagged collapsing from 489 positive predictions under C6 to only 104 under C6+ (of which 71 were correct and 33 false), against 802 genuine problems missed. The two classes have to be read together to see what happened: as recall for the `not_intelligible` class decreased, recall for the `intelligible` class increased from 0.824 to 0.971. This indicates that the model assigned the `intelligible` label to nearly all samples and rarely identified cases as `not_intelligible`. Its apparent increase in precision on the `not_intelligible` class, from 0.593 to 0.683, should therefore be interpreted cautiously. This increase should not be interpreted as evidence of improved separation between the two classes. It occurred alongside a substantial reduction in positive predictions, among which a larger proportion were correct.

The two Qwen models did not show the same pattern. For Qwen2.5-Omni, the transcription requirement left performance on both classes broadly unchanged, with recall for `not_intelligible` increasing slightly. For Qwen2-Audio, the same requirement was associated with a modest increase in accuracy and higher $F_{0.5}$ scores for both classes. The transcription step therefore did not produce a uniform effect across models: it led MiniCPM to assign the `intelligible` label to nearly all samples, while the performance of the two Qwen models remained stable or improved slightly. This is reflected in the macro-F0.5 column: MiniCPM's score falls from 0.580 to 0.452 under C6+, Qwen2.5-Omni's stays roughly flat (0.550 to 0.534), and Qwen2-Audio's rises (0.523 to 0.566), consistent with the per-class patterns above. The differences across models, together with what the `heard_word` field reveals about their transcription behavior, are examined in detail in the error analysis (Chapter 5).

4.5.2 Manual Annotation of Explanations

The final results concern the explanations produced by MiniCPM on the test set. Following the procedure in Section 3.6, 300 MiniCPM-o-2.6 C6 test predictions were sampled evenly across TP, TN, FP, and FN, with 75 cases from each category. Each sample was manually annotated for feedback quality, acoustic grounding, and pedagogical safety. Table 4.7 reports the distribution for feedback quality.

Each of the 300 samples was judged not only on feedback quality but also on two further axes, acoustic grounding and pedagogical safety, and the distributions on those

Table 4.7: Feedback quality of the model's explanations across the 300 annotated test-set samples.

Feedback quality	Count	Share
Appropriate and sensible	90	30.0%
Appropriate but incomplete	24	8.0%
Incorrect but relevant	83	27.7%
Irrelevant or unsupported	103	34.3%
Total	300	100%

two are given in Tables 4.8 and 4.9.

Table 4.8: Acoustic grounding of the model's explanations across the 300 annotated samples.

Grounding of explanation	Count	Share
Clearly based on audible evidence	85	28.3%
Partly based on audible evidence	72	24.0%
Mainly text-based	63	21.0%
Not supported by audio	77	25.7%
Cannot determine	3	1.0%
Total	300	100%

Table 4.9: Pedagogical safety of the model's explanations across the 300 annotated samples.

Pedagogical safety	Count	Share
Safe to show a learner	89	29.7%
Partly safe	56	18.7%
Unsafe	155	51.7%
Total	300	100%

Taken together, the three annotation dimensions point to a consistent weakness in the model's explanations. Only 38.0% of the feedback was judged appropriate, whether fully sensible or incomplete, while 34.3% was classified as irrelevant or unsupported. The grounding results show a similar pattern: only 28.3% of the explanations were clearly based on audible evidence, whereas 46.7% were either mainly text-based or not supported by the audio. This limited grounding was also reflected in the safety ratings, with 155 of the 300 explanations judged unsafe to show a learner and only 89 judged fully safe. The results indicate that many explanations were insufficiently grounded in the recording and were therefore not suitable for direct pedagogical use.

These distributions also differed across the four confusion-matrix outcomes, and the breakdown of each axis by outcome group is given in Tables 4.10, 4.11, and 4.12, with each group containing 75 samples. As a reminder, consistent with the rest of this thesis, the outcomes are labeled with `not_intelligible` as the positive class: a True Positive

(TP) is an unintelligible word correctly flagged, a True Negative (TN) an intelligible word correctly accepted, a False Positive (FP) an intelligible word wrongly flagged as a problem, and a false negative (FN) a genuine problem wrongly let through as intelligible.

Table 4.10: Feedback quality by confusion-matrix outcome (positive class = `not_intelligible`).

Feedback quality	TP	TN	FP	FN
Appropriate and sensible	21	67	0	2
Appropriate but incomplete	17	2	2	3
Incorrect but relevant	4	0	17	62
Irrelevant or unsupported	33	6	56	8

Table 4.11: Acoustic grounding by confusion-matrix outcome (positive class = `not_intelligible`).

Grounding of explanation	TP	TN	FP	FN
Clearly based on audible evidence	20	55	0	10
Partly based on audible evidence	22	18	4	28
Mainly text-based	10	1	17	35
Not supported by audio	21	1	54	1
Cannot determine	2	0	0	1

Table 4.12: Pedagogical safety by confusion-matrix outcome (positive class = `not_intelligible`).

Pedagogical safety	TP	TN	FP	FN
Safe to show a learner	22	65	0	2
Partly safe	31	9	1	15
Unsafe	22	1	74	58

The clearest contrast in all three tables is between the TN and FP groups. For correct acceptances (TN), the explanations were usually appropriate, acoustically grounded, and safe: 67 of the 75 were rated *appropriate and sensible*, 55 were *clearly based on audible evidence*, and 65 were *safe*. The false-positive (FP) group inverts this almost completely, and it is the more revealing of the two. These are cases where the model flagged a perfectly intelligible word as a problem, and because there was no real pronunciation error for it to describe, its explanations were ungrounded: 56 of the 75 were *irrelevant or unsupported*, 54 were *not supported by the audio* at all, and 74 of the 75 were judged *unsafe*. In other words the explanations were at their worst not when the model missed a genuine problem, but when it flagged one that was not there. Forced to justify a rejection of speech that was in fact fine, the model produced feedback that was almost always fabricated and, on the pedagogical-safety axis, almost never fit to show a learner. The breakdown, together with representative examples of each category, is taken up again in the error analysis (Chapter 5).

Chapter 5

Error Analysis

The aggregate evaluation reported in Chapter 4 establishes how the selected system performs overall, but it does not show which recordings were misclassified, what explanations the model produced, whether its confidence scores were meaningful, or whether similar errors appeared across different systems and prompting conditions. This chapter examines these questions using the same predictions reported in Chapter 4.

Section 5.1 will examine variation in class-specific performance across the 53 target words. Section 5.2 will look at whether the model's self-reported confidence and uncertainty carry information about its error rate. Section 5.3 will look at the explanations the model produced and, in Section 5.3.1, will group the errors by the phonetic feature each target word is built to test. Section 5.4 will look at all three candidate systems on the development set, separating errors that every system makes from those specific to a single system. Section 5.5 will provide a deeper analysis of C6+, the diagnostic variant of the best-performing prompt, which requires the model to state the word it heard before assigning a label. Section 5.6 will read the 300 manually annotated explanations in detail, with representative examples for each annotation category. Section 5.7 will draw the recurring error patterns together and hand them to the Discussion.

Part of the analysis focuses on the test set (Sections 5.1 - 5.3.1 and 5.6), while other parts focus on the development set (Sections 5.4 - 5.5). Numbers from the two splits are not comparable and are labeled accordingly. The interpretation of why particular words and phonetic contrasts are persistently difficult is deferred to the Discussion (Chapter 6).

5.1 Word-Level Error Patterns

This section reports on the final system alone: MiniCPM-o-2.6 under condition C6, evaluated on the test set. It works from the predictions that produced the headline results of Chapter 4 and is therefore descriptive rather than a further round of model selection.

As in Chapter 4, the positive class is `not_intelligible`: a *missed problem* is a not-intelligible recording that the system accepts, and a *false rejection* (FP) an intelligible recording that the system flags. Correct acceptances are therefore True Negatives and correct rejections True Positives.

According to the gold annotations, 3,592 of the 8,196 test recordings are not intelligible. The system correctly identified only 1,289 of them, missing the remaining 2,303. It additionally flagged 958 of the 4,604 intelligible recordings as problems. Missed

problems therefore account for 70.6% of the 3,261 errors. The same asymmetry is visible in the class-specific scores: recall is 0.792 for `intelligible` against 0.359 for `not_intelligible`, with corresponding $F_{0.5}$ scores of 0.642 and 0.512.

Table 5.1 reports the per-word gold-label distribution, accuracy, Macro-$F_{0.5}$, and class-specific precision, recall and $F_{0.5}$. The corresponding raw error counts are given for completeness in Appendix Table C.1.

The gold-label distribution varies substantially across words, so accuracy is not directly comparable from one word to the next without considering class balance. For example, 144 of the 160 *bee* recordings are annotated as `intelligible` in the gold data, whereas *pull* is close to balanced, with 75 recordings annotated as `intelligible` and 79 as `not_intelligible`. A classifier that accepted every recording would therefore reach an accuracy of 0.90 on *bee* but only 0.49 on *pull*, without its behavior changing at all.

The majority-class baseline - the accuracy obtained by always predicting a word's more frequent gold label - ranges from 0.51 for the most balanced words to 0.90 for the most skewed, with a mean of 0.68 across the 53 test words. The class-specific recall and $F_{0.5}$ scores, together with Macro-$F_{0.5}$, therefore provide a more informative basis for comparing word-level behavior than accuracy alone.

Table 5.1: Per-word class distributions and classification metrics for MiniCPM-o-2.6 under condition C6 on the test set, ordered by accuracy from lowest to highest. Gold I/NI reports the number of gold `intelligible` and `not_intelligible` recordings, respectively. Macro-$F_{0.5}$ is the unweighted mean of the two class-specific $F_{0.5}$ scores.

					`intelligible`			`not_intelligible`		
Word	n	Gold I/NI	Acc.	Macro-$F^*_{0.5}$	P	R	$F_{0.5}$	P	R	$F_{0.5}$
bird	157	54/103	0.363	0.305	0.345	0.944	0.395	0.667	0.058	0.216
Paul	157	57/100	0.363	0.208	0.363	1.000	0.416	0.000	0.000	0.000
pause	154	47/107	0.364	0.372	0.220	0.426	0.243	0.571	0.336	0.501
seedy	158	61/97	0.386	0.220	0.386	1.000	0.440	0.000	0.000	0.000
paw	157	37/120	0.408	0.421	0.208	0.541	0.238	0.721	0.367	0.604
daft	161	111/50	0.447	0.463	0.683	0.369	0.584	0.307	0.620	0.341
board	152	44/108	0.454	0.486	0.321	0.795	0.365	0.791	0.315	0.607
fur	158	66/92	0.456	0.458	0.384	0.500	0.402	0.542	0.424	0.513
farther	160	67/93	0.469	0.422	0.312	0.224	0.290	0.536	0.645	0.555
putt	160	40/120	0.469	0.463	0.241	0.525	0.271	0.740	0.450	0.655
pour	155	72/83	0.471	0.459	0.456	0.722	0.492	0.512	0.253	0.425
fir	157	61/96	0.478	0.489	0.415	0.836	0.461	0.706	0.250	0.517
bear	159	75/84	0.491	0.480	0.474	0.733	0.510	0.535	0.274	0.449
bard	158	46/112	0.500	0.527	0.346	0.804	0.390	0.824	0.375	0.665
bout	159	70/89	0.503	0.468	0.469	0.971	0.523	0.857	0.135	0.414
fair	156	80/76	0.513	0.284	0.513	1.000	0.568	0.000	0.000	0.000
pull	154	75/79	0.513	0.284	0.000	0.000	0.000	0.513	1.000	0.568
tide	154	62/92	0.519	0.530	0.439	0.694	0.474	0.661	0.402	0.585
pool	159	75/84	0.535	0.476	0.520	0.173	0.371	0.537	0.857	0.581
pit	161	113/48	0.540	0.486	0.693	0.619	0.677	0.283	0.354	0.295
tied	155	58/97	0.548	0.560	0.438	0.724	0.475	0.729	0.443	0.646
meat	153	93/60	0.549	0.482	0.603	0.753	0.628	0.378	0.233	0.337
wait	87	69/18	0.552	0.424	0.750	0.652	0.728	0.111	0.167	0.119
fern	156	92/64	0.571	0.555	0.634	0.641	0.636	0.476	0.469	0.475
doll	158	94/64	0.576	0.525	0.613	0.777	0.640	0.462	0.281	0.409
boat	161	101/60	0.578	0.500	0.632	0.782	0.657	0.389	0.233	0.343

continued on next page

Table 5.1 (continued)

					`intelligible`			`not_intelligible`		
Word	n	Gold I/NI	Acc.	Macro-$F^*_{0.5}$	P	R	$F_{0.5}$	P	R	$F_{0.5}$
pole	157	88/69	0.580	0.533	0.587	0.841	0.625	0.548	0.246	0.440
pore	156	84/72	0.603	0.606	0.677	0.500	0.633	0.553	0.722	0.580
father	161	93/68	0.609	0.539	0.607	0.914	0.651	0.619	0.191	0.428
pat	160	108/52	0.613	0.487	0.674	0.824	0.700	0.321	0.173	0.274
boot	161	103/58	0.627	0.461	0.644	0.932	0.687	0.417	0.086	0.236
cot	157	102/55	0.631	0.582	0.700	0.755	0.710	0.468	0.400	0.453
pot	160	112/48	0.644	0.559	0.731	0.777	0.740	0.390	0.333	0.377
beer	160	88/72	0.650	0.625	0.611	1.000	0.663	1.000	0.222	0.588
knows	155	90/65	0.658	0.605	0.629	1.000	0.680	1.000	0.185	0.531
poor	155	50/105	0.664	0.359	0.000	0.000	0.000	0.673	0.981	0.718
half	158	109/49	0.690	0.368	0.690	1.000	0.735	0.000	0.000	0.000
put	160	112/48	0.694	0.493	0.709	0.955	0.747	0.444	0.083	0.238
plate	75	53/22	0.707	0.375	0.707	1.000	0.751	0.000	0.000	0.000
city	156	112/44	0.718	0.380	0.718	1.000	0.761	0.000	0.000	0.000
dance	158	124/34	0.753	0.558	0.801	0.911	0.821	0.353	0.176	0.294
bee	160	144/16	0.756	0.529	0.913	0.806	0.890	0.152	0.312	0.169
meet	154	118/36	0.766	0.402	0.766	1.000	0.804	0.000	0.000	0.000
pet	161	131/30	0.770	0.589	0.841	0.885	0.849	0.348	0.267	0.328
caught	159	36/123	0.774	0.405	0.000	0.000	0.000	0.774	1.000	0.810
mate	152	124/28	0.783	0.464	0.814	0.952	0.838	0.143	0.036	0.089
buy	160	126/34	0.787	0.411	0.787	1.000	0.822	0.000	0.000	0.000
paws	152	32/120	0.789	0.412	0.000	0.000	0.000	0.789	1.000	0.824
nose	157	125/32	0.796	0.415	0.796	1.000	0.830	0.000	0.000	0.000
bay	162	127/35	0.802	0.576	0.799	1.000	0.832	1.000	0.086	0.319
weight	155	125/30	0.806	0.419	0.806	1.000	0.839	0.000	0.000	0.000
hat	158	133/25	0.810	0.478	0.841	0.955	0.862	0.143	0.040	0.094
boy	161	135/26	0.839	0.433	0.839	1.000	0.866	0.000	0.000	0.000
Total	**8196**	**4604/3592**	**0.602**	**0.577**	**0.613**	**0.792**	**0.642**	**0.574**	**0.359**	**0.512**

* For the fourteen words the model never assigns to a given label (e.g. *Paul*, *seedy*, *pull*), that label's precision is undefined (0/0); following the scikit-learn convention it is reported as 0.000. Recall for the same label is also 0, so its $F_{0.5}$ is 0.000 and Macro-$F_{0.5}$ is correspondingly lower. A fifteenth word, *poor*, reaches a precision of 0.000 for `intelligible` for a different reason: the model returned `intelligible` for only two of its 155 recordings, and both were incorrect.

Fourteen of the 53 target words received a single predicted label across all recordings. Eleven words were always classified as `intelligible`: *Paul*, *seedy*, *fair*, *half*, *city*, *meet*, *buy*, *nose*, *weight*, *boy*, and *plate*. Three words showed the opposite pattern and were always classified as `not_intelligible`: *pull*, *caught*, and *paws*. For each of these words, precision and recall for the unused label are 0.000, which drives the corresponding $F_{0.5}$ to zero and Macro-$F_{0.5}$ down with it, in some cases alongside a high accuracy.

Precision for the `not_intelligible` class deserves particular attention, because in a feedback setting it expresses how often a recording the system flags really does contain a pronunciation problem. In aggregate it is 0.574, but at the word level it is 0.000 for the eleven words the system never rejects, and these words are spread across the entire accuracy range, from 0.363 for *Paul* to 0.839 for *boy*. *Boy* has the highest accuracy of any word in the test set, yet none of its 26 not-intelligible recordings was identified; *Paul* behaves identically, and the two differ only in how many of their recordings are annotated `not_intelligible` in the gold data (26 of 161 for *boy*, 100 of 157 for *Paul*). The gap of almost 48 accuracy points between them therefore reflects their gold-label distributions rather than any difference in what the system did.

Perfect precision is equally uninformative on its own. *Beer*, *knows*, and *bay* reach a precision of 1.000 for `not_intelligible`, so every recording they reject is indeed not intelligible, but their recall for that class is 0.222, 0.185, and 0.086 respectively, giving $F_{0.5}$ scores of 0.588, 0.531, and 0.319. A precision of 1.000 here reflects a system that rejects very few recordings, not one that detects problems reliably. On the metrics that matter most for this task, the system is therefore substantially weaker than its overall accuracy of 0.602 suggests: precision for `not_intelligible` is 0.574 and its $F_{0.5}$ is 0.512, and for eleven of the 53 words no not-intelligible recording is identified at all.

For these fourteen words, the predicted label appears to be driven more by the target word than by the individual recording, suggesting a possible word-level prior. The analyses in the following sections examine this possibility further. The pattern on its own does not establish that the system ignored the acoustic input, and the per-word feature tags used later in this chapter describe pronunciation contrasts only in broad terms rather than analyzing individual recordings acoustically.

5.2 Self-Reported Confidence and Uncertainty

Alongside each label, the model was asked to report how confident it was and whether it was uncertain (Section 3.4.4). For educational applications, these self-reports are useful only if they carry information about the reliability of a prediction, which means their values should differ systematically between correct and incorrect predictions. This section evaluates whether they do, using the frozen MiniCPM-o-2.6 predictions on the test set.

The confidence field takes one of three values: `low`, `medium`, or `high`. Across the 8,196 test predictions the model never returned `high`. It returned `medium` for 8,040 predictions (98.1%) and `low` for the remaining 156 (1.9%). The same pattern holds within every target word: the share of predictions returned with `medium` confidence ranges from 92.3% for *pore* to 100%, with a median of 98.7%, and 19 of the 53 words receive `medium` confidence for every one of their recordings. Because the reported confidence level is almost always the same, it provides little information for distinguishing correct predictions from incorrect ones.

The uncertainty field is binary (`yes`/`no`) and varies considerably more. The model flagged uncertainty in 1,898 of the 8,196 predictions (23.2%). That variation, however, follows the label the model assigned rather than whether the label was correct. Across the 53 target words, the proportion of predictions marked as uncertain is only weakly correlated with the word-level error rate (Pearson $r = 0.30$), but strongly correlated with the proportion of recordings predicted `not_intelligible` (Pearson $r = 0.91$). Table 5.2 shows this by placing the *Rejected* and *Uncertain* columns side by side: the two move together across the whole range, from 0.0% and 0.0% for *Paul* to 98.7% and 91.6% for *poor*, while neither tracks accuracy. *Paul* and *seedy* are among the least accurate words in the test set, with error rates of 63.7% and 61.4%, yet the model never flags uncertainty on either.

Most of the model's errors are of a single kind. As Section 5.1 showed, its most common mistake by far is the missed problem - a recording that is not intelligible, but that the model accepts as intelligible - and missed problems make up roughly seven in ten of all its errors. This pattern is illustrated most clearly by the eleven words the model accepts across the board, giving every one of their recordings the label `intelligible` no matter how it was actually pronounced. If the uncertainty flag did what it is meant

Table 5.2: Word-level accuracy, rejection rate (predicted `not_intelligible`), uncertainty rate, and medium-confidence rate for selected words. The *Medium conf.* column reports the percentage of predictions for each target word returned with `medium` confidence; the remaining predictions were returned with `low` confidence, as the model never returned `high`. Reading the *poor* row: 98.7% of its recordings were predicted `not_intelligible`, 91.6% were flagged as uncertain, and 92.9% were returned with `medium` confidence.

Word	Acc.	Rejected (%)	Uncertain (%)	Medium conf. (%)
Highest-error words				
bird	0.363	5.7	5.1	99.4
Paul	0.363	0.0	0.0	100.0
seedy	0.386	0.0	0.0	100.0
Most-flagged words				
poor	0.664	98.7	91.6	92.9
farther	0.469	70.0	68.1	93.1
caught	0.774	100.0	64.8	100.0
pool	0.535	84.3	56.6	100.0

Note. Across all 53 target words, the uncertainty rate is strongly correlated with the rejection rate ($r = 0.91$), but only weakly correlated with the error rate ($r = 0.30$).

to, this is exactly where we would expect it to be raised. Instead, each of these words has, on average, only 0.9% of its instances labeled as uncertain by the model (Appendix Table D.1 gives the per-word figures). The uncertainty signal is largely absent where the model is most often wrong.

The two hardest words in the test set, *bird* and *Paul*, illustrate the same pattern. The model is wrong on roughly 64% of their recordings, and yet it raises the uncertainty flag on only 5% of *bird*'s recordings and never for *Paul*'s. The reverse holds for the three words the model rejects across the board, classifying every recording as `not_intelligible`: there it flags uncertainty on nearly half of the recordings - 48% on average. Taken together, the flag is raised far more often because the model has decided to reject a recording than because the prediction is actually wrong. The uncertainty flag is therefore largely absent precisely where it would have been most informative: on the missed problems, the one place where a warning would have told the learner something the predicted label did not already convey.

A useful confidence or uncertainty signal would have to be higher on the predictions the model gets wrong than on the ones it gets right, and neither signal here does that: the confidence level is almost always `medium`, so it separates nothing, and the uncertainty flag varies with the label rather than with its correctness. This is the kind of mismatch between stated certainty and empirical correctness that the calibration literature (Section 2.4.3) describes as a calibration gap. Because the confidence field used here is categorical and almost constant, this analysis should not be read as a formal probability-calibration evaluation, but the practical consequence is the same: the model's stated confidence sounds informative and has little to do with how often it is actually correct. The third self-report, the irrelevant-answer-risk flag, is raised on 55 of the 8,196 predictions (0.7%), so it is not examined separately.

5.3 The Model's Explanations

Besides predicting an intelligibility label, the model was required to provide a short explanation together with the audio evidence it claimed to rely on (Section 3.4.4). The distinction between *plausible* and *faithful* explanations was introduced in Section 2.4.2, and the aggregate outcomes of the manual annotation were reported in Chapter 4. This section therefore examines not whether the labels are correct, which Section 5.1 addressed, but whether the explanations are supported by the acoustic information contained in the recording.

Those results are worth restating briefly. The 300 predictions were rated on three axes: whether the feedback was appropriate, whether the explanation was grounded in audible evidence, and whether it was safe to show to a learner. Only 85 of the 300 explanations (28.3%) were judged clearly based on audible evidence, and 155 (51.7%) were judged unsafe to show to a learner. Quality also depended heavily on the outcome. Explanations were best when the model correctly accepted an intelligible word (the True Negatives: 67 of 75 appropriate, 55 clearly grounded, 65 safe) and worst when it incorrectly flagged an intelligible word as a problem (the False Positives: 54 of 75 were not grounded in the audio at all, and 74 of 75 were unsafe). The least trustworthy explanations therefore arose not from missed pronunciation problems, but from unsupported error diagnoses that lacked evidence in the acoustic signal.

To check whether the same tendency holds across the whole test set, and to examine the content of the model's explanations, the `short_reason` and `audio_evidence` fields of all 8,196 predictions were concatenated and matched automatically against seven keyword classes. Matching was case-insensitive and based on simple substring search. The exact match strings used for each class are listed in Appendix Table E.1. A single prediction may match more than one class, so the percentages do not sum to 100. Table 5.3 reports the percentage of predictions matching each class, both overall and broken down by the four confusion-matrix outcomes used in Chapter 4: True Negatives (TN), false negatives, True Positives (TP), and False Positives (FP). Two patterns emerge, and both are consistent with the manual annotation.

First, two generic keyword classes account for the largest share of the matched explanation content, and each is tied to one of the two labels rather than to the audio. When the model accepts a word as intelligible, the combined `short_reason` and `audio_evidence` text most often contains a phrase indicating that no major issue was heard: 29.0% of correct acceptances against 27.8% of missed problems. When the model rejects a word as not intelligible, the combined text most often contains a phrase indicating that the audio was unclear: 63.1% of correct rejections against 68.9% of false rejections. The rates barely differ between correct and incorrect predictions, which indicates that these phrases primarily mirror the predicted label rather than the acoustic characteristics of the recording.

Second, specific phonetic references are uncommon. Across the concatenated `short_reason` and `audio_evidence` fields, none of the 8,196 predictions matched any of the four aspiration strings, and only one prediction matched the rhoticity class. Vowel and consonant-related strings occurred in a small minority of predictions, reaching at most about 10% in any confusion-matrix cell. These broader rates should be interpreted cautiously because the substring matcher may also capture non-phonetic uses of terms such as `quality`, `short`, and `long`. Nevertheless, the near-absence of aspiration- and rhoticity-related language is unlikely to result from an overly restrictive matcher. Al-

Table 5.3: Distribution of keyword classes in the model's `short_reason` and `audio_evidence` fields across all test predictions and by confusion-matrix outcome (MiniCPM-o-2.6, condition C6). Values are percentages; a single prediction may contribute to more than one keyword class. For example, the *unclear audio* row shows that 63.1% of the model's correct rejections (TP) and 68.9% of its false rejections contained a phrase matching this class. A value of 0.0 indicates that no prediction in that cell matched the class, except for the rhoticity row, where a single prediction across the whole test set matched (0.01%).

		Model accepts (`intelligible`)		Model rejects (`not_intelligible`)	
Reason class	Overall	correct (TN)	wrong (FN)	correct (TP)	wrong (FP)
no major issue	20.8	29.0	27.8	0.3	0.2
unclear audio	28.6	15.2	13.6	63.1	68.9
different word heard	3.4	2.7	2.9	5.0	5.1
consonant feature	6.0	7.2	7.2	2.2	3.9
vowel feature	8.6	8.2	8.1	9.2	10.3
aspiration	0.0	0.0	0.0	0.0	0.0
rhoticity (/r/)	0.0	0.0	0.0	0.0	0.0
n	8,196	3,646	2,303	1,289	958

though the target words were selected to contrast features of this kind (Chapter 3), the generated explanations generally restate the decision in coarse terms, such as whether the word or audio was clear, rather than offering a fine-grained diagnosis of the sounds the learner produced.

Taken together with the confidence and uncertainty fields of Section 5.2, this gives a consistent picture of the model's self-reports. Its confidence level, its uncertainty flag, and its reason all follow the label it has chosen rather than the correctness of that label, and they rarely identify the specific acoustic or phonetic detail that would support a pronunciation judgment. A correct classification from this system therefore cannot be taken, on its own, as evidence that the decision was grounded in the recording.

5.3.1 Error Rates by Phonetic Feature

The per-word analysis so far treats every word separately. A complementary question is whether the errors concentrate on particular *phonetic dimensions* - for instance, whether a model that copes with vowel quality nonetheless struggles with initial-stop aspiration. To test this, each of the 53 test words was tagged with the phonetic properties that its identification depends on (initial-stop aspiration for the /p, t, k/ - /b, d, g/ contrasts, a long vowel or diphthong, a final consonant, and whether the word is one member of a minimal pair), and the per-word results were pooled within each group. Table 5.4 reports, for each feature group, the mean per-word $F_{0.5}$ for the `intelligible` class together with the false-rejection and missed-problem rates, which describe the direction of the associated errors.

Initial-stop aspiration shows the largest difference between the feature groups. Words tagged as involving an initial-stop aspiration contrast are handled worse than the rest; their mean $F_{0.5}$ for the intelligible class is 0.47 against 0.63 for words without that tag, and the gap is driven by over-rejection. The model incorrectly rejects 33% of the intelligible recordings in this group, more than double the 15% rate for other words. The

Table 5.4: Test-set performance grouped by phonetic feature (MiniCPM-o-2.6, C6). $F_{0.5}$*(I)* is the mean per-word $F_{0.5}$ for the `intelligible` class; *False-rej. rate* is the share of intelligible recordings that are rejected (the false-positive rate, since `not_intelligible` is the positive class); *Missed-prob. rate* is the share of not-intelligible recordings that are accepted (the false-negative rate, equivalently one minus recall for that class). Groups overlap, because a word may carry several features. Consonant clusters are omitted, as only two words carry that tag.

Feature	Present?	Words	Samples	$F_{0.5}$(I)	False-rej. rate	Missed-prob. rate
Initial-stop aspiration	yes	20	3057	**0.47**	**0.33**	0.52
	no	33	5139	0.63	0.15	0.73
Long vowel / diphthong	yes	30	4649	0.54	0.22	0.64
	no	23	3547	0.61	0.20	0.65
Final consonant	yes	44	6873	0.55	0.22	0.64
	no	9	1323	0.68	0.17	0.67
Minimal-pair member	yes	39	6084	0.56	0.21	0.62
	no	14	2112	0.59	0.21	0.69

missed-problem rate moves in the opposite direction for the same group (0.52 against 0.73), which is what a shift toward rejection on these words would produce. Among the feature groups examined, aspiration-tagged words therefore show both the lowest mean $F_{0.5}$ and the highest false-rejection rate. It is also the group whose feature the model never names: as Table 5.3 showed, aspiration does not appear in a single one of the 8,196 reasons. Final-consonant tagging shows a weaker version of the same pattern ($F_{0.5}$ 0.55 vs 0.68), while a long vowel or diphthong makes only a small difference (0.54 vs 0.61).

Two considerations should be noted when interpreting these results. First, the feature groups are not disjoint; most words carry several tags at once (for example, an aspiration-tagged word is often also a minimal-pair member), so the columns describe overlapping populations rather than independent factors, and the aspiration effect cannot be cleanly isolated from everything the aspiration words also happen to share. Second, the group sizes are small at the word level (20 versus 33 words for aspiration), so the numbers are indicative. The direction of the finding is nonetheless consistent across the two error rates and the $F_{0.5}$ score.

5.4 Shared and Model-Specific Failure Patterns

The analyses in Sections 5.1 to 5.3 focus on a single system: MiniCPM-o-2.6 under condition C6, evaluated on the frozen test set. A separate question is how much of the difficulty found there is specific to that system and how much would be present regardless of which system a practitioner chose. To address this, all three candidate systems evaluated in Chapter 4 were run under condition C6 on the development set, and their predictions were compared at the level of individual recordings. For each of the 1,994 recordings for which all three systems returned a valid prediction, the number of systems that made an error was counted. Because this comparison operates on individual recordings, it separates two distinct sources of difficulty: recordings that all systems find hard, and recordings where one or two systems fail while the others do

not.

The 1,994 recordings divide into four groups. In 726 cases (36.4%) all three systems are correct; in 426 cases (21.4%) all three systems are wrong; in 387 cases (19.4%) exactly two systems are wrong; and in 455 cases (22.8%) exactly one system is wrong. Together, the last two groups - 842 recordings where the systems disagree - show that model choice is not irrelevant: which system is used affects the outcome on roughly 42% of the recordings. Nonetheless, the 426 recordings that all three systems get wrong simultaneously are the most diagnostic subset, because an error that occurs regardless of the system is a stronger candidate for reflecting something about the recording or the word itself than an error that only one system makes.

Of those 426 shared failures, 365 (85.7%) are missed problems - recordings the annotators mark as not intelligible but that all three systems accept - and only 61 (14.3%) are false rejections. When all three systems fail on the same recording, the failure therefore takes the form of over-acceptance in roughly six cases out of seven, which is the same direction as MiniCPM's dominant error on the test set (Section 5.1).

Table 5.5: The ten target words with the largest number of shared failures on the development set (condition C6, three systems). *Any-system* is the proportion of a word's recordings misclassified by at least one system; *all-systems* the proportion misclassified by all three; *shared* is the second as a percentage of the first. Words for which the two rates are nearly equal have failures that are almost entirely shared across the three systems. Reading the *fur* row: 76.9% of its recordings are misclassified by at least one system, 38.5% by all three, so half of its errors are shared.

Word	n	Any-system	All-systems	Shared (%)
Failures almost entirely shared				
board	38	0.711	0.711	100
bird	38	0.658	0.658	100
pull	37	0.486	0.486	100
fair	38	0.474	0.474	100
father	40	0.425	0.400	94
Failures with a system-specific component				
beer	39	0.462	0.385	83
bear	38	0.553	0.368	67
fur	39	0.769	0.385	50
bard	38	0.737	0.368	50
seedy	39	0.744	0.359	48

This table lists the ten words on which the three systems fail together most often, and distinguishes two patterns. For *board*, *bird*, *pull*, *fair* and (nearly) *father*, the any-system and all-systems error rates are almost equal, meaning that every time any system makes an error on a recording of these words, all three systems make the same error. The shared column reaches 100% for the first four: there is not a single recording of *board* or *bird* in this development set that one system gets wrong while the others get right. This pattern suggests that these failures are not specific to one model. They may reflect properties of the recordings or target words, but shared prompting assumptions or common model priors may also contribute. For *fur*, *bard* and *seedy*, the any-system rate is roughly twice the all-systems rate, meaning that about half the errors on these words are made by only one or two systems while the others succeed. These words carry

both a shared difficulty and a system-specific component.

The word-level patterns from the development set align with the test-set findings in Section 5.1, though not uniformly. *Board* and *bird* sit near the bottom of the test-set accuracy ranking and also head the development-set shared-failure ranking, with every error on them made by all three systems at once. *Paul* and *pause* are likewise near the bottom on the test set, and on the development set at least one system is wrong on 89.7% and 97.3% of their recordings respectively; their failures, however, are only partly shared, with 25.6% and 21.6% of recordings missed by all three. The convergence across two held-out splits and three different systems therefore indicates that these words are consistently difficult under the present task and evaluation setup, while leaving open how much of that difficulty is common to every system. What makes them difficult - whether the difficulty lies in the phonetic features being tested, in recording conditions, in the learner population, or in some combination - is a question I return to in Chapter 6.

5.5 Requiring the Model to State What It Heard

Section 5.3 showed that when the model rejects a recording, its reason almost always says something like *the audio is unclear* or *the acoustic evidence does not support identification of the target word.* It rarely names a specific wrong word it heard instead. This leaves open whether the model identifies a specific alternative word but does not report it, or whether its rejection explanations provide little evidence of any such identification.

As described in Section 3.4 and reproduced in full in Appendix A, a diagnostic variant, C6+, was evaluated on the development set. C6+ differs from C6 only in requiring the model to populate the `heard_word` field before assigning a label. If the model's rejection decisions are grounded in the identification of a specific alternative word, requiring that word to be reported should have little effect on its behavior. If the rejection decisions are not based on such an identification, the additional requirement would be expected to alter the model's rejection behavior.

As reported in Chapter 4 (Table 4.6), the three models responded differently to the introduction of the `heard_word` field. For MiniCPM-o-2.6, $F_{0.5}$ for the `not_intelligible` class fell from 0.513 to 0.275, its false rejections (FP) dropped from 199 to 33 and its missed problems rose from 583 to 802. Qwen2.5-Omni changed only marginally (0.462 to 0.445), while Qwen2-Audio improved (0.457 to 0.495). These contrasting responses provide the basis for the error analysis presented below.

What changed at the recording level

Aggregate metrics show the direction of the change but not how individual predictions change between C6 and C6+. Table 5.6 therefore reports instance-level prediction transitions. It records how many recordings changed from `not_intelligible` to `intelligible` (NI→I) and from `intelligible` to `not_intelligible` (I→NI), together with the number of transitions that corrected an earlier mistake (*rescued*) or introduced a new error (*regressed*).

One notable feature of Table 5.6 is the imbalance between the two transition directions for MiniCPM. Most label changes involve predictions moving from `not_intelligible` to `intelligible`, whereas only a handful move in the opposite direction. Under C6 the model rejected 489 of the 2,001 development recordings; 395 of those rejections, or 81%, are retracted under C6+, and only 10 new ones are created. In 1,989 of the 2,001

Table 5.6: Instance-level prediction transitions between C6 and C6+ on the development set (N = 2,001 recordings per model). *Rescued* and *regressed* count recordings whose correctness changed; for the two Qwen models these totals exceed the two transition columns because a small number of recordings also moved between valid and invalid output (25 valid-to-invalid and 3 invalid-to-valid for Qwen2.5-Omni, 4 invalid-to-valid for Qwen2-Audio). Reading the MiniCPM row: 395 recordings moved from `not_intelligible` to `intelligible` and 10 in the opposite direction; of those 405 changes, 176 corrected an earlier error and 229 introduced a new one.

Model	NI→I	I→NI	Rescued	Regressed
MiniCPM-o-2.6	395 (19.7%)	10 (0.5%)	176 (8.8%)	229 (11.4%)
Qwen2.5-Omni	155 (7.8%)	226 (11.3%)	173 (8.7%)	228 (11.4%)
Qwen2-Audio	459 (23.0%)	208 (10.4%)	393 (19.6%)	277 (13.8%)

recordings (99.4%), MiniCPM wrote the target word itself in the `heard_word` field. Across all 405 label changes, 176 were genuine corrections and 229 were new errors, a net loss of 53 recordings, which is exactly the accuracy drop the condition produces (0.609 to 0.583). The decline in $F_{0.5}$ for the `not_intelligible` class is driven by recall rather than precision: precision for that class rises from 0.593 to 0.683, while recall falls from 0.332 to 0.081. The rejections that survive under C6+ are therefore more precise, but far fewer; the dominant effect of the condition is a sharp reduction in the model's willingness to reject at all.

For Qwen2.5-Omni the shifts in the two directions are of similar size (155 NI→I, 226 I→NI), meaning the model both retracts old rejections and creates new ones under C6+. The net correctness outcome is again slightly negative (173 rescued against 228 regressed), but the structure is different from MiniCPM's.

Qwen2-Audio shows a different pattern. Where MiniCPM generated almost no new flags (10 I→NI), Qwen2-Audio generates 208 new flags alongside 459 retractions. This is the most total label movement of any model (667 recordings change label, against 405 for MiniCPM and 381 for Qwen2.5-Omni), and the most retractions of any model in the NI→I direction. Crucially, its correctness balance is positive: 393 recordings rescued against 277 regressed. For this model, committing to `heard_word` first helps more than it hurts. One interpretation consistent with this contrast is that Qwen2-Audio's C6 rejections were already tied, at least in part, to a perceived word identity, so that stating that word organizes the decision rather than disrupting it, whereas MiniCPM's C6 rejections may not have rested on such an identification, so that requiring a transcription leaves little basis for the rejection to stand. The transition analysis on its own cannot separate this reading from other model-specific responses to the schema change, and it is qualified further below.

Which words were most affected

The aggregate decline is not spread evenly across the vocabulary. Table 5.7 lists, for MiniCPM-o-2.6 on the development set, the fifteen target words whose $F_{0.5}$ for the `not_intelligible` class dropped the most between C6 and C6+, together with each word's gold `not_intelligible` rate and its C6 recall and $F_{0.5}$ for the same class. The C6+ $F_{0.5}$ is 0.000 for every word listed because, under C6+, MiniCPM predicted `intelligible` for all recordings of each of them.

Table 5.7: Per-word effect of adding `heard_word` (C6→C6+) for MiniCPM-o-2.6 on the development set, for the fifteen words with the largest drop in $F_{0.5}$ for the `not_intelligible` class. Gold `not_intelligible` is the share of that word's recordings annotated `not_intelligible`.

	Gold `not_intelligible`	C6		C6+	
Word	(%)	Recall	$F_{0.5}$	$F_{0.5}$	$\Delta F_{0.5}$
bard	71.8	0.464	0.774	0.000	−0.774
farther	59.0	0.783	0.709	0.000	−0.709
poor	68.4	0.808	0.700	0.000	−0.700
tied	63.2	0.583	0.648	0.000	−0.648
putt	76.9	0.467	0.636	0.000	−0.636
paw	76.3	0.414	0.619	0.000	−0.619
bear	53.8	0.286	0.612	0.000	−0.612
pause	70.3	0.423	0.585	0.000	−0.585
doll	39.5	0.200	0.556	0.000	−0.556
knows	42.1	0.188	0.536	0.000	−0.536
tide	60.5	0.348	0.533	0.000	−0.533
pore	46.0	0.588	0.515	0.000	−0.515
pour	52.6	0.350	0.515	0.000	−0.515
meat	37.8	0.643	0.500	0.000	−0.500
fur	59.0	0.348	0.482	0.000	−0.482

The table reveals two patterns. First, these are not words on which rejection was optional: eleven of the fifteen have a majority of their recordings annotated `not_intelligible` in the gold data, reaching 76.9% for *putt* and 76.3% for *paw*, so accepting every recording of them means missing most of the problems they contain. Second, they are the words the model had been detecting most actively under C6. *Poor* (recall 0.808), *farther* (0.783) and *meat* (0.643) are among its highest `not_intelligible` recall values on the development set, and *bard* was being rejected with a C6 precision of 0.929. The rejections removed under C6+ were therefore largely the ones the model had been getting right.

What the examples show

Looking at individual recordings where C6+ reversed a MiniCPM rejection reveals a consistent pattern. Of the 395 NI→I retractions, 171 fell on recordings whose gold label is `intelligible` and are genuine rescues, while the remaining 224 fell on recordings whose gold label is `not_intelligible` and count as regressions.

On the rescued cases, the shift is visible in the text itself. Under C6, the `audio_evidence` field says things like *the audio contains indistinct speech with no clear pronunciation* or *the acoustic signal is unclear, making it difficult to identify the spoken word.* Under C6+, the same recording gets `heard_word` = the target word (for example, *daft*, *fir*, *pore*, *tide*), and the `audio_evidence` field reverses: *the acoustic signal distinctly conveys the phonemes of the target word*, or *clear and distinct articulation.*

Because the audio recording remains identical across C6 and C6+, the observed differences arise from the change in output schema rather than from differences in the speech input. Once the model is required to produce a word in the `heard_word` field before assigning a label, the same recording is much more likely to be classified

as `intelligible` rather than `not_intelligible`. It is plausible that MiniCPM's C6 rejection decisions were not consistently accompanied by the identification of a specific alternative word: when required to populate the `heard_word` field, the model almost always outputs the target word, and the rejection decision largely disappears. The present results are consistent with that reading, though they do not establish it directly.

Interpreting the `heard_word` field

High agreement between `heard_word` and the target word does not provide direct evidence of acoustic grounding. Because the target word is included in the prompt, the same pattern could arise if the output were copied from, or strongly influenced by, the prompt. In 99.4% of recordings, MiniCPM outputs the target word in the `heard_word` field, and this is compatible with two different explanations: it may reflect successful identification from the speech signal, or it may reflect reliance on the target word provided in the prompt. C6+ should therefore be interpreted as a diagnostic probe of the effect of requiring an explicit transcription field, rather than as a direct measure of acoustic perception.

Three observations reported earlier in this chapter are relevant here. In Section 5.2, confidence and uncertainty signals were found to track the predicted label more closely than prediction correctness. In Section 5.3, rejection explanations restated the assigned label in generic terms, with little reference to specific acoustic events. Here, requiring an explicit transcription commitment causes 81% of prior rejections to be retracted, with the model outputting the target word in the `heard_word` field. Each observation is independent, and the three point in the same direction: MiniCPM's `not_intelligible` decisions under C6 do not appear to be reliably grounded in the identification of a specific pronunciation error. The contrast with Qwen2-Audio, where the same schema change improved performance, suggests that this pattern may be specific to MiniCPM's rejection behavior rather than to the task or the data.

5.6 A Close Reading of 300 Annotated Test Examples

Chapter 4 reported the distributions of the three explanation-quality axes over the manually annotated sample but deferred the breakdown and the examples to the error analysis. This section reads the annotations, gives representative examples for every class of every axis, and examines one distinction that emerged repeatedly during annotation: whether an explanation is supported by the audio or is mainly driven by the textual context.

Sample and axes

As described in Chapter 3, a manually annotated sample of 300 test-set predictions from the selected system was evaluated along three annotation axes: feedback quality, grounding quality, and pedagogical safety. Table 5.8 reproduces the distributions for reference, while the remainder of this section examines how the annotations differ across the four confusion-matrix outcomes.

In Tables 5.9, 5.10 and 5.11, a consistent pattern across the three tables is that explanation quality tracks the *predicted label*, not correctness. Where the model accepts a recording and is right (correct acceptances) its explanations are mostly good;

Table 5.8: Distribution of the three annotation axes across the 300 test examples (75 per confusion-matrix cell).

Feedback quality	Grounding quality	Pedagogical safety
irrelevant/unsupported: 103	clearly acoustic: 85	unsafe: 155
appropriate/sensible: 90	not supported by audio: 77	safe: 89
incorrect but relevant: 83	partly acoustic: 72	partly safe: 56
appropriate but incomplete: 24	mainly text-based: 63	
	unclear: 3	

everywhere else - and especially where the model rejects a recording - the explanations degrade sharply, whether or not the label is correct.

Table 5.9: Feedback quality by confusion-matrix cell (counts out of 75 per row).

Cell (gold / pred)	appropriate & sensible	appropriate but incomplete	incorrect but relevant	irrelevant or unsupported
Correct acceptance (I / I)	67	2	0	6
Correct rejection (NI / NI)	21	17	4	33
False rejection (FP) (I / NI)	0	2	17	56
Missed problem (FN) (NI / I)	2	3	62	8
All	90	24	83	103

Table 5.10: Grounding quality by confusion-matrix cell (counts out of 75 per row).

Cell (gold / pred)	clearly acoustic	partly acoustic	mainly text-based	not supported by audio	unclear
Correct acceptance (I / I)	55	18	1	1	0
Correct rejection (NI / NI)	20	22	10	21	2
False rejection (FP) (I / NI)	0	4	17	54	0
Missed problem (FN) (NI / I)	10	28	35	1	1
All	85	72	63	77	3

When the model flags a problem that is not there

One prominent failure mode is the false rejection: an intelligible recording that the model rejects. In the annotated sample these are the worst explanations on every axis. Not one of the 75 is rated appropriate and sensible, not one is clearly grounded in the audio, and 74 of the 75 are rated unsafe (Table 5.11). When the model incorrectly tells a learner that an intelligible recording is not intelligible, it almost never has an acoustic reason for doing so, and the resulting feedback would almost always mislead the learner. Two representative cases:

> *bard* - gold `intelligible`, predicted `not_intelligible`.
> *short_reason:* "The acoustic signal is unclear, making it difficult to identify the spoken word."
> *audio_evidence:* "The audio contains noise and lacks clear articulation of sounds that could help in identifying the intended target word."

Table 5.11: Pedagogical safety by confusion-matrix cell (counts out of 75 per row). *Unsafe* marks feedback that would actively mislead a learner.

Cell (gold / pred)	safe	partly safe	unsafe
Correct acceptance (I / I)	65	9	1
Correct rejection (NI / NI)	22	31	22
False rejection (FP) (I / NI)	0	1	74
Missed problem (FN) (NI / I)	2	15	58
All	89	56	155

(feedback: irrelevant/unsupported; grounding: not supported by audio; safety: unsafe)

pause - gold `intelligible`, predicted `not_intelligible`.
short_ reason: "The pronunciation is difficult to identify according to the human label."
audio_ evidence: "Manual listening required before final few-shot use."
(feedback: irrelevant/unsupported; grounding: not supported by audio; safety: unsafe)

In the second example the reason refers to a "human label", although the model had no access to the gold annotation for that recording, and the audio-evidence field contains a researcher-side note about few-shot preparation rather than any description of the recording. Both strings are fragments of the few-shot examples, discussed below.

A correct label is not a reliable explanation

The correct-rejection cell is the reason the annotation protocol included the two correct cells and not only the errors (Section 3.6). When the model correctly rejects an unintelligible recording, the label is right, but the explanation frequently is not. Thirty-three of the 75 correct rejections have *irrelevant or unsupported* feedback and 21 are *not supported by the audio* (Tables 5.9 and 5.10); only 22 are rated *safe*. This contrasts with the correct-acceptance cell, where the explanations are mostly fine (67 of 75 appropriate and sensible, 55 clearly acoustic, 65 safe). The correct label therefore does not certify a trustworthy explanation: a system evaluated on labels alone would score these rejections as successes even though 33 of the 75 are justified by an empty or unfounded reason. One example:

tied - gold `not_intelligible`, predicted `not_intelligible` (label correct).
short_ reason: "The pronunciation is difficult to identify according to the human label."
audio_ evidence: "Manual listening required before final few-shot use."
(feedback: irrelevant/unsupported; grounding: unclear; safety: unsafe)

Missed problems lean on the written word

Missed problems - unintelligible recordings that the model accepts - exhibit a different pattern. False rejections fail by inventing acoustic evidence that is not in the recording (54 of 75 rated *not supported by the audio*), whereas missed problems (FN) fail by leaning on the written target word supplied in the prompt (35 of 75 rated *mainly text-based*, 28

of 75 only partly acoustic, and just one not supported). On the feedback axis, 62 of 75 missed problems are *incorrect but relevant*: the model produces a confident, on-topic description of the target word - it simply describes a word that was not clearly said. This is the concrete form of the *not supported by audio* versus *mainly text-based* distinction that recurred during annotation. A false rejection asserts an acoustic problem that is not there; a missed problem asserts an acoustic clarity that is not there, by leaning on the written target word rather than on the recording. Two missed-problem examples:

> *pet* - gold `not_intelligible`, predicted `intelligible`.
> *short_reason:* "The pronunciation is understandable and the word 'pet' can be reasonably identified."
> *audio_evidence:* "The audio contains a clear, distinct pronunciation of the target word 'pet'. The syllables are pronounced with recognizable intonation and stress patterns."
> (feedback: incorrect but relevant; grounding: mainly text-based; safety: unsafe)

> *wait* - gold `not_intelligible`, predicted `intelligible`.
> *short_reason:* "The spoken word is understandable as 'wait' based on the acoustic evidence."
> *audio_evidence:* "The audio contains a clear and distinct pronunciation of the target word 'wait'. The syllables are well-formed, with no significant distortions or unclear segments that would hinder intelligibility."
> (feedback: appropriate but incomplete; grounding: mainly text-based; safety: partly safe)

Leakage of the few-shot template

Some of the unsupported explanations are exact copies of the few-shot examples placed in the prompt. The few-shot block used placeholder field values - `short_reason` = "The pronunciation is understandable/difficult to identify according to the human label" and `audio_evidence` = "Manual listening required before final few-shot use" - as stand-ins that were never replaced with real per-example text. These strings then reappear, unchanged, in the model's own test outputs. Counting them over all 8,196 valid test predictions, the phrase "according to the human label" appears in the `short_reason` of 1,829 predictions (22.3%), and "Manual listening required before final few-shot use" appears in the `audio_evidence` of 718 predictions (8.8%); one or the other appears in 1,832 predictions (22.4%).

Neither phrase can describe a recording. The model had no access to the test-set gold annotations, although the phrase "according to the human label" was itself present in the few-shot template, and the appearance of one or both strings in nearly a quarter of the outputs indicates that a substantial fraction of the explanations are produced by imitating the template rather than by inspecting the audio. Because the placeholders were never replaced, the explanation-quality results cannot be treated as a clean estimate of the model's explanatory ability under a properly instantiated few-shot prompt; the contamination limits the interpretation of the observed rates and is discussed further among the limitations of the study (Section 6.6).

The vocabulary used across the model's `short_reason` and `audio_evidence` fields was examined over the whole test set in Section 5.3. Table 5.3 shows the same concentration on two generic phrase classes, and the same near-absence of specific phonetic terms, that the annotated sample displays here.

Consistency of `audio_evidence` with the label

The consistency of the `audio_evidence` field with the assigned label was examined by comparing the content of the explanation with the model's prediction. Most inconsistencies take a single form: the evidence asserts a clarity the recording does not have. The field also contains harder contradictions, however, where it argues against its own label. A conservative keyword scan - requiring the evidence to explicitly claim the word is "clear", "distinct" or "recognizable" while the label is `not_intelligible`, or the reverse - flags about 170 of the 8,196 predictions (2.1%); because it only catches unhedged wording, this is a lower bound on the frequency of such contradictions. The manual sample shows the same pattern more sharply, for instance a recording rejected as `not_intelligible` whose evidence nonetheless transcribes it:

> *pole* - gold `intelligible`, predicted `not_intelligible`.
> *audio_evidence:* "Manually transcribed phonemes are 'p', 'o', 'l' which do not correspond to a recognizable word."

Here the field simultaneously reports a successful phoneme-level transcription and denies that a word was recognized, which cannot both hold. Such cases indicate that the evidence field is not functioning as a faithful record of what was heard.

Summary of the annotation findings

Across all three axes, the annotated sample shows a consistent pattern. Explanation quality is governed by the predicted label rather than by correctness or by the audio. Correct acceptances are explained well, while every rejection cell - right or wrong - is dominated by unsupported, non-acoustic, and often unsafe explanations. False rejections assert an acoustic problem that the recording does not contain (not supported by audio), while missed problems assert an acoustic clarity that is not there, drawing on the written target word supplied in the prompt (mainly text-based). A correct label does not imply a reliable reason, the issue vocabulary is largely reduced to generic statements about speech clarity, and roughly a quarter of all explanations still carry fragments of the few-shot template. Together these findings support the same conclusion reached in Sections 5.3 and 5.5 from the aggregate statistics: for this system, the natural-language explanation should be read as a by-product of the label decision, not as evidence that the decision was grounded in the recording.

5.7 Key Error Patterns

The preceding sections examined the errors of the selected system (MiniCPM-o-2.6 under C6) from several angles. Read together, these analyses reveal five recurring error patterns and one cross-cutting distinction, which organize the chapter and motivate the questions taken up in the Discussion.

1. **Over-acceptance (missed problems) is the dominant error.** On the test set the model makes 2,303 missed problems (not-intelligible recordings accepted) against 958 false rejections, a ratio of roughly 2.4 to 1. It therefore produces missed problems (FN) substantially more often than false rejections, though at the word level this pattern can reverse.

2. **False rejections are concentrated in a subset of words, and for some words the model predicts a single label throughout.** False rejections are not spread evenly across the vocabulary: ten of the 53 target words account for 51.8% of them. Within their intelligible recordings the model rejects *pull*, *poor*, *caught* and *paws* 100% of the time, *pool* 83%, *farther* 78% and *daft* 63%, and precision and recall for the `intelligible` class are 0.000 for all four of the first group. For *pull*, *caught* and *paws* the model predicts `not_intelligible` for every recording; *poor* is nearly constant, returning `intelligible` for only two of its 155 recordings, both incorrectly. Taken together, the first two patterns show that the aggregate tendency toward over-acceptance coexists with word-specific patterns in which a single label is assigned to almost every recording of a target word.

3. **Explanations range from incomplete to unsupported.** The annotated sample (Section 5.6) shows explanations ranging from correct-but-incomplete to unsupported. False rejections frequently introduce unsupported acoustic problems, missed problems often attribute an acoustic clarity the reference annotation does not support, and a substantial share simply repeat the few-shot template. The placeholder phrase "according to the human label" appears in 22.3% of all test reasons.

4. **Self-assessment signals carry little information.** For the selected system, confidence is almost always `medium` and never `high`, so it does not separate correct from incorrect predictions (Section 5.2). The same near-constancy holds for the other two systems: on the development set under C6, Qwen2.5-Omni returned `medium` for 99.0% of its predictions and Qwen2-Audio for 100% of them. Confidence is therefore practically constant within each model. The uncertainty flag does vary, but it varies with the predicted label rather than with correctness, and is raised far more often when the model predicts `not_intelligible` than when the prediction is actually wrong (Section 5.2). Neither signal indicates when a prediction can be trusted.

5. **The decision is sensitive to prompt format and probing.** Adding a single output field changes the behavior of the models in different ways. Requiring a `heard_word` transcription (C6+) reduces MiniCPM's $F_{0.5}$ for the `not_intelligible` class from 0.513 to 0.275, changes Qwen2.5-Omni only marginally (0.462 to 0.445), and improves Qwen2-Audio (0.457 to 0.495) (Section 5.5). The effect of this schema modification therefore differs substantially across the three models.

Two sources of difficulty. Cutting across these five patterns is a distinction between two kinds of "hard" word. Some words are difficult for every system in the same way. Ten of the 53 words receive exactly the same accuracy from all three systems on the development set, and *bird* is the clearest case: each system scores 0.342 on it, and each accepts all 38 of its recordings, so none of them ever flags the 25 that are annotated `not_intelligible`. *Board*, *pull* and *fair* behave the same way, which is why their failures were fully shared in Section 5.4. Other words are difficult for only one system: *weight* ranges from 0.816 for MiniCPM-o-2.6 and 0.811 for Qwen2.5-Omni down to 0.184 for Qwen2-Audio, and *put* from 0.692 for both of the first two to 0.256 for the third, spreads of more than 40 accuracy points on the same recordings. The first pattern is consistent with a source of difficulty shared across systems, which may lie in the

recordings, in the target words, in the common prompt, or in priors the models have in common; the second is more consistent with a model-specific component.

Chapter 6

Discussion

This thesis set out to ask whether speech-enabled multimodal LLMs can assess word-level intelligibility of L2 English in a reliable way, both classifying it and explaining it. The three research questions approached that aim from different directions: how a promptable multimodal model compares with a purpose-built ASR pipeline on the same task (RQ1), how far prompt-engineering strategies shape its behavior (RQ2), and to what extent the phonetic explanations it offers are grounded in the audio it was given (RQ3). The results in Chapters 4 and 5 answer these questions in some detail; the task of this chapter is to read those answers together, against the prior work reviewed in Chapter 2.

On the classification task, a multimodal model prompted directly, with no task-specific fine-tuning, reaches approximately the same precision as an ASR pipeline on `not_intelligible` (the class that matters most for teaching), while remaining a markedly more cautious, and therefore less complete, assessor. On the prompting question, no single strategy is best across models, and the effect of a given prompt is model-specific rather than a property of the task. On the explanation question, which is where the study has the most to say, the model's natural-language justifications are governed by the label it has chosen rather than by the recording, so that a correct classification cannot be taken, on its own, as a grounded one. Each of these results is developed below, and the chapter then turns to what they imply for using such systems in pronunciation feedback, for evaluating them, and for the work that would have to follow.

6.1 Multimodal Prompting against an ASR Pipeline (RQ1)

The first research question asks how the multimodal model compares, on this task, with the earlier ASR-based system of Kuan (2025) reviewed in Section 2.3. Based on Table 4.5, the earlier system is the stronger of the two: it reaches a higher macro-F0.5 (0.651 against 0.577) and a higher F0.5 on both classes, on the intelligible class (0.697 against 0.642) and on the not-intelligible class (0.605 against 0.512) alike. It would be easy to read this as a straightforward loss for the multimodal approach.

As described in the methodology, the primary focus is the `not_intelligible` class. Accordingly, the study adopts the precision-weighted $F_{0.5}$ score because the costly mistake is telling a learner that a word they pronounced acceptably is unintelligible. On that precise metric the two systems are almost indistinguishable: 0.581 for the pipeline and 0.574 for the multimodal model. When either system does flag a word as a problem, it is correct about as often as the other. What separates them is not the reliability of a

rejection but how readily each one rejects. The pipeline recovers a large majority of the problems (recall 0.726) while the multimodal model recovers only about a third (0.359), and because F0.5 weights precision more heavily than recall, that recall gap moves the headline score less than it might first appear. The pipeline's advantage is therefore real, and it is driven almost entirely by recall.

The substantive finding for RQ1 follows from this. A general-purpose multimodal model, prompted directly and without task-specific fine-tuning, reaches a similar level of precision on the `not_intelligible` class to a domain-adapted ASR pipeline with a dedicated phonetic-judging stage. For a field in which dedicated pipelines have generally been the standard route to this kind of assessment, that is a finding of some significance: in this study, comparable precision was achieved without task-specific training. The price is recall, and it is a real one. The confusion matrix in Table 4.4 shows the shape of it. Roughly seven in ten of the model's test-set errors are cases where an unintelligible word was let through as acceptable. The dominant failure mode is over-acceptance - the system predicts `intelligible` for about 72.6% of all recordings, appreciably more than the gold distribution warrants - and a tool that misses about two-thirds of the real problems is a cautious one rather than a thorough one. It should also be read with the qualification carried over from the results: the two systems were evaluated on different splits of the same corpus, so the comparison is indicative of the broad shape of the difference and not an exact contrast.

6.2 The Effect of Prompt Design (RQ2)

The second research question concerns prompt engineering. There is no single prompt that is best across models. Reading down Table 4.1, C6 - the condition combining the perception-before-decision framework with few-shot examples - is the strongest for MiniCPM-o-2.6 and for Qwen2.5-Omni, and it was the configuration frozen for the test set. But for Qwen2-Audio the best Macro-$F_{0.5}$ comes from C4 (0.529), which adds few-shot examples but no reasoning framework, with C6 close behind (0.523). The fullest prompt was therefore not the best one for this model, and its worst condition by a wide margin was C2 (0.340), the phonetic-checklist condition. Adding structure did not reliably improve behavior: the checklist condition produced the lowest or near-lowest macro-$F_{0.5}$ for every model, while the perception-before-decision framework combined with few-shot examples was strongest for MiniCPM-o-2.6 and Qwen2.5-Omni but not for Qwen2-Audio.

The same unevenness appears within the conditions, and it is important because it helps explain some of the per-class numbers. Under several conditions, a model predicts almost every sample as belonging to one class, so that its recall on that class approaches one while its recall on the other approaches zero. C2, the phonetic-checklist prompt, shows this pattern most clearly in every model, although the direction differs: MiniCPM and Qwen2.5-Omni predict predominantly `intelligible`, whereas Qwen2-Audio predicts predominantly `not_intelligible`. The same pattern appears again at the level of individual words in Section 5.1, where some words are assigned almost entirely to a single class regardless of the recording.

Two points emerge from these results. First, prompt sensitivity here is model-specific rather than a property of the task: the reasoning framework and the examples that pushed MiniCPM toward a more balanced decision did not transfer to the other models in the same way, and the study does not establish a single prompt-engineering recipe

that can be recommended across speech-enabled models. Second, one finding under this question is methodological rather than about the models at all. As Section 5.6 reports, 22.4% of the test outputs contained at least one fragment of the placeholder text from the few-shot examples: the phrase "according to the human label" appears in the short reason of 1,829 predictions (22.3%) and "Manual listening required before final few-shot use" in the audio-evidence field of 718 (8.8%), with one or the other present in 22.4% of all outputs. Neither phrase can describe a recording, and their presence in nearly a quarter of the outputs suggests that some explanations retained elements of the prompt template rather than producing fully task-specific responses. This template leakage is therefore best understood as a methodological limitation of the prompt design rather than as evidence about the models underlying reasoning. It nevertheless means that the explanation text should be interpreted with care, a point that connects naturally to the grounding analysis presented below.

6.3 Are the Explanations Grounded in the Audio? (RQ3)

The third research question is the one to which this thesis contributes most. The manual annotation of 300 predictions (Section 5.6) reveals the pattern: only 85 of the 300 explanations (28.3%) were judged clearly based on audible evidence, and only 89 (29.7%) safe. Read on its own, that already says the explanations are not fit for their intended use. But the more revealing result is not the rate; it is that explanation quality tracks the predicted label rather than the correctness of that label or the content of the audio, and the four confusion-matrix cells make this visible.

When the model correctly accepts an intelligible word, its explanations are mostly appropriate, acoustically grounded, and safe. The False Positive cases show almost the opposite pattern, and are the more revealing of the two. These are cases where the model incorrectly flags an intelligible word as `not_intelligible`, and the accompanying explanations are unsupported. The model's feedback is therefore weakest not when it misses a real pronunciation problem but when it incorrectly identifies one. The False Negative cases fail in a different way. These are cases where a real `not_intelligible` word is predicted as `intelligible`, and 62 of the 75 explanations were judged *incorrect but relevant*, suggesting that the explanations often remain related to the target word while failing to account for the pronunciation that led to the incorrect decision. The two error types therefore differ in systematic ways: FP explanations tend to be unsupported by the audio, whereas FN explanations are more often relevant to the target word but incorrect. The True Positive cases occupy an intermediate position only on some dimensions. They are safer than either error category, with 22 of the 75 explanations rated safe compared with 2 for FN and none for FP, and they are better grounded than the FP explanations. On the feedback axis, however, they are worse than FN: 33 of the 75 TP explanations are rated irrelevant or unsupported, compared with 8 of the FN explanations. A correct rejection therefore does not necessarily carry a reliable explanation, a point examined in detail in Section 5.6. These differences also have distinct pedagogical implications. From the perspective of computer-assisted pronunciation training, the four confusion-matrix cells are not equally important: the primary value of such a system lies in reliably identifying genuine pronunciation problems and providing feedback that helps learners understand and correct them. Seen in that light, the one cell in which the system produces good feedback buys very little. Telling a learner that a word they pronounced acceptably was indeed acceptable carries far less instructional value

than identifying a word that failed and saying why. The explanations are therefore at their most reliable exactly where they are least needed, and at their least reliable in the two cells a pronunciation tool exists to serve.

The transcription-augmented variant sharpens the picture (Section 5.5). When MiniCPM is required to write down the word it hears before it assigns a label, 395 of its development-set predictions (19.7%) move from `not_intelligible` to `intelligible` while only 10 (0.5%) move the other way, so the great majority of its rejections simply disappear. In 1,989 of the 2,001 recordings (99.4%) it writes the target word itself in the heard-word field, and once it has done so it has no remaining basis on which to reject. This is best interpreted as behavioral evidence arising from the constrained output format rather than as evidence of an internal "transcribe-then-decide" process. The claim here is therefore limited to the observed behavior; making the perceived word explicit substantially changes the model's predictions. The contrast with Qwen2-Audio, for which the same schema change produced an overall improvement (393 rescued against 277 regressed), suggests that this effect is more characteristic of MiniCPM than of the task itself. It also shows that the same intervention can affect different models in different ways.

The self-reported signals provide further evidence in the same direction, and they relate directly to the calibration literature reviewed in Section 2.4.3. The confidence field shows very little variation; across all 53 target words, the model never returns *high* confidence, and the share of predictions returned with *medium* confidence has a median of 98.7% across the 53 words, with the remainder returned as *low*. As a result, the confidence values provide little distinction between correct and incorrect predictions. The uncertainty flag shows greater variation, appearing in 23.2% of predictions. However, it varies more closely with the `not_intelligible` label than with prediction errors. Across the target words, it correlates strongly with the rejection rate ($r = 0.91$) but only weakly with the error rate ($r = 0.30$). The pattern is most evident on the cases where an uncertainty signal would be expected to be most useful. On the eleven words that the model accepts almost uniformly, where every `not_intelligible` recording becomes a False Negative, the uncertainty flag is raised for an average of only 0.9% of recordings. These results suggest that the uncertainty flag is more closely associated with predicting the `not_intelligible` label than with cases in which the model is actually incorrect.

The content of the explanations provides a better perspective. The reasons are dominated by two generic phrases, each associated more closely with a label than with the audio: As Table 5.3 shows, the reason "no major issue" accounts for 27-29% of the model's `intelligible` predictions, while "unclear audio" accounts for 63-69% of its `not_intelligible` predictions. These proportions change very little according to whether the prediction is correct or incorrect. By contrast, the specific phonetic dimensions that the word list was designed to probe are mentioned only rarely. Aspiration and rhoticity appear in essentially none of the 8,196 explanations, and vowel and consonant-level comments appear in at most about 10%, even though the target words were selected to contrast these features. A system asked to explain intelligibility in phonetic terms instead describes its own decision in generic terms, which is the concern raised in Section 1.2 made concrete. These findings align closely with the distinction drawn by Gatto (2024) between the plausibility and the faithfulness of an explanation. In the present study, many explanations appeared plausible but were not well supported by the acoustic signal. The findings are also consistent with the observation of Ma et al. (2025)

that speech foundation models may carry rich acoustic information without necessarily rendering it faithfully in their explanations.

One further point deserves emphasis; the weakness is not confined to the error cells. In the correct-rejection cell, where the label is right, 33 of the 75 explanations are still irrelevant or unsupported and only 22 are judged safe. A correct label, then, does not necessarily imply a trustworthy explanation. Looking only at the predicted labels would count these cases as correct, even though many of the accompanying explanations are unsupported or insufficiently grounded. For the applied goal of the thesis, this has practical implications. An intelligibility label from this system may provide a useful initial indication in some cases, but the accompanying feedback should be interpreted with caution. The manual annotation suggests that it is unsupported by the audio, and judged unsafe, often enough that it is not yet suitable for direct use as learner feedback.

6.4 Which Words and Sounds Are Hard (RQ2-RQ3)

The errors are not spread evenly across the vocabulary, and where they concentrate ties the classification results back to the phonetic contrasts the task was built around. The clearest feature-level difference (Section 5.3.1) concerns words tagged as involving an initial-stop aspiration contrast. These words have a mean $F_{0.5}$ of 0.47 for the `intelligible` class, compared with 0.63 for words without that tag. The difference is accompanied by a higher false-rejection rate: 33% of the intelligible recordings in the aspiration-tagged group are rejected, compared with 15% in the remaining group. The dimension on which the model most often rejects a word a listener would have understood is thus the exact dimension it never names in its explanations, which links this section to the grounding findings: the aspiration contrast is reflected consistently in neither the model's predictions nor its explanations.

The word-level results add a second layer. Accuracy ranges from about 0.36 for the hardest words, *bird* and *Paul*, to above 0.80 for the easiest, and 14 of the 53 words receive a single label across every one of their recordings - eleven always `intelligible`, three always `not_intelligible` - so that for these words the prediction is driven by the target word rather than by anything in the individual recording. The shared-failure analysis (Section 5.4) shows that this difficulty is not confined to a single model. Of the 1,994 development recordings on which all three candidate models returned a valid prediction, 426 (21.4%) are misclassified by all three at once, and 365 of those 426 (85.7%) are missed problems (False Negative); for *board*, *bird*, *pull*, and *fair* the shared error rate reaches 100%, meaning there is not a single recording of these words that one system gets wrong while the others get right. An error that recurs across three different models and two splits is a stronger candidate for reflecting something about the recording or the target word than a property of any one system, which is why these words converge with the hardest words on the test set. The convergence also connects back to the prior ASR-based work reviewed in Section 2.3. Kuan (2025, sec. 5.2.1) reports that words whose recognition depends on the aspiration of a voiceless stop — including *pat* and *pit* — produced many False Positives (false rejections), which were attributed to speaker-to-speaker variation in the aspiration of /p, t, k/ among L2 learners. The multimodal model evaluated here shows a similar pattern, with words tagged as involving an aspiration contrast also receiving a relatively high number of False Positives. Although the two systems were evaluated separately, both identify words tagged as involving an aspiration contrast as particularly challenging. This convergence

is consistent with the possibility that these words are intrinsically more difficult for automatic intelligibility assessment than many of the remaining target words.

One further possibility should be kept open alongside it. The annotation scheme reported in Section 3.1.3 reached 78.9% agreement overall but only 66.3% on the files where a phonetic issue was flagged, and aspiration in particular was noted as difficult for annotators to judge consistently. Some of the words that all three systems find hard may therefore be words that were also more challenging to label consistently, and the same acoustic ambiguity that made them hard to annotate may have pulled the models toward the wrong contrast. The present design cannot separate a genuinely difficult word from an inconsistently labeled one. The caution stated in the results still applies as well: the per-word feature tags provide a broad, heuristic description of each word's phonetic characteristics rather than a detailed acoustic analysis of every recording.

6.5 Implications

Two implications follow, one for the application and one for how models like these should be evaluated. For the educational application, the results support a precision-first design and a cautious view of what the system can currently provide. On the present evidence, the model is better understood as a conservative filter — a first-pass screening tool that achieves precision comparable to that of the earlier ASR-based pipeline when it does flag a problem — than as a generator of pronunciation feedback, because the explanations remain its weakest component and the least reliable part of the output. The FP cases are particularly important in an educational setting; where an intelligible word is classified as `not_intelligible`, and the accompanying explanation is frequently unsupported by the audio and judged unsafe in the manual annotation. Before such feedback could be presented directly to learners, its acoustic grounding and pedagogical safety would need to be improved. On the evidence presented here, that requirement is not yet met.

For the evaluation of multimodal speech models more generally, one implication concerns the relationship between label accuracy and explanation grounding. As the correct rejection (TP) cases showed, correct labels can still be accompanied by explanations that are unsupported or only weakly grounded, even though a label-only evaluation would count these predictions as correct. Treating grounding as a separate dimension of evaluation, rather than assuming that it follows from label accuracy, is a methodological contribution of this thesis. The findings further suggest that, at least on the present task, the two should be evaluated separately.

6.6 Limitations

Several limitations bound these conclusions. The comparison with Kuan (2025) is across different splits of the same corpus, and the figures for that system were recomputed from its reported precision and recall, so it should be interpreted as an indicative side-by-side comparison rather than as a direct evaluation. The per-word feature tags are heuristic and overlapping, so the aspiration-related difference cannot be fully separated from other characteristics shared by the tagged words. The study adopted a single-prompt design to keep the evaluation consistent across models and conditions. More complex strategies, such as dynamic or multi-stage prompting that delays presentation of the target word, were not investigated. Finally, the categorical confidence and uncertainty

self-reports showed little relationship with prediction correctness and do not support a formal calibration analysis.

Placeholder text in the few-shot prompt. The few-shot block used in condition C6 contained placeholder values for the `short_reason` and `audio_evidence` fields that were never replaced with real per-example text before the final runs. These strings reappear verbatim in 22.4% of the model's own test outputs (Section 5.6). Because these placeholders were not replaced, the explanation-quality results cannot be treated as a clean estimate of the model's explanatory ability under a properly instantiated few-shot prompt. The contamination limits the interpretation of the observed rates in both directions: some unsupported explanations may reflect direct imitation of the defective template, while a corrected prompt could also change the model's broader response behavior. Re-running C6 with fully populated few-shot fields is therefore a priority for future work.

Scope of the grounding checks. The acoustic grounding probes reported in Appendix B use a single target word and a small set of hand-constructed control inputs. They establish that a grounding failure can be triggered in every model tested, but they cannot estimate how often it occurs across the corpus.

Single-annotator design. The 300 explanations were annotated by one researcher, with no second annotator and no inter-annotator agreement check. Two of the three axes, grounding quality and pedagogical safety, involve judgments on which annotators could reasonably differ, and the annotator was not blind to the model's predicted label. The distributions reported in Section 5.6 should therefore be read as one careful reader's assessment rather than as a validated measurement, and the direction of the findings is more secure than their exact rates.

Silence padding as a potential confounding factor. The dataset construction preserves up to 1.5 seconds of surrounding silence on either side of each isolated target word to maintain natural acoustic context (Kuan, 2025). Because the spoken word itself is typically much shorter than the full recording, a substantial proportion of each audio sample may consist of background room tone or silence rather than the target pronunciation. This raises the possibility that some generic explanations, particularly *unclear audio*, may partly reflect these non-speech segments rather than the pronunciation of the target word itself. The present study did not isolate the effect of this padding, so its contribution to the observed explanation patterns cannot be determined.

6.7 Future Work

The findings point to several directions for future work, ordered here by how they follow from the results of the thesis. The first, and the most central, concerns acoustic grounding, since grounding-related issues appear across all three research questions. One promising direction at the prompting level is to reduce the model's reliance on the target word, for example by withholding that word until the model has first committed to a transcription of the audio, so that the intelligibility judgment is based more on the acoustic input. More broadly, future work could investigate prompting, training, and evaluation methods that encourage explanations to remain closely tied to the acoustic

signal. The annotation protocol, transcription probe, and grounding-oriented evaluation introduced in this thesis provide one possible starting point for evaluating such approaches. Whether these strategies improve acoustic grounding, however, remains an empirical question.

A second direction concerns the evaluation of grounding itself. The sanity checks used here were small and qualitative, meant only to establish whether a model's behavior changes when the acoustic input is manipulated. They could be extended into a larger and more systematic grounding benchmark, built from a wider range of controlled interventions and applied across several models and prompting conditions, so that grounding behavior could be compared in a reproducible way.

In the present study the confidence and uncertainty signals tracked the predicted label far more closely than the correctness of that label, which left them of little use as indicators of reliability. Future work could look instead to calibration strategies that do not rest on the model's own self-report: confidence estimated from the model's behavior, agreement across several generations or prompting conditions, or external calibration methods applied after the fact.

The prompt comparisons point to two further directions of their own: robustness across models, since no single condition served all three here, and a repair of the few-shot design so that template text can no longer leak into the outputs. The comparison with the earlier ASR-based pipeline invites a direct study of task-specific fine-tuning against prompting, now that prompting alone has been shown to reach comparable precision without it. Finally, any move toward a learner-facing tool would first require a fuller evaluation of both the pedagogical safety and the acoustic grounding of the generated feedback, given how often the explanations produced here were judged unsafe to show.

Chapter 7

Conclusion

This thesis examined whether speech-enabled multimodal large language models can assess word-level intelligibility in L2 English in a reliable way, both by classifying whether a spoken word is intelligible and by explaining that judgment in phonetic terms. To address this, three such models were evaluated on a word-level corpus of L2 English speech across six prompt conditions and a transcription-augmented variant (C6+); a single configuration was selected on a development set and evaluated once on a test set; its labels were compared with an earlier ASR-based system on the same corpus; and a sample of its explanations was annotated for feedback quality, acoustic grounding, and pedagogical safety.

On classification (RQ1), a model prompted directly, with no task-specific fine-tuning, reached about the same precision as the earlier ASR pipeline on the `not_intelligible` class - the class that matters most for pronunciation feedback - while recovering far fewer of the real problems, so that the pipeline remained the stronger of the two systems. On prompting (RQ2), no single prompt was best for all three models; the effect of a prompt was specific to the model it was given to. On explanation (RQ3), the model's stated reason, its confidence, and its uncertainty followed the label it had chosen rather than the recording: only a minority of the explanations were grounded in audible evidence, many were judged unsafe to show a learner, and the specific phonetic contrasts the task was built around were almost never named. A correct label, on its own, was therefore not evidence that the decision had been reached for the right reason.

Taken together, these results suggest that a general multimodal model can match a purpose-built pipeline on the precision a teaching tool most needs, which is a notable outcome given that it required no fine-tuning; but the same model is a cautious classifier that accepts too much, and its explanations are not yet grounded enough to be placed in front of a learner. The thesis therefore makes two main contributions: a systematic evaluation of three current speech-enabled multimodal language models on word-level L2 intelligibility assessment, and an evaluation framework that treats acoustic grounding as a separate dimension of evaluation, independent of label accuracy. The grounding-oriented annotation protocol and the transcription probe introduced in this thesis provide a first step toward evaluating that property more directly.

The main limitations follow from the scope of the study. The evidence on acoustic grounding is based on a single corpus, a fixed target-word list, and a subset of the data annotated by one researcher; and the few-shot block used in the selected condition carried placeholder text that reappears in roughly a fifth of the model's own explanations, so the explanation-quality rates should be read as a lower bound rather than a clean es-

timate. The findings should therefore be interpreted within the scope of this task rather than as general conclusions about multimodal speech assessment. Within that scope, however, the results indicate that a prompted multimodal model can achieve precision comparable to that of a dedicated ASR-based pipeline, whereas the acoustic grounding of its explanations remains substantially weaker. Improving that second aspect is likely to be a prerequisite before such systems can provide reliable pronunciation feedback to learners.

Code and Data Availability

The code developed for this thesis, including the inference notebooks for the three models, the prompt-construction and parsing code, and the analysis scripts used to produce the reported tables, is available at `https://github.com/massoumehjavadi-cloud/speech-llm-for-intelligibility-classification`.

The audio recordings are not included. They are learner speech collected through the MOOC described in Appendix E.3, and the terms under which they were made available do not permit redistribution.

Appendix A

Prompt Texts

This appendix shows the exact prompt text given to the models. All six conditions (C1-C6) and the transcription-augmented variant (C6+) are written here; a shared base instruction, an optional phonetic-checklist block, an optional perception-before-decision (DDCoT) block, an optional set of few-shot examples, and a closing JSON-format instruction. Table A.1 shows which blocks each condition uses; the full text of every block is then reproduced below. Where a block is generated per target word (the phonetic checklist and the few-shot examples), the text shown is the exact text produced for a representative word, with the target word indicated.

Table A.1: Composition of the seven prompt conditions from the shared building blocks.

Condition	Base	Phonetic checklist	DDCoT	Few-shot	JSON format
C1	✓				✓
C2	✓	✓			✓
C3	✓		✓		✓
C4	✓			✓	✓
C5	✓	✓		✓	✓
C6	✓		✓	✓	✓
C6+	✓		✓	✓	✓*

*C6+ is identical to C6 but adds a transcription step and a `heard_word` field, described at the end of this appendix.

Every prompt is assembled by joining the blocks it uses, in the order base instruction, target word, phonetic checklist (if present), DDCoT (if present), few-shot examples (if present), and JSON-format instruction, followed by the final line *"Now evaluate the audio recording and return the JSON object only."*

A.1 Base Instruction (all conditions)

```
You are evaluating word-level intelligibility in L2 English speech.

Task:
Choose exactly one label for the audio:
- intelligible
- not_intelligible

The goal is not native-like pronunciation scoring. The goal is whether an
```

```
average listener could identify the intended target word from the recording.

Use only the acoustic evidence present in the recording. The written target
word indicates what should be evaluated, but it is not evidence that the word
was pronounced correctly.

Do not guess from spelling, lexical expectations, or what the speaker probably
intended. Do not mark a word as "not_intelligible" only because it sounds
non-native or phonetically imperfect.

If the acoustic evidence is weak, ambiguous, incomplete, noisy, or difficult
to interpret, adopt a conservative decision style and explain the uncertainty
in "audio_evidence".
```

This block is immediately followed by the line `Target word: "<word>"`, giving the word under evaluation.

A.2 Phonetic Checklist Block (C2, C5)

This block is generated per target word; the relevant phonetic dimensions and the lexical contrasts are selected automatically from the target word. The text below is the exact block produced for the target word *pat*.

```
Use a targeted phonetic checklist before deciding.

For this target word, pay attention to these potentially relevant phonetic
dimensions:
aspiration, vowel_quality, final_consonant, stress

Possible lexical contrasts to keep in mind:
bat / pet / pit

These lexical contrasts are examples only. They are not a closed list. If the
audio sounds like another plausible English word, mention that in
"audio_evidence".

Use the checklist as attention guidance, not as a scoring rubric.

Decision rules:
- Choose "intelligible" if the spoken word can reasonably be understood as the
  target word, even if it sounds non-native or phonetically imperfect.
- Choose "not_intelligible" if the spoken word sounds like a different word,
↪  if
  an important lexical contrast is lost, or if the target word cannot be
  recognized from the audio.
- Do not mark a word as "not_intelligible" only because it differs from
  native-speaker pronunciation.
- Do not invent a phonetic issue that is not clearly audible.
- Do not choose a phonetic explanation only because that feature appears in
↪  the
  checklist.

If "predicted_label" is "not_intelligible", the "short_reason" must begin with
one of these annotation-style issue labels:
```

```
no_major_issue, rhoticity_missing, vowel_length_shortening,
vowel_length_lengthening, vowel_quality_substitution,
initial_consonant_missing_aspiration, initial_consonant_substitution,
middle_consonant_substitution, final_consonant_substitution,
extra_sound_initial, extra_sound_middle, extra_sound_end,
missing_sound_initial, missing_sound_middle, missing_sound_end,
unclear_audio, unknown_issue

Use these issue labels as follows:
- "rhoticity_missing": use when an expected /r/ is missing or unclear.
- "vowel_length_shortening" or "vowel_length_lengthening": use when vowel
  duration may cause lexical confusion.
- "vowel_quality_substitution": use when the vowel sounds like a different
↪  vowel
  and may change the perceived word.
- "initial_consonant_missing_aspiration": use only for missing aspiration of
  initial stressed /p/, /t/, or /k/.
- "initial_consonant_substitution", "middle_consonant_substitution", or
  "final_consonant_substitution": use when a consonant sounds replaced by
  another consonant.
- "extra_sound_initial", "extra_sound_middle", or "extra_sound_end": use when
↪  an
  additional unwanted sound is audible.
- "missing_sound_initial", "missing_sound_middle", or "missing_sound_end": use
  when an expected sound is absent.
- "unclear_audio": use when the recording is too weak, noisy, clipped,
  incomplete, or ambiguous to identify the issue.
- "unknown_issue": use when the pronunciation seems problematic but the issue
  does not fit the listed labels.

If "predicted_label" is "intelligible", the "short_reason" may begin with
"no_major_issue" unless there is a minor issue that does not block
↪  recognition.

In "audio_evidence", briefly mention the audible evidence supporting the
↪  label.
Mention only what is actually heard.
If the audio is weak, noisy, incomplete, or unclear, say this in
"audio_evidence" and set "uncertainty_flag" to "yes".
```

A.3 Perception-Before-Decision Block (C3, C6, C6+)

```
Use a perception-before-decision procedure.

Step 1:
Attend only to the acoustic signal. Briefly identify what is clearly audible
in the recording. Describe only evidence that is directly supported by the
audio.

Step 2:
Do not infer pronunciation details that are not clearly audible. Do not rely
on the written target word when forming the acoustic description.

Step 3:
After completing the acoustic inspection, decide whether the spoken word can
```

```
reasonably be understood as the target word.

Your final decision must be consistent with the evidence reported in the
"audio_evidence" field. Do not introduce additional evidence during the
decision stage.

The purpose of the acoustic description is to document the evidence supporting
the intelligibility judgment, not to provide a detailed phonetic analysis.
```

A.4 Few-Shot Examples Block (C4, C5, C6, C6+)

The few-shot block presents a fixed set of manually selected examples, each shown as a target word paired with the desired JSON output. The examples are reproduced below exactly as they appeared in the prompt. Every example carried the same field values (confidence `medium`, the same `short_reason` and `audio_evidence` text, and both flags set to `no`), differing only in the target word and the label, so Example 1 is shown in full and the remainder are abbreviated with "…" for the shared fields.

```
Few-shot examples. These show the desired decision style and JSON format.
Evaluate the new audio independently. Do not copy the example label unless the
new audio supports it.
Example 1: target_word="mate" -> {"predicted_label": "intelligible",
  "confidence": "medium", "short_reason": "The pronunciation is understandable
  according to the human label.", "audio_evidence": "Manual listening required
  before final few-shot use.", "uncertainty_flag": "no",
  "possible_irrelevent_answer_risk": "no"}
Example 2: target_word="boy"    -> {"predicted_label": "intelligible", ...}
Example 3: target_word="pool"   -> {"predicted_label": "not_intelligible",
↪  ...}
Example 4: target_word="caught" -> {"predicted_label": "not_intelligible",
↪  ...}
Example 5: target_word="nose"   -> {"predicted_label": "intelligible", ...}
Example 6: target_word="paws"   -> {"predicted_label": "not_intelligible",
↪  ...}
Example 7: target_word="pull"   -> {"predicted_label": "not_intelligible",
↪  ...}
Example 8: target_word="seedy"  -> {"predicted_label": "intelligible", ...}
```

A.5 JSON-Format Instruction (all conditions)

```
Return EXACTLY one valid JSON object.

Do not return a Python dictionary. Do not use single quotation marks. Use
double quotation marks (") for all JSON keys and all string values.
Do not use markdown formatting. Do not wrap the JSON inside code fences. Do
not include explanations before or after the JSON object.

You must choose EXACTLY ONE value for "predicted_label":
- "intelligible"
- "not_intelligible"

The JSON object must contain exactly these keys:
{"predicted_label": "...", "confidence": "...", "short_reason": "...",
```

```
 "audio_evidence": "...", "uncertainty_flag": "...",
 "possible_irrelevent_answer_risk": "..."}

Allowed values:
"predicted_label" must be exactly one of: "intelligible", "not_intelligible".
"confidence" must be exactly one of: "low", "medium", "high".
"uncertainty_flag" must be exactly one of: "yes", "no".
"possible_irrelevent_answer_risk" must be exactly one of: "yes", "no".

Do not use any other keys, labels, or value names.
```

A.6 Transcription-Augmented Extension (C6+)

C6+ uses the same blocks as C6 (base, DDCoT, few-shot, JSON) but inserts a transcription step after the DDCoT block, and replaces the JSON-format instruction with a version that adds a `heard_word` field at the front of the required output. The transcription step inserted after the DDCoT block reads:

```
Transcription step (for analysis):
Before deciding the label, write down the single English word you actually
hear in the recording, based only on the acoustic signal. Report it in the
"heard_word" field. If you cannot make out any word, write "unintelligible".
Fill in "heard_word" first, before the label.
```

The JSON-format instruction is then modified so that `heard_word` is required first:

```
Fill the fields in the order given. The FIRST field must be "heard_word": the
word you actually heard, before you decide the label.

The JSON object must contain exactly these keys, in this order:
{"heard_word": "...", "predicted_label": "...", "confidence": "...",
 "short_reason": "...", "audio_evidence": "...", "uncertainty_flag": "...",
 "possible_irrelevent_answer_risk": "..."}

Allowed values:
"heard_word" is a single English word you heard, or "unintelligible".
```

The remaining allowed-value constraints are identical to those of the standard JSON-format instruction in Section A.5.

Appendix B

Acoustic Grounding Checks

Because every prompt names the target word, a model that never attended to the recording could still return a confident, plausible-looking judgment purely from the word it was told to expect. A correct label, on its own, therefore does not show that a decision was grounded in the audio. This appendix collects a set of small, early diagnostic checks that probe this directly. They were run before the final model and prompt were chosen, applied uniformly to all three models under the simplest condition (C1). Being based on a handful of hand-constructed inputs, they are diagnostic sanity checks rather than results, which is why they are reported here rather than in the main text. A larger and more systematic treatment of acoustic grounding, based on the transcription-augmented C6+ variant, is given in the error analysis (Section 5.5).

B.1 Rationale and Design

The logic of the checks is to hold the prompt fixed - including the named target word - while removing or disrupting the acoustic evidence, and then to observe whether the model's answer changes. If a judgment is grounded in the recording, replacing the recording with silence, white noise, or the audio of a different word should change the answer. If the judgment is driven by the prompt, it should not. Each model was presented with a genuine reference recording and with a set of control inputs derived from it: silent audio, white noise, a mismatched recording (the audio of a different target word, presented under the original prompt word), a mismatched prompt (the original recording, presented under a different target word), a raw-transcription request, and a repeated presentation of the reference recording used to check output stability.

Two overlapping probes were run. The first is built around the word *daft* (gold label *not intelligible*) and reports the predicted label under five control conditions. The second is built around the word *put* under condition C1 and additionally records the confidence the model attached to each prediction; it subsumes an earlier four-input check (reference, silence, white noise, and a mismatched target) that produced the same pattern. The two probes do not use an identical set of controls: the raw-transcription request was run only in the first probe, where its purpose was to check whether a model would report anything at all about the recording, and it was replaced in the second probe by a reference-repeat condition once output stability became the more informative question. Because both probes test the same question with the same logic, they are reported together below.

B.2 Results

Tables B.1 and B.2 report the predicted labels for each probe. The clearest single observation is that white noise - an input carrying no linguistic content at all - was accepted as a clearly spoken word by all three models. The models diverge only once the input is silence or a mismatched word.

Table B.1: Predicted labels under the *daft* grounding probe (gold label: not intelligible).

Control input	Qwen2-Audio	Qwen2.5-Omni	MiniCPM-o-2.6
Genuine audio, target "daft"	Int.	Int.	Int.
Mismatched target "banana"	Int.	Int.	Int.
Silent audio	Int.	Int.	NI
White noise	Int.	Int.	Int.
Raw transcription	"unspecified"	"unclear"	fabricated sentence

Table B.2: Predicted labels and confidence under the *put* grounding probe (condition C1; gold label: intelligible).

Control input	Qwen2-Audio	Qwen2.5-Omni	MiniCPM-o-2.6
Reference recording	Int., medium	Int., high	Int., high
Mismatched audio	Int., medium	Int., high	Int., high
Silent audio	Int., medium	Int., high	NI, low
White noise	Int., medium	Int., high	Int., high
Wrong target word	Int., medium	Int., high	NI, low
Reference repeat	Int., medium	Int., high	Int., high

MiniCPM-o-2.6. MiniCPM was the only model whose predicted label changed under any control. It labeled the silent recording *not intelligible* in both probes and, in the *put* probe, also assigned the mismatched-prompt condition (the audio of *put* presented under the prompt word *paw*) the label *not intelligible* with low confidence. The change was not a confident negative judgment: rather than switching decisively, its confidence dropped to low, which suggests that the missing or conflicting signal introduced uncertainty rather than a clear rejection. Other controls produced no such change - white noise was still labeled intelligible, and the raw-transcription output for *daft* was an unrelated sentence rather than a single-word transcription.

Qwen2.5-Omni-7B. Qwen2.5-Omni showed little variation across the control conditions, returning the intelligible label with high confidence for every input, including silent audio and white noise. The accompanying `audio_evidence` field also remained broadly similar across conditions.

Qwen2-Audio-7B. Qwen2-Audio behaved similarly, assigning the intelligible label with medium confidence in every condition, including silent audio and white noise. Its `audio_evidence` field continued to describe speech content even when the input contained no speech at all.

Output stability. On the reference-repeat control, all three models returned the same label and confidence as on the first presentation, so the responses observed here are reproducible within these probes and are not an artifact of run-to-run variation.

B.3 Interpretation and Scope

These are anecdotal controls, not a benchmark. Each probe is built around a single target word and a small set of hand-constructed inputs, so the absolute numbers carry no statistical weight and no rate can be estimated from them. Their value is qualitative: on inputs where the correct answer is unambiguous - silence and white noise are not speech, and a mismatched recording is not the target word - two of the three models never use the acoustic signal, and the third uses it only weakly and without confidence. The checks should therefore be read as an existence proof that a grounding failure can be triggered in every model tested, not as a measurement of how often it would occur across the corpus. They also motivate the systematic, corpus-scale grounding analysis carried out with the C6+ variant in Section 5.5, which reaches the same conclusion on the full development set.

A fourth model, Qwen3-Omni-30B, was evaluated only in the *daft* probe and is not included in Table B.1. Its behavior was mixed: it rejected both the mismatched "banana" target and the white-noise recording, but accepted the silent recording as intelligible. It was not carried into the main evaluation for feasibility reasons and is noted here only for completeness.

Appendix C

Per-Word Error Counts

This appendix reports the raw per-word error counts underlying the class-specific metrics presented in Table 5.1. The table is included for completeness; the main analysis in Section 5.1 focuses on recall and $F_{0.5}$ because the gold-label distribution varies across target words.

Table C.1: Supplementary per-word error counts for MiniCPM-o-2.6 under condition C6 on the test set. *Missed problems* are gold `not_intelligible` recordings classified as `intelligible`; *false rejections* are gold `intelligible` recordings classified as `not_intelligible`.

Target word	n	Acc.	Gold NI	Missed	Gold I	False rej.
bird	157	0.363	103	97	54	3
Paul	157	0.363	100	100	57	0
pause	154	0.364	107	71	47	27
seedy	158	0.386	97	97	61	0
paw	157	0.408	120	76	37	17
daft	161	0.447	50	19	111	70
board	152	0.454	108	74	44	9
fur	158	0.456	92	53	66	33
farther	160	0.469	93	33	67	52
putt	160	0.469	120	66	40	19
pour	155	0.471	83	62	72	20
fir	157	0.478	96	72	61	10
bear	159	0.491	84	61	75	20
bard	158	0.500	112	70	46	9
bout	159	0.503	89	77	70	2
fair	156	0.513	76	76	80	0
pull	154	0.513	79	0	75	75
tide	154	0.519	92	55	62	19
pool	159	0.535	84	12	75	62
pit	161	0.540	48	31	113	43
tied	155	0.548	97	54	58	16
meat	153	0.549	60	46	93	23
wait	87	0.552	18	15	69	24
fern	156	0.571	64	34	92	33
doll	158	0.576	64	46	94	21
boat	161	0.578	60	46	101	22
pole	157	0.580	69	52	88	14
pore	156	0.603	72	20	84	42
father	161	0.609	68	55	93	8
pat	160	0.613	52	43	108	19

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Table C.1 (continued)

Target word	n	Acc.	Gold NI	Missed	Gold I	False rej.
boot	161	0.627	58	53	103	7
cot	157	0.631	55	33	102	25
pot	160	0.644	48	32	112	25
beer	160	0.650	72	56	88	0
knows	155	0.658	65	53	90	0
poor	155	0.664	105	2	50	50
half	158	0.690	49	49	109	0
put	160	0.694	48	44	112	5
plate	75	0.707	22	22	53	0
city	156	0.718	44	44	112	0
dance	158	0.753	34	28	124	11
bee	160	0.756	16	11	144	28
meet	154	0.766	36	36	118	0
pet	161	0.770	30	22	131	15
caught	159	0.774	123	0	36	36
mate	152	0.783	28	27	124	6
buy	160	0.787	34	34	126	0
paws	152	0.789	120	0	32	32
nose	157	0.796	32	32	125	0
bay	162	0.802	35	32	127	0
weight	155	0.806	30	30	125	0
hat	158	0.810	25	24	133	6
boy	161	0.839	26	26	135	0
Total	**8196**	**0.602**	**3592**	**2303**	**4604**	**958**

Appendix D

Per-Word Uncertainty Behavior

This appendix reports the per-word uncertainty rates underlying the analysis in Section 5.2.

Table D.1: Per-word uncertainty behavior across the 53 target words for MiniCPM-o-2.6 under condition C6 on the test set. *Missed* and *False rej.* denote not-intelligible recordings accepted as intelligible and intelligible recordings flagged as not intelligible, respectively. *Pred. not int.* is the proportion of recordings classified as `not_intelligible`, and *Uncertainty* the proportion flagged as uncertain. Reading the *poor* row: of its 155 recordings, 98.7% were classified `not_intelligible` and 91.6% were flagged as uncertain.

Word	n	Acc	Missed	False rej.	Pred. not int. (%)	Uncertainty (%)
bird	157	0.363	97	3	5.7	5.1
Paul	157	0.363	100	0	0.0	0.0
pause	154	0.364	71	27	40.9	35.7
seedy	158	0.386	97	0	0.0	0.0
paw	157	0.408	76	17	38.9	34.4
daft	161	0.447	19	70	62.7	59.0
board	152	0.454	74	9	28.3	28.3
fur	158	0.456	53	33	45.6	40.5
farther	160	0.469	33	52	70.0	68.1
putt	160	0.469	66	19	45.6	42.5
pour	155	0.471	62	20	26.5	26.5
fir	157	0.478	72	10	21.7	20.4
bear	159	0.491	61	20	27.0	26.4
bard	158	0.500	70	9	32.3	31.0
bout	159	0.503	77	2	8.8	9.4
fair	156	0.513	76	0	0.0	0.6
pull	154	0.513	0	75	100.0	36.4
tide	154	0.519	55	19	36.4	31.8
pool	159	0.535	12	62	84.3	56.6
pit	161	0.540	31	43	37.3	35.4
tied	155	0.548	54	16	38.1	36.1
meat	153	0.549	46	23	24.2	22.9
wait	87	0.552	15	24	31.0	32.2
fern	156	0.571	34	33	40.4	36.5
doll	158	0.576	46	21	24.7	22.8
boat	161	0.578	46	22	22.4	21.7
pole	157	0.580	52	14	19.7	17.2
pore	156	0.603	20	42	60.3	56.4

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Table D.1 (continued)

Word	n	Acc	Missed	False rej.	Pred. not int. (%)	Uncertainty (%)
father	161	0.609	55	8	13.0	12.4
pat	160	0.613	43	19	17.5	19.4
boot	161	0.627	53	7	7.5	9.9
cot	157	0.631	33	25	29.9	31.2
pot	160	0.644	32	25	25.6	23.1
beer	160	0.650	56	0	10.0	10.6
knows	155	0.658	53	0	7.7	10.3
poor	155	0.664	2	50	98.7	91.6
half	158	0.690	49	0	0.0	0.6
put	160	0.694	44	5	5.6	5.0
plate	75	0.707	22	0	0.0	1.3
city	156	0.718	44	0	0.0	1.9
dance	158	0.753	28	11	10.8	12.0
bee	160	0.756	11	28	20.6	18.8
meet	154	0.766	36	0	0.0	1.9
pet	161	0.770	22	15	14.3	13.7
caught	159	0.774	0	36	100.0	64.8
mate	152	0.783	27	6	4.6	5.3
buy	160	0.787	34	0	0.0	1.9
paws	152	0.789	0	32	100.0	42.8
nose	157	0.796	32	0	0.0	0.0
bay	162	0.802	32	0	1.9	1.8
weight	155	0.806	30	0	0.0	1.9
hat	158	0.810	24	6	4.4	5.1
boy	161	0.839	26	0	0.0	0.0
Overall	**8196**	**0.602**	**2303**	**958**	**27.4**	**23.2**

Appendix E

Keyword Classes Used to Categorize the Model's Reasons

The keyword audit reported in Section 5.3 gives a coarse map of which explanation themes appear in the model's own text. It is not a substitute for the manual annotation of Section 5.6; it is a whole-corpus check on whether the themes found in the annotated sample generalize to all 8,196 valid test predictions.

E.1 Matching Procedure

For each prediction, the `short_reason` and `audio_evidence` fields were concatenated into a single string, lower-cased, and padded with one space at each end. The padded string was then tested for the presence of any of the match strings belonging to each class. Matching is plain case-insensitive substring containment; no regular expressions and no word-boundary constraints were used, except for the space-padded `" r "` pattern in the rhoticity class, where the surrounding spaces act as a manual word boundary. The exact code was:

```
def contains_any_keyword(text, keywords):
    text = ' ' + str(text).lower() + ' '
    return any(keyword in text for keyword in keywords)
```

A prediction is credited to a class once if any of that class's strings is present, regardless of how many times or in which of the two fields. Because a single explanation can satisfy several classes, the percentages reported in Table 5.3 do not sum to 100.

E.2 Match Strings

Table E.1: The complete set of match strings used for each of the seven keyword classes, reproduced from the analysis notebook. Strings are matched as case-insensitive substrings of the concatenated `short_reason` and `audio_evidence` fields.

Class (as reported)	Internal name	Match strings
no major issue	`no_major_issue`	`no major issue`, `understandable`, `clearly intelligible`, `can be understood`
unclear audio	`unclear_audio`	`unclear`, `ambiguous`, `weak`, `noisy`, `noise`, `clipped`, `incomplete`, `difficult to hear`
different word heard	`different_word`	`different word`, `sounds like`, `could be heard as`, `perceived as`, `not the target`
consonant feature	`consonant`	`consonant`, `final consonant`, `initial consonant`, `middle consonant`
vowel feature	`vowel`	`vowel`, `quality`, `duration`, `length`, `short`, `long`, `diphthong`
aspiration	`aspiration`	`aspiration`, `aspirated`, `puff`, `burst`
rhoticity (/r/)	`r_or_rhoticity`	`rhotic`, `" r "` (a single `r` with a space on each side), `/r/`, `r-color`, `r color`

E.3 Limitations of the Keyword Audit

The keyword audit was designed as a broad descriptive scan rather than as a precise linguistic annotation. First, some strings are semantically broad. For example, `quality` is included in the vowel class and may therefore match a phrase such as *poor audio quality*, even when the text does not refer to vowel quality. Second, matching used simple substring search without general word boundaries. Consequently, strings such as `long` and `short` may also occur inside longer words. Third, the matcher did not account for negation, so a phrase such as *not understandable* would still match the `no_major_issue` class. The reported rates for broad classes such as vowel and no major issue should therefore be interpreted as approximate upper bounds rather than exact linguistic frequencies.

None of these properties undermines the two absence-based findings for which the audit is used. The `aspiration` class matched none of the 8,196 predictions on any of its four strings, and the `r_or_rhoticity` class matched exactly one. Because the matching procedure is relatively inclusive, these near-absences are unlikely to be artifacts of an overly restrictive matcher. They should nevertheless be interpreted as properties of the generated text rather than as direct evidence about the acoustic signal itself.

Data Statement

This data statement documents the corpus on which the internship and this thesis were carried out. The purpose of this document is to describe the data such as where the recordings came from, who produced them, who labeled them, under what conditions, what is known about those people and what is not, and what may be concluded from the result. The corpus was still being built while this thesis was written and no data statement had been prepared for it before.

A Curation Rationale

A.1 Origin

The data is a learner speech corpus collected through the Massive Open Online Course *English Pronunciation in a Global World*, which is produced by the Centre for Global English (CGE) at Vrije Universiteit Amsterdam and hosted on FutureLearn, where it has attracted more than 139,000 learners (FutureLearn, 2026). The course was created and is taught by Dr. Laura Rupp, head of the CGE and a specialist in English phonology.

Spoken submission is one component of the course. Learners record themselves and upload the audio to the course server so that it can be given pedagogical feedback. The assignment has more than one part, reading aloud a fixed word list is just a part of it. Only the word-list material is used in this thesis.

A.2 Original purpose

The recordings were not collected in order to build or evaluate computational systems. They were collected as pronunciation practice within a teaching course, and the possibility of using them for machine learning arose only afterwards, as a way of eventually improving the course itself. Most of the properties documented in this statement follow from that origin: the sparse metadata, the uncontrolled recording conditions, the self-selected submissions and the absence of a pre-existing data statement are all consequences of a collection designed for teaching rather than for research. The use of the corpus in this thesis, as in the two prior theses on it, is therefore a secondary use of data gathered for a different original purpose.

A.3 Two releases

MOOC-2024. 2,115 word-level clips derived from 41 participant-level recordings. This is the release used by Zou (2024).

MOOC-2025. Approximately 10,200 word-level clips recorded and labeled by the CGE after the 2024 release. This is the release used by Kuan (2025) and by the present

thesis; the material reconstructed here comprises 10,197 clips.

The corpus should be understood as a living resource; not a fixed one. The CGE has continued to record and label material, and the annotation of MOOC-2025 was not closed at the time this thesis was written.

A.4 The word list as an instrument

The word list is not a sample of English but a designed instrument. Dr. Rupp adapted it from an existing word list; the source of that original list was not obtained during this project, so the list cannot at present be traced to a published reference.

The words predominantly follow a consonant(-cluster)-vowel-consonant(-cluster) pattern, and they are arranged into minimal and near-minimal pairs so that a small number of phonetic contrasts recur across the list. Vowel length is contrasted through pairs such as *pit*/*Pete*, *city*/*seedy*, *pull*/*pool*, *cot*/*caught* and *meet*/*mid*; rhoticity through *father*/*farther* and *fir*/*fear*; and initial-stop aspiration through *pit*/*bit* and *pull*/*bull*. The principle behind the design is that a word becomes unintelligible when a phonetic difference makes it indistinguishable from one of its minimal pairs, so that the listener may recover a different word from the one intended, and the list is constructed so that this can happen.

Two consequences follow from this. The distribution of phonetic difficulty within the corpus is a property of the instrument rather than of English. And per-word results describe how a system behaves on these constructed contrasts; they do not estimate how often such contrasts cause a communication failure in unconstrained speech.

The annotation instrument was designed for this list in particular. The guidelines note that the feature set "is designed specifically with the current word list in mind and might thus not be optimal for all other words", while allowing for later expansion.

B Language Variety

The recordings are productions of English by second-language speakers, and no native target variety is assumed. The course does not teach a British or American standard; its criterion is global intelligibility. Rhoticity illustrates the difference most clearly. Learners are encouraged to pronounce /r/ where it is written, not because a non-rhotic pronunciation is wrong, but because a pronounced /r/ is more widely recoverable across listeners with different first languages.

The labeling criteria were developed with Dr. Rupp on an intelligibility-focused framework drawing on Jenkins's work on English as an International Language. That framework is used to identify which phonetic features may matter for mutual understanding rather than to impose a fixed definition of intelligibility or a native-speaker standard. The project is background-agnostic by design: instead of asking how one particular language group pronounces English, it asks which features support mutual understanding across a diverse global population.

C Speaker Demographics

No speaker demographic metadata is available for this corpus, and this is the property that most constrains what can be claimed from it. No demographic information was

collected at enrolment, and no form or agreement covering demographic data was in place when learners registered, so age, gender, race or ethnicity, first language, nationality, country of origin and educational background are all unavailable. For this reason no claim anywhere in this thesis is conditioned on a speaker attribute.

C.1 Self-selection

Learners controlled the recording and submission process themselves, so they were able to practice a word, record several attempts and upload the version they preferred. Some are likely to have done this, while others may have submitted a first or practice take, and no statistics on the number of attempts, repetitions or trials exist for any learner. The uploaded file is therefore not necessarily a first attempt, and the corpus carries a degree of self-selection that cannot be measured. The guidelines describe intelligibility as the outcome of a rule-based system in which human-defined if-then rules are applied to annotated phonetic features. The binary label is therefore not an independent holistic judgment made by a listener after hearing the recording, but the result of applying those rules to the feature annotations. This way, the final label reflects the relative importance assigned to each phonetic feature by the rule set; and the binary label should not be interpreted as a direct measure of human perception, but as the output of a predefined decision procedure built on annotated phonetic features.

D Annotator Demographics

D.1 Gold intelligibility labels

Segmentation and labeling of MOOC-2025 were carried out by two student interns at the CGE, Lianne Dekker and Cecilia Montenegro Pérez. They worked from written guidelines developed with Dr. Rupp, and ambiguous cases were documented and reviewed with her so that the guidelines were applied consistently. The annotation scheme was written with listener variation in mind. It asks annotators to mark observable properties, such as whether aspiration is present, rather than to give holistic impressions, and the stated aim of this is to reduce listener-specific bias.

E Speech Situation

- **Time and place.** Recorded by learners in their own homes and environments. Exact recording dates are not preserved in the material available for this thesis.
- **Supervision.** None. The recording process was not controlled or observed by anyone from the course or from the project, so this is not a laboratory collection.
- **Modality.** Spoken, read aloud, and non-interactive. Each clip contains a single word.
- **Synchronicity.** None. No interlocutor was present, so there was no communicative feedback of any kind and no opportunity for repair.
- **Intended audience.** The course, for pronunciation feedback. Learners were not producing speech for an automatic system and had no reason to accommodate to one.

- **Equipment.** Learners own devices. No equipment was specified or controlled.
- **Participation.** Voluntary, and within an online educational course rather than a research experiment. Learners chose to enroll, chose to submit, and chose what to submit.

F Preprocessing

Segments containing abnormal breathing or unrelated speech were handled separately during segmentation, and a filename suffix (`0audio`) was used to flag problematic recordings, meaning clips with excessive background noise, poor microphone quality, or extremely soft volume. No further processing was applied for this thesis. No normalization, denoising, resampling or trimming was performed, and clips were passed to the models exactly as they were received.

G Corpus Statistics

Word list. The version used here contains 53 target words, whereas the guidelines document a list of 52. The additional item is *wait*, which occurs in only part of the corpus and is best understood as a variant of *weight* arising in some course runs: it has 87 test clips against a median of about 157 for the other words. *Plate* is similarly under-represented at 75 test clips and is marked as a special case in the guidelines.

Filenames and reconstructed metadata. There was no metadata file distinguishing development and test material, so the split, the target word and the gold label were reconstructed from the filenames, which follow the schema `series<NN>-s<ID>-<word>-<label>.mp3`, where the final segment is `1` for intelligible and `0` for not intelligible. Nineteen test filenames contained a stray space before the label segment and were parsed separately. The resulting metadata table records, for each clip, the filename, the filepath, the course series, the student identifier, the target word, the gold label and the duration in seconds.

Splits and label balance.

Split	N	*intelligible*	*not_intelligible*	% not int.
Development	2,001	1,128	873	43.6
Test	8,196	4,604	3,592	43.8
Total	10,197	5,732	4,465	43.8

The class balance is nearly identical across the two splits, though it is not uniform across words: the per-word share of intelligible clips ranges from 21.1 % for *paws* to 90.0 % for *bee*, and per-word clip counts range from 75 to 162 (Table 3.1).

Split construction and speaker overlap. For each of the 53 target words, both development and test sets contain every target word and no individual clip appears in both. The split is therefore disjoint at the level of clips but not at the level of speakers, since the same learner may contribute clips to both sets. Test-set results obtained on this corpus should accordingly be read as generalization to unseen recordings, not to unseen speakers.

H Recording Quality

The audio is not acoustically uniform, and it was never intended to be. Hardware, recording environment, distance from the microphone, ambient noise and the care taken by the speaker all vary from one recording to the next. The corpus reflects the conditions under which learners actually submit speech to an online course, and a system intended for that setting has to cope with them. It does mean, however, that recording quality cannot be separated from pronunciation quality: where a system makes an error, part of that error may originate in the recording rather than in the learner's production. Apart from the `0audio` filename flag, the corpus provides no independent measure of audio quality against which such a claim could be checked.

I Provenance of the Labels - Annotation Scheme

Feature categories. Annotators did not mark intelligibility directly; they marked phonetic features on eight categories.

Category	Labels
rhoticity	MISSING
vowel length	SHORTENING, LENGTHENING
vowel quality	UNACCEPTABLE SUBSTITUTION, OTHER SUBSTITUTION
consonant initial	MISSING ASPIRATION, SUBSTITUTION
consonant middle	SUBSTITUTION
consonant final	SUBSTITUTION
extra sound	INITIAL, MIDDLE, END
missing sound	INITIAL, MIDDLE, END

J Consent, Privacy and Distribution

Learners gave consent for recording audio to the course server. Participation was voluntary throughout and took place within an educational course. All data was anonymized before use. Direct learner identities are not available to the researcher; the speaker identifiers in the filenames are pseudonymous.

The work was carried out under a VU Amsterdam work placement agreement with the Faculty of Humanities, English Linguistics, running from 1 April to 30 June 2026. That agreement requires all information accessed during the placement to be treated as confidential and not made available to third parties, and this requirement remains in force after the placement has ended. The audio is therefore not redistributed with this thesis and is not archived alongside it.

The prompt texts (Appendix A), the annotation scheme and the derived per-word statistics reported in this thesis are sufficient to follow and check the analyses without access to the audio itself. The code that produced these analyses is available separately[1].

[1] `https://github.com/massoumehjavadi-cloud/speech-llm-for-intelligibility-classification`

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